

A geometric model for instantaneous phase and frequency of multivariate signals

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Abstract

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Keywords: Multivariate time series, non-stationarity, instantaneous phase, fiber bundle, geometric phase.

1 Introduction

The notion of phase is as old as wave mechanics and wave optics. It has been recognised as a central concept with the development of the mathematical physics of waves during the 19th century, and its fundamental importance in many aspects of science has become ubiquitous in the 20th century, in quantum mechanics, in the theory of dynamical systems, in the theory of oscillations and waves, in observational astronomy, in signal and image processing, etc.

In signal processing, the notion of phase was mathematically developed after WWII by Gabor and Ville; It was shown to rely on the notion of analytic signal [1, 2]. Its importance nowadays ranges from fundamental analysis (delay estimation, correlation or spectral analysis) to filtering, together with its daily usage in numerous applications (radar, imaging, remote sensing, telecommunications, etc.). For non-stationary

signals, instantaneous phase and amplitude are of major interest to track time evolution of the signal characteristics, and is at the roots of time-frequency analysis. The importance of the notion of phase is well understood and documented, especially for univariate signals. Basically, the decomposition of a univariate signal into instantaneous amplitude and phase relies on the complex polar form $x(t) = a_x(t)e^{j\varphi_x(t)}$ where $a_x(t)$ is the instantaneous amplitude and $\varphi_x(t)$ the instantaneous phase. For a real valued signal, as developed in [3], the instantaneous phase is defined as the argument of the analytic signal associated to the real signal, and the instantaneous amplitude its modulus. Under some assumptions (Bedrosian conditions), the instantaneous frequency defined as the time derivative of the instantaneous phase provides a well mathematically grounded interpretation of the local frequency content of the signal.

For multivariate signals $\mathbf{x}(t) \in \mathbb{C}^n$ however, the notion of instantaneous phase is not as well understood. A clear definition is even absent in classical textbooks. Interesting developments have been proposed by Lilly and his coauthors in a series of papers [4–7]. They study a particular class of multivariate signals, called *modulated elliptical signals*, and propose a definition of instantaneous phase and frequency for such signals. It is based on a AM-FM decomposition of the multivariate signal, which is a natural generalization of the univariate case. The decomposition they arrive at writes $\mathbf{x}(t) = \mathbf{a}_x(t)e^{j\varphi_x(t)}\mathbf{p}_x(t)$, where $\mathbf{a}_x(t) \in \mathbb{R}^+$ is the instantaneous amplitude, $\varphi_x(t)$ is a common phase and $\mathbf{p}_x(t) \in \mathbb{C}^n$ is a normalized n -dimensional complex-valued vector containing the remaining parameters of $\mathbf{x}(t)$. In their work, vector $\mathbf{p}_x(t)$ takes a particular parametrization. However, the decomposition remains ambiguous, and one aim of the present work is to clarify this ambiguity.

Multivariate functions $\mathbf{x}(t) \in \mathbb{C}^n$ occur in other fields of science, such as quantum mechanics, where they represent wave functions, whose square modulus represent probability densities. In this context, the notion of phase is also central, and has been studied in depth. In particular, it has been shown that the phase can be decomposed into two components: a dynamical phase and a geometric phase [8–10]. The geometric phase is a global property of the signal, which depends on the path taken by the signal in its parameter space. It is related to the curvature of the parameter space, and can be interpreted as a holonomy effect. The geometric phase has been observed in many physical systems, and has important implications for quantum mechanics and quantum information theory.

The aim of the present work is to show that the notion of geometric phase also occurs in multivariate signal processing, and that it has important implications for the definition of instantaneous phase and frequency. The main contributions of the work are the following:

- We clarify the notion of instantaneous phase for multivariate signals, and show that it can be decomposed into the sum of a dynamical phase and a geometric phase. A first proof relies on signal processing arguments. We also show the impact of this findings on the notion of instantaneous frequency.
- Using differential geometry tools, we provide another proof and show that the geometric phase can be computed from the path taken by the signal in its parameter space.

- For the particular case of AM-FM signals, we provide explicit calculations of the different phases for different equivalent parametrizations of the signals; We particularly detail the bivariate and trivariate cases, and show how they are related. We also introduce the Majorana stellar representation in order to handle n dimensional signals as $n - 1$ bivariate interacting signals.
- We provide a practical algorithm for the estimation of the geometric phase of a multivariate signal, based on the computation of the so-called Bargmann invariants. We show that the algorithm can be applied to real-world signals, and that it provides an accurate estimation of the geometric phase.
- We provide several illustrative examples of multivariate signals, and show how the geometric phase can be computed and interpreted in each case. We also show how the geometric phase can affect the estimation of instantaneous frequency, and how it can be used to improve the analysis of multivariate signals. Finally, we discuss potential applications of the existence of geometric phases in array processing.

The main results are the following. A complex square integrable multivariate signal $\mathbf{x}(t) \in \mathbb{C}^n$ is written in the form $\mathbf{x}(t) = \mathbf{a}_x(t)\mathbf{p}_x(t)$ where $\mathbf{a}_x(t) \in \mathbb{R}^+$ is a real scalar amplitude and $\mathbf{p}_x(t) \in \mathbb{C}^n$ a normalized n -dimensional complex-valued vector. This decomposition is not unique, since $\mathbf{x}(t) = \mathbf{a}_x(t)\mathbf{q}_x(t)$ with $\mathbf{q}_x(t) = e^{j\theta(t)}\mathbf{p}_x(t)$ is also a valid decomposition.

The instantaneous phase $\phi_{\mathbf{x},t_0}(t)$ relative to a reference time t_0 is defined as the argument of the inner product $\mathbf{x}(t_0)^\dagger \mathbf{x}(t)$. It can be decomposed as

$$\phi_{\mathbf{x},t_0}(t) = \phi_{d,\mathbf{x},t_0}(t) + \phi_{g,\mathbf{x},t_0}(t) \text{ where} \quad (1)$$

$$\phi_{d,\mathbf{x},t_0}(t) = \int_{t_0}^t \frac{\text{Im}(\mathbf{x}(\tau)^\dagger \dot{\mathbf{x}}(\tau))}{\|\mathbf{x}(\tau)\|^2} d\tau \quad (2)$$

$$\phi_{g,\mathbf{x},t_0}(t) = \arg(\mathbf{p}_x(t_0)^\dagger \mathbf{p}_x(t)) - \int_{t_0}^t \text{Im}(\mathbf{p}_x(\tau)^\dagger \dot{\mathbf{p}}_x(\tau)) d\tau \quad (3)$$

$\phi_{d,\mathbf{x},t_0}(t)$ is called the *dynamical phase* and depends on the choice of the particular decomposition of $\mathbf{x}(t)$. The instantaneous frequency is defined as the time derivative of the dynamical phase.

$\phi_{g,\mathbf{x},t_0}(t)$ is called the *geometric phase* and is independent of the choice of the particular decomposition of $\mathbf{x}(t)$, and is thus a global property of the signal. It is a pure geometric quantity as it only depends on the path $\rho_x(t) = \mathbf{x}(t)\mathbf{x}(t)^\dagger / \|\mathbf{x}(t)\|^2$ created by projecting the signal into a space called the base space (the space of density matrices in quantum mechanics). The main properties of the geometric phase are the following:

- *Reference and time interval*: The geometric phase is defined with respect to a reference time t_0 and is a function of time t (integration from t_0 to t). The geometric phase is a quantity accumulated over a time interval and depends on the origin t_0 .
- *Gauge invariance*: Signals $\mathbf{x}(t)$ and $\mathbf{x}(t)e^{j\alpha(t)}$ accumulate the same geometric phase over a time interval. It is a pure geometric quantity as it only depends on the path in the base space.

- *Time reparametrization*: For a given smooth bijective function $s : [t_0, t] \rightarrow s([t_0, t])$, $\mathbf{x}(t)$ and $\mathbf{x}(s(t))$ have the same geometric phase accumulated at time t , meaning that it does not matter how fast or slow the path $\rho_{\mathbf{x}}(t)$ is followed in the base space. This also implies that the geometric phase is independent of the carrying frequency of the signal. Said differently, two signals having different carrying frequency accumulate the same geometric phase if their initial and final points are the same in the base space
- *Geodesic path*: The geometric phase accumulated along a geodesic path is null.
- *Single point path*: If the path in the projective space reduces to a single point, then there is no geometric phase accumulated. This corresponds to a constant value. A geometric phase can only appear if a change occurs in $\mathbf{p}_{\mathbf{x}}(t)$.
- *Area surrounded by the path $\mathcal{C}$ drawn by $\rho(t)$* : If the path $\mathcal{C}$ between $t = 0$ and $t = 1$ is a *loop* ($\rho_{\mathbf{x}}(0) = \rho_{\mathbf{x}}(1)$), then the geometric phase can be obtained from the area inside the loop [9, chapter 10]. If the path is *open* (such that $\rho_{\mathbf{x}}(1) \neq \rho_{\mathbf{x}}(0)$), then it is possible to close the loop by a geodesic from $\rho_{\mathbf{x}}(1)$ and $\rho_{\mathbf{x}}(0)$ as geodesic paths has null geodesic contribution, and the area surrounded is still connected to the geometric phase.

The structure of the paper is as follows: Section 2 illustrates how the instantaneous phase and frequency concepts have to be carefully handled in the multivariate case and how they may incorporate a geometric component. Section 3 provides a mathematical description of the geometric phase concept that can affect multivariate complex signals using the language of fiber bundles. Section 4.5.3 treats the particular case of modulated multivariate signals, and provides explicit calculations of the different phases; especially for the bivariate and trivariate cases. In section 5, the practical issue of estimating the geometric phase of a multivariate signal is addressed, while section 6 introduces several illustrative examples where geometric phases occur, for bivariate and multivariate signals.

1.1 Notations and conventions

Throughout the paper, a multivariate signal is a map $\mathbf{x} : I \rightarrow \mathbb{C}^n$, $t \mapsto \mathbf{x}(t)$ defined on a time interval $I \subseteq \mathbb{R}$ and assumed continuously differentiable with $\mathbf{x}(t) \neq 0$ for all $t \in I$. Following signal processing tradition, we write $\mathbf{x}$ or $\mathbf{x}(t)$ for the signal interchangeably. Pointwise statements such as $\mathbf{x}(t) \in \mathbb{C}^n$ are understood for all $t \in I$. Univariate signals are denoted in standard italic font, e.g., x . Table 1 collects the main notations used in the paper.

JF:Add a statement about our convention of $\arg$ as a continuous determination in t .

2 The instantaneous phase of a multivariate signal

The concept of phase is well known in signal and image processing. As recalled in the Introduction it is a fundamental notion central in many applications. To discuss the concept for multivariate signals, it is interesting to first examine the case of scalar signals and recall some basic ideas which allows the understanding of phases. As well

Symbol.	Description
$\mathbb{R}, \mathbb{C}$	set of real and complex numbers
$\mathbb{CP}^{n-1}$	complex projective space of $\mathbb{C}^n$
$\mathcal{S}^{2n-1}$	unit sphere in $\mathbb{C}^n$
$x(t)$	univariate signal taking values in $\mathbb{C}$
$\mathbf{x}(t)$	multivariate signal taking values in $\mathbb{C}^n$
$\mathbf{x}^\dagger(t)$	transposed-conjugate of $\mathbf{x}(t)$
$\langle \mathbf{x}, \mathbf{y} \rangle$	hermitian inner product $\langle \mathbf{x}, \mathbf{y} \rangle := \mathbf{x}^\dagger \mathbf{y}$
$\ \mathbf{x}\ $	associated norm
$\rho_{\mathbf{x}}(t)$	projection operator of $\mathbf{x}(t)$
$A_{\mathbf{x}}$	linear form of Ehresmann connection, Berry connection
$\dot{\mathbf{x}}(t)$	time derivative of $\mathbf{x}(t)$
$a_{\mathbf{x}}(t)$	amplitude of $\mathbf{x}(t)$
$\mathbf{p}_{\mathbf{x}}(t)$	unit vector $\mathbf{x}(t)/a_{\mathbf{x}}(t)$
$\phi_{\mathbf{x}, t_0}(t)$	instantaneous phase at t relative to t_0
$\phi_{d, \mathbf{x}, t_0}(t)$	dynamical phase at t relative to t_0
$\phi_{g, \mathbf{x}, t_0}(t)$	geometrical phase accumulated at t relative to t_0

Table 1 Summary of notations.

known since the 19th century, using the complex representation of signals permits an easy manipulation of phases, as angles between vectors. This particular point puts also forward the central idea of the relativity of phases: The phase has a meaning only when there is a reference.

We elaborate in this section on these two basic ideas, first recalling known facts in the scalar case, and then on developping the notion of instantaneous phase for multivariate signals and on examining the consequence of the multivariate character on the notion.

2.1 The univariate case

Consider a univariate complex-valued signal $t \mapsto x(t)$, typically constructed as the *analytic signal* of a real-valued signal. We assume that x is smooth (at least, continuously differentiable) throughout. On any interval where $x(t) \neq 0$, the signal x admits a polar form $x(t) = a_x(t) \exp(j\varphi_x(t))$ with $a_x(t) = |x(t)| > 0$ and $\varphi_x(t)$ (the unwrapped argument of x) continuous; in particular a_x is unique and φ_x is unique up to an additive constant $2k\pi$. Picking an arbitrary time t_0 as a reference, the *instantaneous phase relative to t_0* is defined as

$$\phi_{x, t_0}(t) = \varphi_x(t) - \varphi_x(t_0) = \arg(x^*(t_0)x(t)) \quad (4)$$

where throughout the paper, the argument of a time-dependent quantity denotes its continuous determination in t vanishing at $t = t_0$, a procedure also referred to as unwrapping. Note that ϕ_{x, t_0} does not depend on the $2k\pi$ ambiguity of φ_x .

When x is the analytic signal of a real signal $A(t) \cos \Psi(t)$, a natural question is whether the quantities a_x and φ_x coincide with A and Ψ . This is only possible under conditions on the amplitude and frequency/phase modulation (AM-FM), such as a slow-modulation assumption [2] or disjoint spectral support [11]; see also [3] for further discussion. For this reason, the quantities a_x and φ_x are usually referred to as instantaneous amplitude and phase, respectively. To avoid confusion with ϕ_{x,t_0} we will simply refer to φ_x as the argument of x . Note finally that none of the above conditions are needed for the results of this section.

The phase relative to t_0 can be obtained in two ways: globally, by comparing $x(t)$ with the reference $x(t_0)$ as in (4), or locally, by accumulating the phase increments of x between t_0 and t . Since $x^* \dot{x}/|x|^2 = \dot{x}/x = \dot{a}_x/a_x + j\dot{\varphi}_x$, we have $\text{Im}(x^* \dot{x})/|x|^2 = \dot{\varphi}_x$, and therefore

$$\phi_{x,t_0}(t) = \int_{t_0}^t \frac{\text{Im}(x^*(s)\dot{x}(s))}{|x(s)|^2} ds. \quad (5)$$

In the univariate case, global and local approaches thus give the same result. The integrand defines the instantaneous frequency of x ,

$$\nu_x(t) = \frac{\text{Im}(x^*(t)\dot{x}(t))}{2\pi|x(t)|^2} = \frac{1}{2\pi}\dot{\varphi}_x(t) = \frac{1}{2\pi}\dot{\phi}_{x,t_0}(t), \quad (6)$$

denoted ω_x (up to a factor 2π) in [4, 6]. It does not depend on t_0 and coincides with the usual definition as the derivative of φ_x . However, unlike $\dot{\varphi}_x$, the first expression in (6) requires no polar form; it also generalizes to multivariate signals, where it does no longer coincides with the derivative of the phase, as explained in Section 2.4.

The instantaneous frequency is also related to the spectral content of x . Let $X(\nu)$ be the Fourier transform of x and assume, without loss of generality, that x has unit energy $\int_{\mathbb{R}} |x(t)|^2 dt = \int_{\mathbb{R}} |X(\nu)|^2 d\nu = 1$. Then, classical manipulations using Fourier transform properties [1] yield

$$\int_{\mathbb{R}} \nu |X(\nu)|^2 d\nu = \int_{\mathbb{R}} \nu_x(t) |x(t)|^2 dt, \quad (7)$$

i.e., the mean frequency of x is the average of ν_x weighted by the instantaneous power $|x(t)|^2$. The identity (7) does not allow to determine ν_x uniquely; a stronger result is that $\nu_x(t)$ is the conditional mean frequency at time t of the Wigner–Ville distribution of x [1, 2].

For multivariate signals, we will show, perhaps surprisingly, that the global and local approaches differ generally by an additional term, the geometric phase.

2.2 In-phase multivariate signals

As seen above, a phase is only defined relative to a reference. For univariate signals this is unambiguous, as two signals $x(t)$ and $y(t)$ can only differ by a positive scaling and phase shift at each time instant. In particular, taking $y(t) = x(t_0)$ permits to define the instantaneous phase ϕ_{x,t_0} of x with respect to t_0 , see (4). For two complex-valued multivariate signals $\mathbf{x}(t)$ and $\mathbf{y}(t)$, defining the notion of *phase shift* (or relative phase) is ambiguous *a priori*, since at a given time t , they may differ by

a relative phase factor but also by their direction. For bivariate signals, i.e., signals in $\mathbb{C}^2$ one can address this question by leveraging their interpretation as time-dependent Jones polarization vectors in optics (see for instance [12] for an introduction). Indeed, in this context, the same exact question was addressed by Pancharatnam in 1956. He proposed the following definition [13, p. 252]:

[...] the phase advance of one polarised beam over another (not necessarily in the same state of polarisation) is the amount by which its phase must be retarded relative to the second, in order that the intensity resulting from their mutual interference may be a maximum.

In signal terms, the phase advance of $\mathbf{y}(t)$ over $\mathbf{x}(t)$ is the function $\alpha(t)$ maximizing, at each time t , the norm of the superposition $\mathbf{x}(t) + e^{-j\alpha}\mathbf{y}(t)$. Since

$$\|\mathbf{x}(t) + e^{-j\alpha(t)}\mathbf{y}(t)\|^2 = \|\mathbf{x}(t)\|^2 + \|\mathbf{y}(t)\|^2 + 2\operatorname{Re}\left(e^{-j\alpha(t)}\mathbf{x}^\dagger(t)\mathbf{y}(t)\right), \quad (8)$$

the maximum is reached at $\alpha(t) = \arg(\mathbf{x}^\dagger(t)\mathbf{y}(t))$, provided $\mathbf{x}^\dagger(t)\mathbf{y}(t) \neq 0$; if, for a given t , $\mathbf{x}^\dagger(t)\mathbf{y}(t) = 0$ the norm does not depend on α and the phase shift is undefined.

A few observations are in order. First, Pancharatnam's argument only involves the Hermitian inner product; its extension to multivariate signals in $\mathbb{C}^n$ is therefore straightforward. Second, the value of $\theta(t) := \arg(\mathbf{x}^\dagger(t)\mathbf{y}(t))$ defines, at time t , the phase of $\mathbf{y}(t)$ relative to $\mathbf{x}(t)$. Third, we can observe that two signals $\mathbf{x}$ and $\mathbf{y}$ are *in phase* at time t if

$$\operatorname{Im}(\mathbf{x}^\dagger(t)\mathbf{y}(t)) = 0 \quad \text{and} \quad \operatorname{Re}(\mathbf{x}^\dagger(t)\mathbf{y}(t)) > 0. \quad (9)$$

For $n = 1$ this reduces to the univariate relative phase (4). However, unlike the univariate case, this relation is not transitive. For instance, at time t , suppose on the one hand that signals $\mathbf{x}(t)$ and $\mathbf{y}(t)$ are in phase, and on the other hand that $\mathbf{y}(t)$ and $\mathbf{z}(t)$ are in phase; then, in general this does not imply that $\mathbf{x}(t)$ is in phase with $\mathbf{z}(t)$. This lack of transitivity is the origin of the geometric phase. **JF:mention this is actually the idea behind Bargman's invariant? ref?**

2.3 Instantaneous phase of a multivariate signal

We now turn our attention to the definition of the instantaneous phase of a multivariate signal $\mathbf{x}$. As in the univariate case, such definition is relative and uses the value of the signal at a particular time t_0 as reference.

Definition 1 (Instantaneous phase of a multivariate signal). *Let t_0 be such that $\mathbf{x}(t_0) \neq 0$. The instantaneous phase of $\mathbf{x}(t)$ with respect to t_0 is defined as*

$$\phi_{\mathbf{x},t_0}(t) = \arg(\mathbf{x}^\dagger(t_0)\mathbf{x}(t)) \quad (10)$$

*as long as $\mathbf{x}^\dagger(t_0)\mathbf{x}(t) \neq 0$. **JF:how to handle the case of $\mathbf{x}(t_0)^\dagger\mathbf{x}(t) = 0$?***

This definition extends the *global* univariate case (4) to the general case by using Pancharatnam's notion of relative phase introduced in Section 2.2. Indeed $\phi_{\mathbf{x},t_0}$ is

precisely the relative phase between $\mathbf{x}(t)$ and the fixed reference $\mathbf{x}(t_0)$. Moreover, the quantity $\phi_{\mathbf{x},t_0}$ can be viewed as the usual univariate phase (i.e., the complex argument) of the univariate signal $z(t)$ defined by the (Hermitian) inner product $z(t) = \mathbf{x}^\dagger(t_0)\mathbf{x}(t)$. The instantaneous phase satisfies the following properties, which follow directly from (10):

- *invariance under amplitude modulations*: let $\alpha(t) > 0$ and define $\tilde{\mathbf{x}}(t) = \alpha(t)\mathbf{x}(t)$. Then $\phi_{\tilde{\mathbf{x}},t_0} = \phi_{\mathbf{x},t_0}$.
- *covariance under global phase modulations*: let $\beta(t)$ be any smooth real-valued signal, and define $\tilde{\mathbf{x}}(t) = \exp(j\beta(t))\mathbf{x}(t)$. Then $\phi_{\tilde{\mathbf{x}},t_0} = \phi_{\mathbf{x},t_0} + \beta(t) - \beta(t_0)$.
- *unitary invariance*: let $\mathbf{U} \in \mathbb{C}^{n \times n}$ be a constant unitary matrix (i.e., such that $\mathbf{U}^\dagger \mathbf{U} = \mathbf{U} \mathbf{U}^\dagger = \mathbf{I}_n$) and define the unitary transformed signal $\tilde{\mathbf{x}}(t) = \mathbf{U} \mathbf{x}(t)$. Then $\phi_{\tilde{\mathbf{x}},t_0} = \phi_{\mathbf{x},t_0}$.

However, unlike the univariate case (4), changing the reference time from t_0 to t_1 does not result into a constant shift: in general, $\phi_{\mathbf{x},t_1}(t) - \phi_{\mathbf{x},t_0}(t)$ depends on t . This is a direct consequence of the lack of transitivity of the in-phase relation.

2.4 Dynamical phase, geometric phase, and instantaneous frequency

The objective of this subsection is to develop an alternative *local* counterpart to the *global* approach taken in Definition 1 to define the instantaneous phase of a multivariate signal. As explained below, the local approach leads to different result than the global one, differing by the *geometric phase*.

Any non-vanishing multivariate signal $\mathbf{x}(t)$ can be written as

$$\mathbf{x}(t) = a_{\mathbf{x}}(t) e^{j\varphi_{\mathbf{x}}(t)} \mathbf{p}_{\mathbf{x}}(t), \quad (11)$$

where $a_{\mathbf{x}}(t) > 0$, $\varphi_{\mathbf{x}}(t)$ is continuous and where $\mathbf{p}_{\mathbf{x}}(t)$ is a unit normalized signal such that $\|\mathbf{p}_{\mathbf{x}}(t)\| = 1$ for every t . The amplitude can be defined unambiguously as $a_{\mathbf{x}}(t) = \|\mathbf{x}(t)\|$; however there is an infinity of choices for the couple $(\varphi_{\mathbf{x}}, \mathbf{p}_{\mathbf{x}})$. We will come back later on this fundamental problem.

For the moment, let us chose one of these couples, say $(\varphi_{\mathbf{x}}, \mathbf{p}_{\mathbf{x}})$ and elaborate on the implications of Definition 1. From (10), the instantaneous phase of $\mathbf{x}(t)$ with respect to t_0 reads

$$\phi_{\mathbf{x},t_0}(t) = \arg(\mathbf{x}(t_0)^\dagger \mathbf{x}(t)) = \varphi_{\mathbf{x}}(t) - \varphi_{\mathbf{x}}(t_0) + \arg(\mathbf{p}_{\mathbf{x}}(t_0)^\dagger \mathbf{p}_{\mathbf{x}}(t)), \quad (12)$$

where the extra term arises from the covariance under phase modulations property of the instantaneous phase. Now, let us define the quantity

$$\nu_{\mathbf{x}}(t) = \text{Im} \left(\frac{\mathbf{x}^\dagger(t) \dot{\mathbf{x}}(t)}{2\pi \|\mathbf{x}(t)\|^2} \right). \quad (13)$$

Using similar considerations than in the univariate case, one can show that $\nu_{\mathbf{x}}(t)$ has a natural interpretation as an instantaneous moment since

$$\langle \nu_{\mathbf{x}} \rangle = \int_{\mathbb{R}} \nu \|\mathbf{X}(\nu)\|^2 d\nu = \int_{\mathbb{R}} \nu_{\mathbf{x}}(t) \|\mathbf{x}(t)\|^2 dt, \quad (14)$$

assuming without loss of generality that $\mathbf{x}(t)$ is of unit energy. As in (7), the mean frequency $\langle \nu_{\mathbf{x}} \rangle$ of the spectral density $\|\mathbf{X}(\nu)\|^2$ is the average of $\nu_{\mathbf{x}}$ weighted by the instantaneous power $\|\mathbf{x}(t)\|^2$. However, unlike the univariate case, accumulating $\nu_{\mathbf{x}}(t)$ from t_0 to t does not yield the instantaneous phase $\phi_{\mathbf{x},t_0}(t)$, i.e., Eq. (5) does no longer holds. Instead, using the parametrization (11) we have $\mathbf{x}^\dagger \dot{\mathbf{x}} / \|\mathbf{x}\|^2 = \dot{\mathbf{a}}_{\mathbf{x}} / a_{\mathbf{x}} + \mathbf{j} \dot{\phi}_{\mathbf{x}} + \mathbf{p}_{\mathbf{x}}^\dagger \dot{\mathbf{p}}_{\mathbf{x}}$, where $\mathbf{p}_{\mathbf{x}}^\dagger \dot{\mathbf{p}}_{\mathbf{x}}$ is purely imaginary.¹ Hence one gets

$$\int_{t_0}^t \frac{\text{Im}(\mathbf{x}^\dagger(s) \dot{\mathbf{x}}(s))}{\|\mathbf{x}(s)\|^2} ds = \varphi_{\mathbf{x}}(t) - \varphi_{\mathbf{x}}(t_0) + \int_{t_0}^t \text{Im}(\mathbf{p}_{\mathbf{x}}^\dagger(s) \dot{\mathbf{p}}_{\mathbf{x}}(s)) ds, \quad (15)$$

and therefore, combining with (12)

$$\phi_{\mathbf{x},t_0}(t) = \int_{t_0}^t \frac{\text{Im}(\mathbf{x}^\dagger(s) \dot{\mathbf{x}}(s))}{\|\mathbf{x}(s)\|^2} ds + \left(\arg(\mathbf{p}_{\mathbf{x}}(t_0)^\dagger \mathbf{p}_{\mathbf{x}}(t)) - \int_{t_0}^t \text{Im}(\mathbf{p}_{\mathbf{x}}^\dagger(s) \dot{\mathbf{p}}_{\mathbf{x}}(s)) ds \right). \quad (16)$$

Compared to (5), the *local approach* now include a second term. This term is independent of the particular parametrization $(\varphi_{\mathbf{x}}, \mathbf{p}_{\mathbf{x}})$ chosen. To verify this, observe that the second term in (16) can be expressed in terms $\phi_{\mathbf{x},t_0}(t)$ and the integral of $\text{Im}(\mathbf{x}^\dagger \dot{\mathbf{x}} / \|\mathbf{x}\|^2)$, which depend of $\mathbf{x}(t)$ only: hence it is independent from the parameterization.

Following the terminology of [14], we name the two terms as follows.

Definition 2 (Dynamical and geometric phases). *Let t_0 be such that $\mathbf{x}(t_0) \neq 0$. The dynamical phase $\phi_{d,\mathbf{x},t_0}(t)$ and the geometric phase $\phi_{g,\mathbf{x}}(t)$ of $\mathbf{x}$ relative to t_0 are*

$$\phi_{d,\mathbf{x},t_0}(t) = \int_{t_0}^t \frac{\text{Im}(\mathbf{x}^\dagger(s) \dot{\mathbf{x}}(s))}{\|\mathbf{x}(s)\|^2} ds, \quad (17a)$$

$$\phi_{g,\mathbf{x},t_0}(t) = \phi_{\mathbf{x},t_0}(t) - \phi_{d,\mathbf{x},t_0}(t), \quad (17b)$$

for all t such that $\phi_{\mathbf{x},t_0}(t)$ is defined.

Both phases can be computed directly from $\mathbf{x}$, without choosing a parametrization. Given one, Eq. (16) provides the equivalent expression

$$\phi_{g,\mathbf{x},t_0}(t) = \arg(\mathbf{p}_{\mathbf{x}}(t_0)^\dagger \mathbf{p}_{\mathbf{x}}(t)) - \int_{t_0}^t \text{Im}(\mathbf{p}_{\mathbf{x}}^\dagger(s) \dot{\mathbf{p}}_{\mathbf{x}}(s)) ds, \quad (18)$$

¹Indeed, since $\|\mathbf{p}_{\mathbf{x}}(t)\|^2 = 1$, differentiating both sides $\dot{\mathbf{p}}_{\mathbf{x}}^\dagger \mathbf{p}_{\mathbf{x}} + \mathbf{p}_{\mathbf{x}}^\dagger \dot{\mathbf{p}}_{\mathbf{x}} = 0$ or equivalently, $\text{Re}(\mathbf{p}_{\mathbf{x}}^\dagger \dot{\mathbf{p}}_{\mathbf{x}}) = 0$.

which will be convenient for explicit computations in Section 4. For $n = 1$, (5) gives $\phi_{g,x,t_0}(t) = 0$ for every t : the geometric phase is a genuine multivariate quantity. Moreover, if $\mathbf{x}$ depends on time only through its scalar part, i.e., $\mathbf{x}(t) = a_{\mathbf{x}}(t)e^{j\varphi_{\mathbf{x}}(t)}\mathbf{p}_0$ where $\mathbf{p}_0$ is a constant complex unit vector, Eq. (18) shows that the geometric phase is equal to zero. This case corresponds to the “single point path” that will be described in Section 3.8. The converse is false: the geometric phase also vanishes along some non-constant paths, such as geodesics (see Section 3).

The name *geometric* is justified by the following properties, which follow from those of $\phi_{\mathbf{x},t_0}$ listed in Section 2.3.

- *invariance under scalar modulations*: let $c(t) \neq 0$ be a smooth complex-valued signal and $\tilde{\mathbf{x}}(t) = c(t)\mathbf{x}(t)$. Writing $c(t) = b(t)e^{j\beta(t)}$ with $b(t) > 0$, one has $\tilde{\mathbf{x}}^\dagger \dot{\tilde{\mathbf{x}}} / \|\tilde{\mathbf{x}}\|^2 = \dot{b}/b + \dot{\mathbf{j}}\beta + \mathbf{x}^\dagger \dot{\mathbf{x}} / \|\mathbf{x}\|^2$, hence $\phi_{d,\tilde{\mathbf{x}},t_0} = \phi_{d,\mathbf{x},t_0} + \beta(t) - \beta(t_0)$. Since $\phi_{\tilde{\mathbf{x}},t_0}$ is shifted by the same amount, $\phi_{g,\tilde{\mathbf{x}},t_0} = \phi_{g,\mathbf{x},t_0}$.
- *invariance under time reparametrization*: let τ be a smooth strictly increasing function with $\tau(s_0) = t_0$, and $\tilde{\mathbf{x}}(s) = \mathbf{x}(\tau(s))$. Then $\phi_{\tilde{\mathbf{x}},s_0}(s) = \phi_{\mathbf{x},t_0}(\tau(s))$ and, by a change of variables in (17a), $\phi_{d,\tilde{\mathbf{x}},s_0}(s) = \phi_{d,\mathbf{x},t_0}(\tau(s))$; hence $\phi_{g,\tilde{\mathbf{x}},s_0}(s) = \phi_{g,\mathbf{x},t_0}(\tau(s))$.
- *unitary invariance*: let $\tilde{\mathbf{x}} = \mathbf{U}\mathbf{x}$ for a constant unitary matrix $\mathbf{U}$. Then $\phi_{g,\tilde{\mathbf{x}},t_0} = \phi_{g,\mathbf{x},t_0}$, since $\phi_{\mathbf{x},t_0}$ and $\phi_{d,\mathbf{x},t_0}$ are invariant under unitary transforms.

The first property states that $\phi_{g,\mathbf{x},t_0}$ is invariant with respect to the usual AM-FM modulations applied *simultaneously* to all entries of $\mathbf{x}(t)$. The second means that the speed at which the unit vector $\mathbf{p}_{\mathbf{x}}(t)$ evolves does not matter, only its *path*. This property will be illustrated in Section 6.1.2.

JF:The part below needs to be rediscussed. We now extend the notion of instantaneous frequency to multivariate signals. Differentiating (17) gives

$$\frac{1}{2\pi} \dot{\phi}_{\mathbf{x},t_0}(t) = \nu_{\mathbf{x}}(t) + \frac{1}{2\pi} \dot{\phi}_{g,\mathbf{x},t_0}(t), \quad (19)$$

with $\nu_{\mathbf{x}}$ defined in (13). For $n = 1$ the geometric term vanishes and both sides reduce to (6); for $n > 1$ they differ in general.

Definition 3 (Instantaneous, dynamical and geometric frequencies). *Let t_0 be such that $\mathbf{x}(t_0) \neq 0$. For all t such that $\mathbf{x}^\dagger(t_0)\mathbf{x}(t) \neq 0$, the instantaneous frequency $f_{\mathbf{x},t_0}$, the dynamical frequency $\nu_{\mathbf{x}}$ and the geometric frequency $f_{g,\mathbf{x},t_0}$ of $\mathbf{x}$ relative to t_0 are*

$$f_{\mathbf{x},t_0}(t) = \frac{1}{2\pi} \dot{\phi}_{\mathbf{x},t_0}(t) = \frac{1}{2\pi} \text{Im} \left(\frac{\mathbf{x}^\dagger(t_0)\dot{\mathbf{x}}(t)}{\mathbf{x}^\dagger(t_0)\mathbf{x}(t)} \right), \quad (20a)$$

$$\nu_{\mathbf{x}}(t) = \frac{1}{2\pi} \dot{\phi}_{d,\mathbf{x},t_0}(t) = \frac{\text{Im}(\mathbf{x}^\dagger(t)\dot{\mathbf{x}}(t))}{2\pi\|\mathbf{x}(t)\|^2}, \quad (20b)$$

$$f_{g,\mathbf{x},t_0}(t) = \frac{1}{2\pi} \dot{\phi}_{g,\mathbf{x},t_0}(t) = f_{\mathbf{x},t_0}(t) - \nu_{\mathbf{x}}(t). \quad (20c)$$

Moreover, by (19), one has $f_{\mathbf{x},t_0}(t) = \nu_{\mathbf{x}}(t) + f_{g,\mathbf{x},t_0}(t)$.

Our definition for the instantaneous frequency $f_{\mathbf{x},t_0}(t)$ departs from [4, 6], where $\nu_{\mathbf{x}}$ (denoted $\omega_{\mathbf{x}}$) is called the instantaneous frequency; the two coincide only when the geometric frequency vanishes. **JF:**This choice is motivated here by **JF:**I wonder whether we should stick to the instantaneous frequency being the first moment because 1) it agrees with lilly, 2) it is the only one that is "local" 3) I believe that we can show a similar result as for univariate Wigner-Ville, i.e., show that the $\nu_{\mathbf{x}}(t)$ as defined is the conditional mean of the trace of the WV distribution. We could have relative frequency, instantatneous freq, geometric freq.

We also note that, while the dynamical frequency is independent from t_0 , the instantaneous frequency $f_{\mathbf{x},t_0}$ and the geometric frequency $f_{g,\mathbf{x},t_0}$ depend on the reference time t_0 , since $\phi_{\mathbf{x},t_1} - \phi_{\mathbf{x},t_0}$ is not constant in general (Section 2.3). These notions are illustrated in Section 6.2.

The next sections are organized as follows. Section 3 explains the geometric origin of this phase, and Section 4 specializes these results to modulated multivariate signals, for which explicit expressions are available.

JF:J'ai laissé le texte ci-dessous pour commenter un point qui me semble important.

Therefore, to remain consistent with what is known in the scalar case, we propose the following terminology:

- The **instantaneous frequency** $f_{\mathbf{x}}(t)$ of a multivariate signal is the derivative of the instantaneous phase at time t relative to t_0 :

$$f_{\mathbf{x}}(t) = \frac{1}{2\pi} \dot{\phi}_{\mathbf{x},t_0}(t)$$

Since changing the reference time t_0 to t_1 modifies the instantaneous phase by the constant $\arg \mathbf{x}^\dagger(t_1)\mathbf{x}(t_0)$, this definition does indeed only depends on t . **JF:**Je pense qu'il y a un problème ici. La phase instantanée elle-meme dépend de t_0 (c'est notre définition) je ne vois pas comment dans le cas $n \geq 2$ sa dérivée s'affranchirait de cela. Surtout qu'elle contient la dérivée de la phase géo, qui dépend de t_0 .. The instantaneous frequency is the sum of the two following frequencies:

- The **dynamical frequency** $\nu_{\mathbf{x}}(t)$ of a multivariate signal is the derivative of the dynamical phase. It is given by the instantaneous moment

$$\nu_{\mathbf{x}}(t) = \frac{1}{2\pi} \dot{\phi}_{d,\mathbf{x},t_0}(t) = \text{Im} \left(\frac{\mathbf{x}^\dagger(t) \dot{\mathbf{x}}(t)}{2\pi \|\mathbf{x}(t)\|^2} \right)$$

- The **geometric frequency** $f_{g,\mathbf{x}}(t)$ of a multivariate signal is the derivative of the geometric phase. It writes

$$\begin{aligned} f_{g,\mathbf{x}}(t) &= \frac{1}{2\pi} \dot{\phi}_{g,\mathbf{x},t_0}(t) \\ &= f_{\mathbf{x}}(t) - \nu_{\mathbf{x}}(t) \end{aligned}$$

These notions will be illustrated in Section 6.

Having shown the existence of a geometric phase for multivariate signals, we now turn to describing the mathematical causes of why this occurs. After a detailed

derivation in Section 3, understanding of the geometry induced by the peculiar amplitude-modulation multivariate signal model is helpful and will be developed in Section 4.

3 A geometric description of the concept of phase

As recalled and exploited in the last section, phase is a relative notion. The phase of a complex number is evidently dependent on the basis chosen to define the real and imaginary parts. This means that once the basis is chosen, the phase of a complex number is covariant under a multiplication by a unit norm complex number, or equivalently covariant under an action of the unitary group $U(1)$. This remains valid in the multivariate case: The definition adopted after Pantcharatnam of the instantaneous phase $\phi_{\mathbf{x},t_0}(t)$ relative to an origin t_0 is covariant under the action of $U(1)$ since if $\mathbf{y}(t) = \exp(j\varphi(t))\mathbf{x}(t)$ then

$$\phi_{\mathbf{y},t_0}(t) = \varphi(t) - \varphi(t_0) + \phi_{\mathbf{x},t_0}(t)$$

This allows to conclude that the theory of phase in multivariate signals can only be done up to a global phase, and that solving this ambiguity requires to work in the quotient space $\mathbb{C}^n/U(1)$, as developed below.

3.1 Ambiguity and equivalent classes

Let us decompose $\mathbf{x}(t)$ as $\mathbf{x}(t) = a_{\mathbf{x}}(t)\mathbf{p}_{\mathbf{x}}(t)$, where $a_{\mathbf{x}}(t) = (\mathbf{x}(t)^\dagger \mathbf{x}(t))^{1/2} \in \mathbb{R}^+$ is the instantaneous amplitude (unambiguously defined), and $\mathbf{p}_{\mathbf{x}}(t) \in \mathcal{S}^{2n-1}$ is a unit norm complex vector.

The decomposition above is ambiguous because vector $\mathbf{p}_{\mathbf{x}}(t)$ is defined up to a global phase. In fact, the equivalence relation $\sim$ defined on $\mathcal{S}^{2n-1}$ by

$$(\mathbf{p}_{\mathbf{x}}(t) \sim \mathbf{q}_{\mathbf{x}}(t)) \iff (\mathbf{p}_{\mathbf{x}}(t) = e^{j\varphi(t)} \mathbf{q}_{\mathbf{x}}(t))$$

implies that $\mathbf{x}(t)$ is as well written $a_{\mathbf{x}}(t)\mathbf{p}_{\mathbf{x}}(t)$ as $a_{\mathbf{x}}(t)\mathbf{q}_{\mathbf{x}}(t)$ if $(\mathbf{p}_{\mathbf{x}}(t) \sim \mathbf{q}_{\mathbf{x}}(t))$.

The decomposition becomes unambiguous if we impose to $\mathbf{p}_{\mathbf{x}}(t)$ to belong to the quotient $\mathcal{S}^{2n-1}/U(1)$. The space $\mathcal{S}^{2n-1}/U(1)$ is known to be the complex projective space $\mathbb{CP}^{n-1}$.

3.2 Principal fiber bundle structure of $\mathcal{S}^{2n-1}$ and $\mathbb{C}^n$

NLB:je pense qu'on devrait clarifier dès le début la distinction $\mathcal{S}^{2n-1}$ et $\mathbb{C}^n$. Là on oscille un peu entre les deux dans la section. C'est un peu confusant.

Fiber bundles appear in many physical models [9]. Here, we describe the space $\mathcal{S}^{2n-1}$ of normalized complex signals as a fiber bundle. This fiber bundle is denoted $\mathcal{S}^{2n-1}(\mathbf{U}(1), \mathbb{CP}^{n-1})$. $\mathcal{S}^{2n-1}$ is called the *total space*; $\mathbb{CP}^{n-1}$ is the *base space* and is the set of all equivalent classes of $\mathcal{S}^{2n-1}$ under the equivalence relation $\sim$. A *fiber* above an element $[\mathbf{p}] \in \mathbb{CP}^{n-1}$ is an orbit of $\mathbf{p}$ under the action of the group $U(1)$, that is the set $\mathcal{F}_{\mathbf{p}} = \{\exp(j\psi)\mathbf{p}, \forall \exp(j\psi) \in U(1)\}$.

The bundle structure comes with a so-called projection $\pi : \mathcal{S}^{2n-1}(\mathbf{U}(1), \mathbb{CP}^{n-1}) \longrightarrow \mathbb{CP}^{n-1}$ which associates to an element of the total space an element of the base space, with the property that $\mathcal{F}_{\mathbf{p}} = \pi^{-1}([\mathbf{p}])$ for any $[\mathbf{p}] \in \mathbb{CP}^{n-1}$. Since a fiber is a subset of a vector space, we can use the usual projector for $\mathbf{p} \in \mathcal{S}^{2n-1}$ to find its equivalence class in $\mathbb{CP}^{n-1}$, or $\pi(\mathbf{p}) = \mathbf{p}\mathbf{p}^\dagger = [\mathbf{p}]$ since obviously it is invariant under action of $U(1)$. We will use the notation $\rho_{\mathbf{p}} = [\mathbf{p}]$ from now on, and mention the fact that $\rho_{\mathbf{p}}$ is the coherence matrix of vector $\mathbf{p}$ (also called the density operator in quantum mechanics [14]).

Two important remarks must be made here:

- Firstly, for technical reasons, $\{0\}$ must be excluded from the total space: This ensures the manifold structures we are going to consider ([15, 16] for details—freeness of group actions). This is assumed in the sequel, but is not explicitly mentioned to avoid heavier notations.
- Secondly, the total space considered up to here is the $2n - 1$ dimensional sphere, and the fiber at $\rho_{\mathbf{p}}$ is the set of all $\exp(j\psi)\mathbf{p}$. But the construction can be easily generalized by considering $\mathbb{C}^n$ as the total space and $\mathbb{CP}^{n-1}$ as the *base space*, the fibers being defined as $\mathcal{F}_{\mathbf{x}} = \{\lambda\mathbf{x}, \forall \lambda \in \mathbb{C}\}$ for any $\mathbf{x} \in \mathbb{C}^n$ along with the projection

$$\rho_{\mathbf{x}} = \frac{\mathbf{x}\mathbf{x}^\dagger}{\mathbf{x}^\dagger\mathbf{x}} = \mathbf{p}_{\mathbf{x}}\mathbf{p}_{\mathbf{x}}^\dagger \in \mathbb{CP}^{n-1} \quad (21)$$

However, in the sequel, since we can unambiguously extract the instantaneous amplitude, we only deal with the bundle structure of $\mathcal{S}^{2n-1}$ and concentrate on the properties of $\mathbf{p}_{\mathbf{x}}(t)$ in the decomposition $\mathbf{x}(t) = a_{\mathbf{x}}(t)\mathbf{p}_{\mathbf{x}}(t)$.

A signal $\mathbf{x}(t)$ is a trivialized section of the bundle, it is to say an application from the base $\mathbb{CP}^{n-1}$ to the total space $\mathbb{C}^n$ together with a continuous basis of $\mathcal{F}_{\mathbf{x}}$. The particular sections we consider here are continuous, differentiable and of finite energy. They are parametrized by time $t \in [t_0, T]$ (each bound may be infinite), by an instantaneous positive amplitude $a_{\mathbf{x}}(t) \in L^2(\mathbb{R})$ and a unit norm complex vector $\mathbf{p}_{\mathbf{x}}(t)$. We assume both to be continuous and differentiable. $\pi(\mathbf{x}(t))$ then draws a curve on the manifold $\mathbb{CP}^{n-1}$. Conversely, to each continuous and differentiable curve $\rho_{\mathbf{x}}(t) \in \mathbb{CP}^{n-1}$ corresponds an infinite number of possible signals in the total space, each differing by some unit norm complex multiplicative function $\exp(j\lambda(t))$: Each of these indeed project onto $\rho_{\mathbf{x}}(t)$ by π . These signals are called *liftings* of the curve $\rho_{\mathbf{x}}(t)$. Likewise, a curve $\rho_{\mathbf{x}}(t)$ may be lifted in different sections of $\mathcal{S}^{2n-1}$, each differing by some complex multiplicative function $\exp(j\alpha(t))$. The existence of different possible liftings is at the root of the existence of the geometrical phase. Figure 1 illustrates the principal fiber bundle structure of $\mathcal{S}^{2n-1}$.

3.3 Ehresmann connection

In order to study the dynamics of multivariate signals, and to understand and describe the time evolution of their instantaneous NLB:attributes (Pas défini avant ici, mais repris dans la section 4.1... soit on dit ce que c'est avant, soit on dit ici que c'est la phase et la fréquence instantanée et les moments peut être ?), a key concept is the tangent

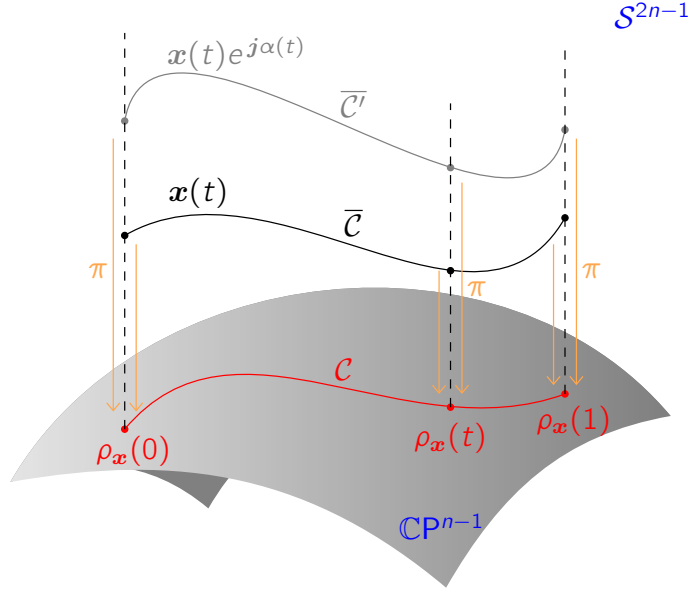


Fig. 1 Illustration of the fiber bundle $\mathcal{S}^{2n-1}(\mathbf{U}(1), \mathbb{CP}^{n-1})$. A trajectory $\mathcal{C}$ (in red) in the base space $\mathbb{CP}^{n-1}$ corresponds to the time evolution of the projection $\rho_{\mathbf{x}}(t)$ and can be lifted to a curve in $\mathcal{S}^{2n-1}$ ($\bar{\mathcal{C}}$ or $\mathcal{C}'$ for example) which is a section of the bundle and corresponds to the time evolution of a signal.

space to the total space $\mathcal{S}^{2n-1}$. The Ehresmann connection provides a decomposition of this tangent space into two orthogonal vector spaces, the so-called *horizontal* and *vertical* spaces. It also provides a 1-form (or linear form) allowing to connect the fibers.

NLB: On rajouterait pas à un moment que quand on parle de $\mathbf{x}$, c'est $\mathbf{x}(t)$ pris à un instant arbitraire ?? (pas sûr que ce soit nécessaire...)

All the spaces considered up to here are manifolds. Obviously $\mathcal{S}^{2n-1}$ is a submanifold of $\mathbb{C}^n/\{0\}$, itself a submanifold of $\mathbb{C}^n$. Since $\mathbb{C}^n$ is of finite dimension n , its tangent space at $\mathbf{x}$ is also of finite dimension, and is thus isomorphic to $\mathbb{C}^n$: It is therefore identified to it. Likewise, $\mathbb{C}^n/\{0\}$ is of dimension n and its tangent space $T_{\mathbf{x}}\mathbb{C}^n/\{0\}$ at $\mathbf{x}$ is identified with $\mathbb{C}^n$. However, $\mathcal{S}^{2n-1}$ is of dimension $n-1$ and its tangent space $T_{\mathbf{x}}\mathcal{S}^{2n-1}$ at $\mathbf{x}$ is of dimension $n-1$. Furthermore, a fiber at $[\mathbf{x}]$ is identified with the circle $\mathcal{S}^1$ and is also a manifold.

Consider a smooth curve $\mathbf{x}(t) \in \mathcal{S}^{2n-1}$. Since $\mathbf{x}(t)^\dagger \mathbf{x}(t) = 1 \forall t$, derivating with respect to t shows that $\text{Re}(\mathbf{x}(t)^\dagger \dot{\mathbf{x}}(t)) = 0$ and thus that $T_{\mathbf{x}}\mathcal{S}^{2n-1}$ at $\mathbf{x}$ is the set of vectors $\mathbf{v}$ of $\mathbb{C}^n$ such that $\text{Re}(\mathbf{x}^\dagger \mathbf{v}) = 0$.

We want to link the tangent space at $\mathbf{x} \in \mathcal{S}^{2n-1}$ with the tangent space at $[\mathbf{x}] = \pi(\mathbf{x}) \in \mathbb{CP}^{n-1}$. However, there is a difficulty since the tangent space at $\mathbf{x}$ can be given two definitions, depending on the choice $\mathbf{x} \in \mathcal{S}^{2n-1}$ or $\mathbf{x} \in \mathcal{F}_{[\mathbf{x}]}$. Indeed, to define a tangent vector at $\mathbf{x}$, we need a curve lying in the space to which $\mathbf{x}$ belongs. But obviously the fiber at $\mathbf{x}$ is included in $\mathcal{S}^{2n-1}$ so that $T_{\mathbf{x}}\mathcal{F}_{[\mathbf{x}]} \subset T_{\mathbf{x}}\mathcal{S}^{2n-1}$. Thus, the

tangent space at $\mathbf{x}$ in $\mathcal{S}^{2n-1}$ can be decomposed as a direct sum, the tangent space $T_{\mathbf{x}}\mathcal{F}_{[\mathbf{x}]}$ plus its orthogonal complement.

In the literature, the tangent space to the fiber at $\mathbf{x}$ is called the vertical space $V_{\mathbf{x}}$. In the case studied here, it is the set $V_{\mathbf{x}}\mathcal{S}^{2n-1} = \{j a \mathbf{x}, a \in \mathbb{R}\}$. Indeed, consider a curve $\mathbf{y}(t) \in \mathcal{F}_{[\mathbf{x}]}$ passing through $\mathbf{x}$ at $t = 0$. Since the fiber is the set of all equivalent vectors to $\mathbf{x}$, we can parametrize $\mathbf{y}(t)$ by $\mathbf{y}(t) = e^{j\alpha(t)}\mathbf{x}(t)$ with $\alpha(0) = 0$. Deriving $\mathbf{y}$ at $t = 0$ gives the tangent vector $j\dot{\alpha}_0\mathbf{x}$, which proves the assertion. The orthogonal complement $H_{\mathbf{x}}\mathcal{S}^{2n-1}$ to $V_{\mathbf{x}}\mathcal{S}^{2n-1}$ in $T_{\mathbf{x}}\mathcal{S}^{2n-1}$ is called the horizontal space. Note that $V_{\mathbf{x}}\mathcal{S}^{2n-1}$ is also the kernel of the differential of π at $\mathbf{x}$. $D\pi_{\mathbf{x}} : T_{\mathbf{x}}\mathcal{S}^{2n-1} \rightarrow T_{[\mathbf{x}]} \mathbb{CP}^{n-1}$ associates a tangent vector $\mathbf{v}_{\mathbf{x}}$ at $\mathbf{x}$ to a tangent vector $\mathbf{w}_{[\mathbf{x}]}$ at $[\mathbf{x}] \in \mathbb{CP}^{n-1}$. $\mathbf{v}_{\mathbf{x}}$ is by definition the velocity vector of a curve in $\mathcal{S}^{2n-1}$, and $\mathbf{w}_{[\mathbf{x}]}$ is a velocity vector of the image of the curve by π . Now NLB:is if $\mathbf{v}_{\mathbf{x}}$ lies in $V_{\mathbf{x}}\mathcal{S}^{2n-1}$, then the curve is included in $\mathcal{F}_{[\mathbf{x}]}$, and its image by π is a constant curve, with zero velocity. Thus if $\mathbf{v}_{\mathbf{x}} \in V_{\mathbf{x}}\mathcal{S}^{2n-1}$, $D\pi_{\mathbf{x}}(\mathbf{v}_{\mathbf{x}}) = 0$. This explains the term vertical space.

To characterize $H_{\mathbf{x}}\mathcal{S}^{2n-1}$, let us consider $\mathbf{y} \in T_{\mathbf{x}}\mathcal{S}^{2n-1}$. It can be approximated by a vector of $V_{\mathbf{x}}\mathcal{S}^{2n-1}$ plus an error term χ orthogonal to $V_{\mathbf{x}}\mathcal{S}^{2n-1}$, whence in $H_{\mathbf{x}}\mathcal{S}^{2n-1}$. Thus $\mathbf{y} = j b \mathbf{x} + \chi$, $b \in \mathbb{R}$, such that χ is orthogonal to $V_{\mathbf{x}}\mathcal{S}^{2n-1}$, that is to all $j a \mathbf{x}$ with a real. Therefore $-j a \mathbf{x}^\dagger \chi = 0 = -j a \mathbf{x}^\dagger \mathbf{y} - a b \mathbf{x}^\dagger \mathbf{x}$ and then $b = -j \mathbf{x}^\dagger \mathbf{y}$ which must be real by hypothesis. Hence $\mathbf{y}$ can be written as $\mathbf{y} = j \text{Im}(\mathbf{x}^\dagger \mathbf{y}) \mathbf{x} + \chi$ and $\mathbf{y}$ lies in $H_{\mathbf{x}}\mathcal{S}^{2n-1}$ if and only if $\text{Im}(\mathbf{x}^\dagger \mathbf{y}) = 0$. Consider the following linear form

$$\begin{aligned} A_{\mathbf{x}} : T_{\mathbf{x}}\mathcal{S}^{2n-1} &\longrightarrow \mathbb{R} \\ \mathbf{y} &\longmapsto A_{\mathbf{x}}(\mathbf{y}) = \text{Im}(\mathbf{x}^\dagger \mathbf{y}) \end{aligned} \quad (22)$$

called Berry connection in the physics literature. The preceding discussion shows that $H_{\mathbf{x}}\mathcal{S}^{2n-1} = \ker A_{\mathbf{x}}$ NLB:as it is the set of vectors $\mathbf{y} \in T_{\mathbf{x}}\mathcal{S}^{2n-1}$ such that $\text{Im}(\mathbf{x}^\dagger \mathbf{y}) = 0$.

To sum up, the Ehresmann connexion can be defined as follows: For all $\mathbf{x} \in \mathcal{S}^{2n-1} = (\mathbf{U}(1), \mathbb{CP}^{n-1})$:

- $A_{\mathbf{x}} : T_{\mathbf{x}}\mathcal{S}^{2n-1} \longrightarrow \mathbb{R}$ defined by $A_{\mathbf{x}}(\mathbf{y}) = \text{Im}(\mathbf{x}^\dagger \mathbf{y})$
- $T_{\mathbf{x}}\mathcal{S}^{2n-1} = V_{\mathbf{x}}\mathcal{S}^{2n-1} \oplus H_{\mathbf{x}}\mathcal{S}^{2n-1}$ where

$$\begin{aligned} V_{\mathbf{x}}\mathcal{S}^{2n-1} &= \{j a \mathbf{x}, a \in \mathbb{R}\} = \ker D\pi_{\mathbf{x}} \\ H_{\mathbf{x}}\mathcal{S}^{2n-1} &= \{\mathbf{v} \in T_{\mathbf{x}}\mathcal{S}^{2n-1} / \text{Re}(\mathbf{x}^\dagger \mathbf{v}) = 0\} = \ker A_{\mathbf{x}} \end{aligned}$$

NLB:Est-ce qu'on ferait pas un schema pour visualiser les espaces verticaux et horizontaux ???

3.4 Horizontal lifting

We just saw that the differential $D\pi_{\mathbf{x}}$ of π at $\mathbf{x}$ has the vertical space at $\mathbf{x}$ as kernel. Therefore, the restriction of $D\pi_{\mathbf{x}}$ to $H_{\mathbf{x}}$ is a bijective application from $H_{\mathbf{x}}\mathcal{S}^{2n-1}$ to $T_{[\mathbf{x}]} \mathbb{CP}^{n-1}$. Its inverse $D\pi_{\mathbf{x}}^{-1}$ uniquely associates a tangent vector $\mathbf{v}_{[\mathbf{x}]}$ at $[\mathbf{x}]$ to a tangent vector $\mathbf{w}_{\mathbf{x}}$ at $\mathbf{x}$, called the *horizontal lift* of $\mathbf{v}_{[\mathbf{x}]}$. Note that $\mathbf{v}_{[\mathbf{x}]}$ can be lifted to a unique vector $\mathbf{w}_{\mathbf{y}}$ at any $\mathbf{y}$ in the fiber $\mathcal{F}_{[\mathbf{x}]}$.

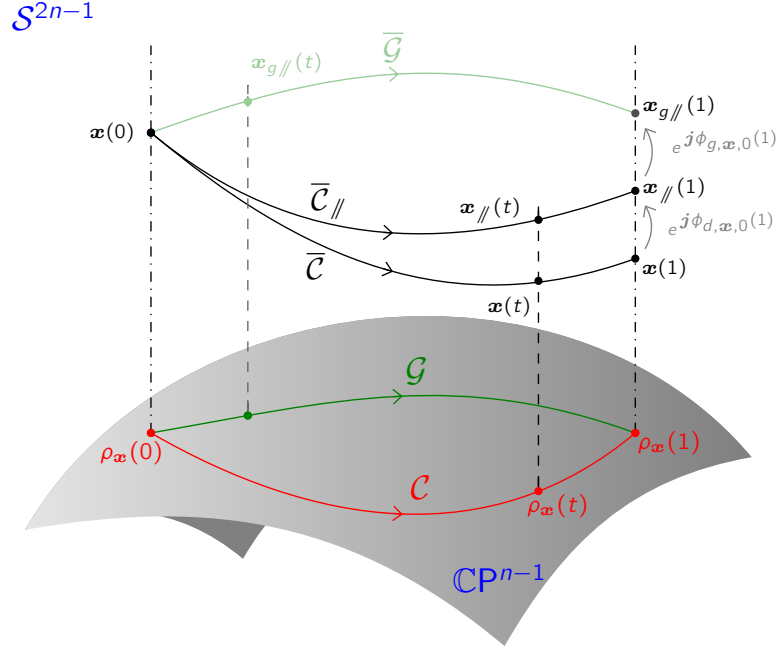


Fig. 2 to be completed. The path of $\mathbf{x}_g(t)$ in $\mathcal{S}^{2n-1}$ projects as a geodesic $\mathcal{G}$ in the base space. It is also a geodesic in the total space. Among all these geodesic, the horizontal geodesic remains in phase with $\mathbf{x}(0)$. The trajectory $\mathbf{x}_{\parallel}(t)$ is the *horizontal lift* of $\mathcal{C}$. The initial point $\mathbf{x}(0)$ lives in the fiber above $\rho(0)$. The final point $\mathbf{x}(1)$ lives in the fiber above $\rho(1)$. It is related to $\mathbf{x}(0)$ by a pure phase term (an element of $U(1)$) which depends on the trajectory followed during the time interval.

Consider now a path $\mathcal{C} = \{\rho(t), t \in [0, T]\}$ in the base space $\mathbb{CP}^{n-1}$, starting at point $\rho(0)$, as represented in Figure 1. To this path in the base space corresponds many possible paths in the total space $\mathcal{S}^{2n-1}$. This means that there exists infinitely many multivariate signals that may have their projected trajectory over time that is exactly the one followed by $\rho(t)$. However, some peculiar trajectories exist. Using for example the fact that $D\pi_{\mathbf{x}}^{-1}$ uniquely associates a tangent vector $\mathbf{v}_{[\mathbf{x}]}$ at $[\mathbf{x}]$ to a tangent vector $\mathbf{w}_{\mathbf{x}}$ at $\mathbf{x}$, $D\pi_{\mathbf{x}}^{-1}(\dot{\mathcal{C}}) \in \mathcal{S}^{2n-1}$ defines a curve such that at each time, the velocity is the horizontal lift of the corresponding velocity $\dot{\rho}(t)$ tangent to the curve $\mathcal{C} \in \mathbb{CP}^{n-1}$. Equivalently, using Ehresmann connection, this curve is defined by the signal $\mathbf{x}(t)$ such that $A_{\mathbf{x}(t)}(\dot{\mathbf{x}}(t)) = \text{Im}(\mathbf{x}^H(t)\dot{\mathbf{x}}(t)) = 0, \forall t$. It is called the *horizontal lift* of $\mathcal{C}$.

In Figure 2, the horizontal lift of the path $\mathcal{C}$ is the trajectory denoted $\mathbf{x}_{\parallel}(t)$. It corresponds to a trajectory where the vector $\dot{\mathbf{x}}_{\parallel}(t)$ tangent to the trajectory is parallel transported.

3.5 Turning manifolds into Riemannian manifolds, geodesics

We have already encountered a metric above when dealing with unit norm vectors in $\mathbb{C}^n$. The metric considered at $\mathbf{x}$ in $T_{\mathbf{x}}\mathcal{S}^{2n-1}$ is provided by the usual euclidean inner product of $\mathbb{C}^n$, $\mathbf{v}^\dagger \mathbf{w}$ where $\mathbf{v}$ and $\mathbf{w}$ are tangent vectors at $\mathbf{x} \in \mathcal{S}^{2n-1}$. This is provided by the identification of $T_{\mathbf{x}}\mathcal{S}^{2n-1}$ with $\mathbb{C}^{n-1}$. The metric allows to define the

Levi-Civita connection, which in this situation, is simply the orthogonal projection of the usual differential onto the tangent space of the fiber, that is the vertical space. Precisely, if $\mathbf{x}(t)$ is a curve for t in some interval, the covariant derivative of a tangent vector $\mathbf{v}(t)$ along the curve is given by

$$\frac{D\mathbf{v}(t)}{dt} = \mathbf{x}(t)\mathbf{x}^\dagger(t)\mathbf{v}(t)$$

But the euclidean metric can not be used directly for the base space (or quotient) $\mathbb{CP}^{n-1}$ because it is not invariant under $\sim$. However, extracting from a tangent vector at $\mathbf{x}$ its horizontal part may provide a solution. Indeed, let $\mathbf{v} \in T_{\mathbf{x}}\mathcal{S}^{2n-1}$. Then, its projection onto $H_{\mathbf{x}}\mathcal{S}^{2n-1}$ writes $\mathbf{v}_\perp = \mathbf{v} - \mathbf{j}A_{\mathbf{x}}(\mathbf{v})\mathbf{x}$ and can be shown to be covariant to $\sim$. Thus the metric $\langle \mathbf{v}|\mathbf{w} \rangle_{\mathbf{x}} = \mathbf{v}_\perp^\dagger \mathbf{w}_\perp$ is invariant to $\sim$ and can be used as a metric in the quotient manifold $\mathbb{CP}^{n-1}$ to compare the tangent vectors at $[\mathbf{x}]$, $d\pi_{\mathbf{x}}(\mathbf{v})$ and $d\pi_{\mathbf{x}}(\mathbf{w})$ [14, 15].

This also allows to study geodesics in the quotient manifold. Since the metric in the quotient manifold is evaluated by the metric in the horizontal lifted space, then a geodesic in the quotient manifold will be lifted to an horizontal geodesic in the lifted space. Importantly, a geodesic in the quotient manifold can be lifted to another geodesic which is not horizontal, but whose vertical component does not count for the length in the quotient space.

The expression of the geodesics in $\mathcal{S}^{2n-1}$ are known [14, 15]. Since this manifold is the sphere in $\mathbb{C}^n$, the geodesics can be shown to be

$$\mathbf{x}_t = \mathbf{x}_0 \cos(\omega t) + \dot{\mathbf{x}}_0 \frac{\sin(\omega t)}{\omega}$$

where $\omega^2 = \|\dot{\mathbf{x}}_0\|^2$, and with $\mathbf{x}_0^\dagger \mathbf{x}_0 = 1$. It remains horizontal if $\mathbf{x}_0^\dagger \dot{\mathbf{x}}_0 = 0$. Projecting this equation using π would allow to write down the equation of geodesics in the projective space.

Figure 2 displays the geodesic $\mathcal{G}$ as an alternate trajectory between points $\rho(0)$ and $\rho(1)$ in the base space. This trajectory is the equivalent of the straight line for flat spaces. Here, the base space is curved, the geodesic $\mathcal{G}$ between initial and final point is the curve with zero acceleration [9]. Its horizontal lift in the total space is denoted $\mathbf{x}_{\mathcal{G}}(t)$ and has no contribution in the vertical space $V_{\mathbf{x}}\mathcal{S}^{2n-1}$ while the velocity vector is also parallel transported along the path.

3.6 A geometric view of the phases

As shown using signal processing arguments, the main point of the paper is to show that the phase $\phi_{\mathbf{x},t_0}(t) = \arg \mathbf{x}(t_0)^\dagger \mathbf{x}(t)$ is the sum of two terms, the first one being the *dynamical phase*, a phase acquired along time with respect to a particular reference signal; The second one, overlooked in signal theory up to now, is the *geometrical phase*, a phase existing only for multivariate signals, reflecting purely geometrical aspects of the signals and reminiscent of the underlying base space.

We now provide a differential geometry proof of the result, using the developments presented above. Even if inspired by proofs of the existence of the Berry phase in quantum mechanics ([14] for example), the proof presented here is new.

Basically the dynamical phase comes from the behavior of the signal in the total space $\mathcal{S}^{2n-1}$ whereas the geometrical phase shows up from the behavior of the equivalent class of the signal in the base space, *i.e.* the quotient $\mathbb{CP}^{n-1}$.

To develop our presentation, we consider as in figure (2), two states of a physical system in $\mathbb{CP}^{n-1}$, and study what can happen for the physical measurements in $\mathcal{S}^{2n-1}$, depending on how the states are joined by a curve along time and by how we lift this curve to the physical measurement $\mathbf{x}(t)$.

Thus, we select two times, $t_0 = 0$ and t , such that the physical states at those two times are $\rho(0) \in \mathbb{CP}^{n-1}$ and $\rho(t) \in \mathbb{CP}^{n-1}$. Then the physical measurements at 0 and t are elements of the fibers $\mathcal{F}_{[\rho(0)]}$ and $\mathcal{F}_{[\rho(t)]}$ respectively. But these measurements are not unique in their fiber and differ within a fiber by a pure phase term. In other words, if $\mathcal{C} = \{\rho(u), u \in [0, t]\}$ is the curve in $\mathbb{CP}^{n-1}$ followed by the state of the system, a possible measurement consists in a lifted curve $\bar{\mathcal{C}} \in \mathcal{S}^{2n-1}$, leaving $\mathcal{F}_{[\rho(0)]}$ at time 0 and arriving at $\mathcal{F}_{[\rho(t)]}$ at time t . Any signal $\mathbf{x}(t)$ such that $\{\pi(\mathbf{x}(u)) = \rho(u), u \in [0, t], \pi(\mathbf{x}(0)) = \rho(0), \pi(\mathbf{x}(t)) = \rho(t)\}$ is a possible physical measurement.

We will however fix the initial condition of the measurement to $\mathbf{x}(0) \in \mathcal{F}_{[\rho(0)]}$, so that all possible signals mentioned above start at $\mathbf{x}(0)$ at time 0.

We now show that two different phases appear in a generic physical measurement: The first one is purely geometric and comes from the geometry of the curve $\mathcal{C}$ traced in the base space from $\rho(0)$ to $\rho(t)$; The second one is purely dynamical and reflects the behavior of the lifted curve $\bar{\mathcal{C}}$ in the physical space.

1. Let $\mathcal{C}_g$ be a geodesic in $\mathbb{CP}^{n-1}$ joining $\rho(0)$ to $\rho(t)$. Then we saw that a lifting of this curve is a geodesic and hence will be described by $\mathbf{x}_g(u), \mathcal{S}^{2n-1}$ for $u \in [0, t]$, and thus writes $\mathbf{x}_g(u) = \mathbf{x}(0) \cos(\omega u) + \hat{\mathbf{x}}_0 \frac{\sin(\omega u)}{\omega}$ with $\text{Re}(\mathbf{x}(0)^\dagger \hat{\mathbf{x}}_0) = 0$. Note that in general this measurement is valid only at time 0 and time t , since there is not reason that $\pi(\mathbf{x}_g(u)) = \rho(u)$ for $u \in (0, t)$. We will see later why this is the root of the geometrical aspect of the geometrical phase.

As we saw above, $\mathbf{x}_g(u)$ can be chosen to remain horizontal if $\text{Im}(\mathbf{x}(0)^\dagger \hat{\mathbf{x}}_0) = 0$. In this situation we denote it as $\mathbf{x}_{g//}(u)$, and the curve $\mathcal{C}_g$ is lifted to the horizontal geodesic $\bar{\mathcal{C}}_{g//}$ traced out by the section $\mathbf{x}_{g//u}, u \in [0, t]$. Therefore, using the equation of $\mathbf{x}_{g//u}$ we obtain $\mathbf{x}_{g//}^\dagger(u) \mathbf{x}_{g//}(0) \in \mathbb{R}^+, \forall u \in [0, t]$ and $\mathbf{x}_{g//}(t)$ remains in phase with $\mathbf{x}_{g//}(0) = \mathbf{x}(0)$ or $\phi_{\mathbf{x}_{g//},0}(u) = 0, \forall u \in [0, t]$. Thus $\mathbf{x}_{g//}(u), u \in [0, t]$ provides a perfect reference signal to compare measurements having a same physical state.

2. Suppose that $\mathcal{C}$ is not a geodesic in $\mathbb{CP}^{n-1}$ joining $\rho(0)$ to $\rho(t)$.

Consider $\mathbf{x}_{//}(u), u \in [0, t]$ the horizontal lift such that $\mathbf{x}_{//}(0) = \mathbf{x}_0 = \mathbf{x}_{g//}(0) = \mathbf{x}(0)$, which traces out the curve $\bar{\mathcal{C}}_{//}$. Then at time t , $\mathbf{x}_{//}(t)$ and $\mathbf{x}_{g//}(t)$ are in the same fiber $\mathcal{F}_{[\rho(t)]}$ since they project down to the same point $\rho(t)$ in $\mathbb{CP}^{n-1}$. Thus there exist a phase $\varphi_g(t)$ such that $\mathbf{x}_{//}(t) = e^{j\varphi_g(t)} \mathbf{x}_{g//}(t)$. We call $\varphi_g(t)$ the geometrical phase at time t .

3. Consider now $\mathbf{x}(u), u \in [0, t]$ a non horizontal lift of $\mathcal{C}$. $\mathbf{x}(u)$ and $\mathbf{x}_{//}(u)$ have thus the same projection by π , the same initial condition, but differ by a pure phase

term (unit complex number) in the total space at all time $u \in [0, t]$, or $\mathbf{x}(u) = e^{j\psi(u)} \mathbf{x}_{//}(u)$, with $\psi_0 = 0$. If we now evaluate the covariant derivative of $\mathbf{x}(u)$ we get

$$\begin{aligned} \frac{D\mathbf{x}(u)}{du} &\stackrel{\text{def.}}{=} \mathbf{x}(u) \mathbf{x}(u)^\dagger \dot{\mathbf{x}}(u) \\ &\stackrel{\text{Leibnitz rule}}{=} j \dot{\psi}(u) \mathbf{x}(u) + e^{j\psi(u)} \frac{D\mathbf{x}_{//}(u)}{du} \end{aligned}$$

Since $\mathbf{x}_{//}(u)$ follows the *parallel transport*, its covariant derivative is zero

$$\frac{D\mathbf{x}_{//}(u)}{du} = 0$$

and $\psi(u) \in \mathbb{R}$ is then determined by the differential equation

$$\dot{\psi}(u) = -j \mathbf{x}(u)^\dagger \dot{\mathbf{x}}(u) = A_{\mathbf{x}(u)}(\dot{\mathbf{x}}(u))$$

whose solution reads

$$\psi(t) = \int_0^t A_{\mathbf{x}(u)}(\dot{\mathbf{x}}(u)) du$$

We insist again on the fact that $\mathbf{x}(u)$ and $\mathbf{x}_{//}(u)$ are equivalent according to $\sim$: In the bivariate setting for example, these signals could be two electromagnetics waves having the same polarization state over time.

$\psi(t)$ only depends on the curve followed, not on the speed at which it is traced out. Indeed, the previous integral is invariant under a reparametrization $\mathbf{x} \mapsto \mathbf{y}$ given by any bijection $u \mapsto s(u)$ such that $\mathbf{y}(s) = \mathbf{x}(s(u))$. Hence, we call this phase *the dynamical phase* and change a bit the notation to $\varphi_d(t) = \psi(t)$. *The dynamical phase hence measures the departure of the path followed by $\mathbf{x}(t)$ from the path followed by the parallel transport $\mathbf{x}_{//}(t)$.*

Back to the instantaneous phase. We previously showed that at all time $u \in [0, t]$, $\mathbf{x}(u) = e^{j\varphi_d(u)} \mathbf{x}_{//}(u)$ and also that $\mathbf{x}_{//}(t) = e^{j\varphi_g(t)} \mathbf{x}_{g//}(t)$. We thus end up with

$$\mathbf{x}(t) = e^{j(\varphi_d(t) + \varphi_g(t))} \mathbf{x}_{g//}(t)$$

Therefore, the instantaneous phase of $\mathbf{x}(t)$ relative to $t = 0$ is given by

$$\begin{aligned} \phi_{\mathbf{x},0}(t) &= \arg(\mathbf{x}^\dagger(0) \mathbf{x}(t)) = \varphi_g(t) + \varphi_d(t) + \arg(\mathbf{x}_{g//}^\dagger(0) \mathbf{x}_{g//}(t)) \\ &= \varphi_g(t) + \varphi_d(t) \end{aligned}$$

because we saw $\mathbf{x}_{g//}^\dagger(0) \mathbf{x}_{g//}(u) \in \mathbb{R}^+, \forall u \in [0, t]$.

This concludes the proof of the main statement. It remains to give an explicit form to the geometric phase. As already shown in the previous section, we can have two definitions for it. Let $\mathbf{x}(t)$ be factored as $a_{\mathbf{x}}(t)\mathbf{p}_{\mathbf{x}}(t)$ where $\mathbf{p}_{\mathbf{x}}(t)$ is a unit norm vector in $\mathbb{C}^n$ at all time. Let signal $\mathbf{x}$ be observed on the interval $[t_0, t]$. Then we have

$$\begin{aligned}\varphi_d(t) &= \phi_{d,\mathbf{x},t_0}(t) = \int_{t_0}^t A_{\mathbf{x}(u)}(\dot{\mathbf{x}}(u))du = \int_{t_0}^t \text{Im} \left(\frac{\mathbf{x}^\dagger(u)\dot{\mathbf{x}}(u)}{\mathbf{x}^\dagger(u)\mathbf{x}(u)} \right) du \\ \varphi_g(t) &= \phi_{g,\mathbf{x},t_0}(t) = \phi_{\mathbf{x},t_0}(t) - \phi_{d,\mathbf{x},t_0}(t)\end{aligned}$$

which are definitions in agreement with equations (17).

3.7 The geometrical phase as the holonomy of a closed curve

Physicist readers familiar with the geometric or Berry phase may complain that usually the geometric phase is defined as the holonomy of a loop traced by a physical system in the base space. However, our proof above apparently does not close any loop. In fact, the loop is closed by the choice of the reference horizontal lifted geodesic. We develop this now.

Let $\bar{\mathcal{C}}_t$ be any curve in the fiber $\mathcal{F}_{[\rho(t)]}$ joining $\mathbf{x}_t \in \mathcal{F}_{[\rho(t)]}$ to $\mathbf{x}_{g//}(t) \in \mathcal{F}_{[\rho(t)]}$. Let the closed curve $\bar{\mathcal{C}}_o = \bar{\mathcal{C}} \cup \bar{\mathcal{C}}_t \cup \bar{\mathcal{C}}_{g//}$. We now relate the geometric phase to $\oint_{\bar{\mathcal{C}}_o} A_{\mathbf{x}}$.

The curve $\bar{\mathcal{C}}_o$ can be parametrized by a function $\mathbf{y}(u)$ from an interval $I_T = [0, T]$ in $\mathbb{R}$ to $\mathcal{S}^{2n-1}$ such that : $t \in I_T$, $\mathbf{y}(u) = \mathbf{x}(u), \forall u \in [0, t]$, $\mathbf{y}(u) = e^{j\alpha(u)}\mathbf{x}(t)$ for $u \in [t, s]$ (for some $T > s \geq t$ and some $\alpha(t)$ ensuring $\mathbf{y}$ to remain continuous and differentiable), and finally $\mathbf{y}(t) = \mathbf{x}_{g//}(T - u)$ for $u \in [s, T]$.

In other words, $\mathbf{y}(u)$ traces the closed curve following the physical measurement $\mathbf{x}(t)$ from 0 to t , then evolving from t to s in the fiber $\mathcal{F}_{[\rho(t)]}$, joining $\mathbf{x}(t)$ to $\mathbf{x}_{g//}(t)$, and then going back to the initial measurement $\mathbf{x}(0)$ at time T .

Integrating the 1-form along the close curve $\bar{\mathcal{C}}_o$ we get

$$\oint_{\bar{\mathcal{C}}_o} A_{\mathbf{y}}(u) = \int_{\bar{\mathcal{C}}} A_{\mathbf{y}}(u) + \int_{\bar{\mathcal{C}}_t} A_{\mathbf{y}}(u) + \int_{-\bar{\mathcal{C}}_g} A_{\mathbf{y}}(u)$$

As shown before $\int_{\bar{\mathcal{C}}} A_{\mathbf{y}} = \varphi_d(t)$ and $\int_{-\bar{\mathcal{C}}_g} A_{\mathbf{y}} = 0$. For the contribution on $\bar{\mathcal{C}}_t$ in the fiber, the signal reads $\mathbf{y}(u) = e^{j\alpha(u)}\mathbf{x}(t)$ for $u \in [t, s]$ for some phase $\alpha(u)$. Since $\mathbf{y}(t) = \mathbf{x}(t)$ we need $\alpha(t) = 0$. Furthermore, since $\mathbf{y}(s) = e^{j\alpha(s)}\mathbf{x}(s) = \mathbf{x}_{g//}(t)$ we also need $\alpha(s) = -(\varphi_g(t) + \varphi_d(t))$.

Now, we have $\dot{\mathbf{y}}(u) = j\dot{\alpha}(u)\mathbf{y}(u)$ so that $\alpha(s) - \alpha(t) = \int_{\bar{\mathcal{C}}_t} A_{\mathbf{y}}$. Adding up the three integrals then leads to

$$\varphi_g(t) = - \oint_{\bar{\mathcal{C}}_o} A_{\mathbf{x}}$$

It is evident that the sign above comes from the direction of time: Reversing the direction, *i.e.* tracing $\bar{\mathcal{C}}_o$ in the opposite direction, would reverse the sign.

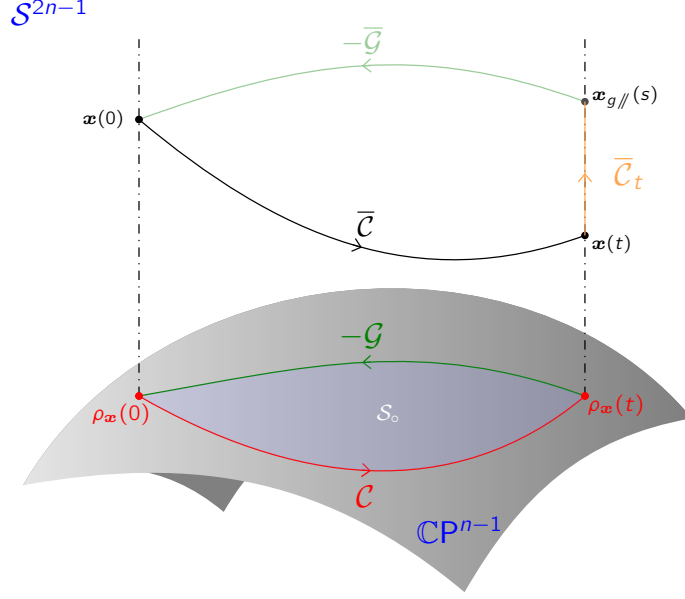


Fig. 3 NLB:To be finished...

We would like the same kind of result in $\mathbb{CP}^{n-1}$. Using results on integration in manifolds from [16], we would have an immediate result if $A_{\mathbf{x}}$ was the pullback by π of a 1-form on $\mathbb{CP}^{n-1}$. But, as noticed by Mukunda & Simon [14], this is not the case. In fact, as shown below, φ_g can be calculated along curves in $\mathbb{CP}^{n-1}$ but not in an additive way (with respect to curve splitting), excluding the possibility to calculate it as the integral of any 1-form along a curve.

However, this conclusion is reversed when considering 2-forms. Stokes theorem [16] stipulates that

$$\varphi_g(t) = - \oint_{\bar{\mathcal{C}}_o} A_{\mathbf{x}} = - \iint_{\bar{\mathcal{S}}_o} dA_{\mathbf{x}}$$

where $\bar{\mathcal{S}}_o$ is any surface such that $\bar{\mathcal{C}}_o = \partial \bar{\mathcal{S}}_o$, and where $dA_{\mathbf{x}}$ is the exterior derivative of $A_{\mathbf{x}}$. It is a 2-form, that is an alternating second order differential form or equivalently a second order antisymmetric covariant tensor. As shown by Mukunda & Simon [14], there exists a 2-form Ω on $\mathbb{CP}^{n-1}$ such that $dA_{\mathbf{x}}$ is the pullback by the natural projection π of Ω . Using results from Lee [16] allows to conclude that

$$\varphi_g(t) = \phi_{g,\mathbf{x},t_0}(t) = - \iint_{\bar{\mathcal{S}}_o} dA_{\mathbf{x}} = - \iint_{\mathcal{S}_o} \Omega$$

where $\mathcal{S}_o$ is any surface in $\mathbb{CP}^{n-1}$ such that $\partial \mathcal{S}_o = \pi(\bar{\mathcal{C}}_o) = \mathcal{C}_o$. This quantity is called the holonomy of the curve.

3.8 To sum up...

To end these general theoretical sections, we recall and emphasize some of the main properties of the geometric phase already known in quantum mechanics [14], but here presented in a signal processing language:

- Reference and time interval: Geometric phase is defined with respect to a reference time t_0 and is a function of time t (integration from t_0 to t). The geometric phase is a quantity accumulated over a time period.
- Gauge invariance: Signals $\mathbf{x}(t)$ and $\mathbf{x}(t)e^{j\alpha(t)}$ may accumulate the same geometric phase over a time interval. It is a pure geometric quantity as it only depends on the path in the base space.
- Time reparametrization: For a given smooth bijective function $s : [t_0, t] \rightarrow s([t_0, t])$, $\mathbf{x}(t)$ and $\mathbf{x}(s(t))$ have the same geometric phase accumulated at time t , meaning that it does not matter how fast or slow the path is followed. This also implies that the geometric phase is independent of the carrying frequency of the signal. Said differently, two signals having different carrying frequency accumulate the same geometric phase if their initial and final points are the same in the projective space.
- Geodesic path: The geometric phase accumulated along a geodesic path is null. For a geodesic, the tangent vector to $\bar{\mathcal{C}}_g$ has no component in the vertical space (it is an horizontal lift), and also $\text{Im}(\mathbf{p}^\dagger(t)\dot{\mathbf{p}}(t)) = 0 \forall t$. There is however a subtlety, as the geodesics are great circles on the sphere, and may cross the opposite pole. If $\rho(t)$ follows a great circle on the sphere, then the signal has no geometric phase accumulated along time, as soon as the great circle does not cross the opposite pole. In this case, the closing geodesic is indeed itself and the closed loop encloses a zero surface. In the contrary, the closing geodesic completes the great circle, and the closed loop encloses a non zero surface (in the bivariate case, this surface will be shown to be a half sphere, and the geometric phase experiences a π jump at each crossing of an antipodal point (see the item below on the area surrounded by $\mathcal{C}$)).
- Single point path: If the path in the projective space reduces to a single point, then there is no geometric phase accumulated. This corresponds to a constant value for $\rho(t)$, forcing the geometric phase term $\text{Im}(\mathbf{p}^H(t)\dot{\mathbf{p}}(t))$ to vanish, as $\mathbf{p}(t)$ is a constant. In the bivariate case, it means that the polarization state of the signal is constant, meaning that a geometric phase can only appear for a bivariate signal if a change occurs in its polarization state $\mathbf{p}(t)$ along time.
- Area surrounded by $\mathcal{C}$: The geometric phase is related to the area surrounded by the path $\mathcal{C}$. If the path $\mathcal{C}$ between $t = 0$ and $t = 1$ is a *loop* ($\rho(0) = \rho(1)$), then the geometric phase can be obtained from the area inside the loop [9, chapter 10]. If the path is *open* (such that $\rho(1) \neq \rho(0)$), then it is possible to close the loop by a geodesic from $\rho(1)$ and $\rho(0)$ as geodesic paths has null geodesic contribution, so that the area surrounded is still connected to the geometric phase.

We turn in the next section to the application of this theory to the case of modulated multivariate signals, making the link with the work of Lilly.

4 The geometric phase of modulated multivariate signals

Waves are described in physics by multivariate complex fields in space and time. These can be seismic waves, electromagnetic waves such as light, or even gravitational waves. We focus here on a measurement at one point in space, and therefore we restrict ourselves to multivariate signals taking complex values. We also assume that the evolution is mostly related to an amplitude modulated and frequency modulated (AM-FM) description, meaning that the measurement can be represented as a complex time dependent vector modulated by a complex scalar signal. The aim of the section is to describe the occurrence and meaning of a geometric phase in this particular case. We begin however by a brief overview of an important existing framework developed by Lilly and co-workers.

4.1 Modulated multivariate signals: Lilly's approach

The AM-FM case was developed in several papers by Lilly and his co-authors and we refer to these papers for a more detailed presentation [4, 6, 7]. They worked at developing a model for oscillatory multivariate signals, with a focus on instantaneous attributes (amplitude, phase and frequency). This is particularly adapted to signals obtained from measurement of wave phenomena. Lilly's approach led to a nice definition of AM-FM signals in the multivariate setting. We briefly recall the setting and show the link with the developments above.

Lilly studied multidimensional measurements of AM-FM signals, *e.g.* $(b_1(t) \cos(f_1(t)), \dots, b_n(t) \cos(f_n(t)))^\top$ where the Bedrosian conditions² are satisfied for all components. A complex signal can be created by using the analytic signal component wise and writes $\mathbf{x}(t) = (a_1(t) \exp(j\varphi_1(t)) \dots a_n(t) \exp(j\varphi_n(t)))^\top$. The pairs (a_k, φ_k) are called canonical pairs ([3]) for the components.

Lilly exposed how to extract from these multiple measurements a common phase which allows to write the analytic signal as

$$\mathbf{x}(t) = a_{\mathbf{x}}(t) e^{j\varphi_{\mathbf{x}}(t)} \mathbf{p}_{\mathbf{x}}(t) \quad (23)$$

where the extraction means that the parameterization is (almost) unambiguous. The idea is to consider the notion of a modulated multivariate oscillation as multicomponents sharing locally a same carrier.

To begin with, consider a typical signal $\mathbf{x}(t) = \exp(j\omega_0 t) \mathbf{x}_0$, where $\mathbf{x}_0$ is a constant vector. Then $\mathbf{x}(t + \tau) = \exp(j\omega_0 \tau) \mathbf{x}(t)$ such that $\mathbf{x}(t + \tau)$ is obtained from $\mathbf{x}(t)$ by a phase shift. This obviously cannot be true for general multivariate oscillations, but gives a hint to approach locally $\mathbf{x}(t + \tau)$.

²Recall that for one component $y(t) = b(t) \cos(f(t))$ Bedrosian conditions consider that $b(t)$ is a low frequency signal, whereas $\cos(f(t))$ is a high frequency signal, and that the frequency bands of the two signals do not overlap. This implies that the Hilbert transform $\mathcal{H}(y(t))$ can be written as $b(t) \mathcal{H}(\cos(f(t)))$ and thus the analytic signal of $y(t)$ written as $b(t) (\cos(f(t)) + j\mathcal{H}(\cos(f(t)))) = \rho(t) \exp(j\varphi(t))$. As recalled in [3] there is no reason for having $\mathcal{H}(\cos(f(t))) = \sin(f(t))$ and hence having $\varphi(t) = f(t)$ and $\rho(t) = b(t)$. $(\rho(t), \varphi(t))$ is called a canonical pair.

Write $\mathbf{x}(t + \tau) = \exp(j\omega(t)\tau)\tilde{\mathbf{x}}(t, \tau)$ where $\tilde{\mathbf{x}}(t, \tau) = \exp(-j\omega(t)\tau)\mathbf{x}(t + \tau)$ is a local demodulation (or phase shift) of $\mathbf{x}(t + \tau)$. $\omega(t)$ is *a priori* chosen arbitrarily for the moment. Lilly then proposes that locally (*i.e.* for small τ) $\tilde{\mathbf{x}}(t, \tau)$ should be close to $\mathbf{x}(t)$. Defining $\mathbf{x}_i(t, \omega_t) = \partial^i \tilde{\mathbf{x}}(t, \tau) / \partial \tau^i|_{\tau=0}$, Taylor expanding $\tilde{\mathbf{x}}(t, \tau)$ leads to the approximation

$$\mathbf{x}(t + \tau) = e^{j\omega(t)\tau} \mathbf{x}_t + e^{j\omega(t)\tau} \left(\mathbf{x}_1(t, \omega(t)) + \frac{\tau^2}{2} \mathbf{x}_2(t, \omega(t)) + O(\tau^3) \right)$$

If $\omega(t)$ is chosen equal to the constant ω_0 , the approximation is perfect if $\mathbf{x}_1(t, \omega(t)) = \dot{\mathbf{x}}(t) - j\omega(t)\mathbf{x}(t) = 0$, an ODE whose solution is hopefully $e^{j\omega_0\tau}$!

For general multivariate oscillations, the idea of Lilly is to choose $\omega(t)$ so as to minimize the first order reminder. The relative quadratic error to minimize is given by

$$v_{\mathbf{x}}^2(t) = \frac{\|\mathbf{x}_1(t, \omega(t))\|^2}{\mathbf{x}(t)^\dagger \mathbf{x}(t)} = \frac{\|\dot{\mathbf{x}}(t) - j\omega(t)\mathbf{x}(t)\|^2}{\mathbf{x}(t)^\dagger \mathbf{x}(t)}$$

This is the leading order error in τ when approximating $\mathbf{x}(t + \tau)$ by $e^{-j\omega(t)\tau}\mathbf{x}(t)$. The optimal phase which minimizes $v_{\mathbf{x}}(t)$ is shown to be

$$\omega_{\mathbf{x}}(t) = \frac{\text{Im}(\mathbf{x}^\dagger(t)\dot{\mathbf{x}}(t))}{\mathbf{x}(t)^\dagger \mathbf{x}(t)}$$

This extension comes from a direct generalization to the bivariate case of what is known for the scalar case [6]. Indeed, the idea is to keep the duality between the time and the frequency domains in the definition of a mean frequency. The mean frequency $\langle \nu_{\mathbf{x}} \rangle$ can be defined as

$$\langle \nu_{\mathbf{x}} \rangle = E_{\mathbf{x}}^{-1} \int \nu \|\mathbf{X}(\nu)\|^2 d\nu$$

where $E_{\mathbf{x}} = \int \|\mathbf{X}(\nu)\|^2 d\nu = \int \|\mathbf{x}(t)\|^2 dt$ is the energy of the signal. An instantaneous frequency $\nu_{\mathbf{x}}(t)$ can be defined as a time dependent quantity such that

$$\langle \nu_{\mathbf{x}} \rangle = E_{\mathbf{x}}^{-1} \int \nu_{\mathbf{x}}(t) \|\mathbf{x}(t)\|^2 dt$$

In the scalar case, the instantaneous frequency is the imaginary part of the derivative of the log of the signal. It can also be written as the imaginary part of $x^*(t)\dot{x}(t)/|x|^2$, whence the generalisation which satisfies the previous equation

$$2\pi\nu_{\mathbf{x}}(t) = \frac{\text{Im}(\mathbf{x}^\dagger(t)\dot{\mathbf{x}}(t))}{\mathbf{x}(t)^\dagger \mathbf{x}(t)} = A_{\mathbf{x}(t)}(\dot{\mathbf{x}}(t))$$

where $A_{\mathbf{x}}$ is the linear form defined in equation (22). This is really done by generalizing the scalar case to the multivariate case. Furthermore, the dispersion of energy $\langle \sigma_{\mathbf{x}}^2 \rangle$

around the mean frequency can also have a time counterpart. An instantaneous energy can be defined as the function $\sigma_{\mathbf{x}}^2(t)$ which satisfies

$$\langle \sigma_{\mathbf{x}}^2 \rangle = E_{\mathbf{x}}^{-1} \int (\nu - \overline{\nu_{\mathbf{x}}})^2 \|\mathbf{X}(\nu)\|^2 d\nu = E_{\mathbf{x}}^{-1} \int \sigma_{\mathbf{x}}^2(t) \|\mathbf{x}(t)\|^2 dt$$

A solution to this is

$$\sigma_{\mathbf{x}}^2(t) = \frac{\|\dot{\mathbf{x}}_t - 2\pi j \overline{\nu_{\mathbf{x}}} \mathbf{x}(t)\|^2}{\mathbf{x}(t)^\dagger \mathbf{x}(t)}$$

which is linked to the above $v_{\mathbf{x}}^2$ by

$$\min_{\omega_t} v_{\mathbf{x}}^2(t) = \sigma_{\mathbf{x}}^2(t) - (2\pi \nu_{\mathbf{x}}(t) - 2\pi \langle \nu_{\mathbf{x}} \rangle)^2$$

Lilly calls this the instantaneous bandwidth which corresponds to the part of the second order moments not accounted for by the deviation of the instantaneous frequency from the mean frequency. Or as seen as an estimation measure, it relates the MMSE to the variance and the bias.

The interesting point made here, is the fact that the instantaneous frequency as defined by Lilly corresponds to the definition coming from differential geometry which leads to the notion of dynamical phase. Since the approach of Lilly considers constant vectors $\mathbf{p}_x$ in equation (23), the geometric phase is zero and Lilly was not able to put it in light in full generality. Note however than in his 2011 paper [7, eq (43)] for the case $n = 3$, Lilly decomposes the phase³ into “two terms, (i) phase progression of the “particle” along the ellipse periphery and (ii) the combined effect of internal precession and external precession of the ellipse.”, which clearly corresponds to the dynamical phase (i) and the geometrical phase (ii), giving by the way an alternate form to the analytical description to be presented in section 4.5 below.

4.2 Modulated multivariate signals: instantaneous frequency and geometric phase.

We now apply the general setting developped in Section 3 to the particular case of AM-FM modulated signal. Suppose that the analytic signal measured writes

$$\mathbf{x}(t) = a_{\mathbf{x}}(t) e^{j\varphi_{\mathbf{x}}(t)} \mathbf{p}_{\mathbf{x}}(t) \quad (24)$$

where $\mathbf{p}_{\mathbf{x}}(t)$ is a unit norm complex vector. The phase ambiguity remains, but this allows to choose the parametrization of $\mathbf{p}_{\mathbf{x}}(t)$ as we wish as soon as it remains in $\mathcal{S}^{2n-1}$. We can for example decide that its first component is real. Any other choice leads to a modification of $\varphi_{\mathbf{x}}(t)$.

³The instantaneous frequency in Lilly’s paper.

As shown in Section 2, the dynamical and geometrical phases of $\mathbf{x}(t)$ read

$$\begin{aligned}\phi_{d,\mathbf{x},t_0}(t) &= \int_{t_0}^t \frac{\text{Im}(\mathbf{x}^\dagger(s)\dot{\mathbf{x}}(s))}{\|\mathbf{x}(s)\|^2} ds \\ \phi_{g,\mathbf{x},t_0}(t) &= \arg(\mathbf{p}_{\mathbf{x}}(t_0)^\dagger \mathbf{p}_{\mathbf{x}}(t)) - \int_{t_0}^t \text{Im}(\mathbf{p}_{\mathbf{x}}^\dagger(s)\dot{\mathbf{p}}_{\mathbf{x}}(s)) ds\end{aligned}$$

and the instantaneous phase relative to t_0 is their sum $\phi_{\mathbf{x},t_0}(t) = \phi_{d,\mathbf{x},t_0}(t) + \phi_{g,\mathbf{x},t_0}(t)$.

As pointed in Section 2.4, any fluctuation over time of the vector $\mathbf{p}_{\mathbf{x}}(t)$ is likely to add a contribution to the instantaneous frequency. This was actually noted by Cano in [17, Section 1.4] while defining the moments of a quaternion time-frequency density for AM-FM bivariate signals, but not identified as a geometric phase at that time. A detailed derivation of the result in [17, Section 1.4] is given in Appendix B.

4.3 Hyperspherical parametrization

Explicit calculations may have an interest in some situations, but lead to cumbersome interpretations. An example of applications where this could be useful is array processing in order to study how fluctuations of the sensors may be monitored by tracking geometrical phases. We keep this idea for further work, even if we propose a simple example of the idea in section 6 where we detect a fluctuation of one sensor in an array using the geometric phase.

For the moment we illustrate how we can parametrize vector $\mathbf{p}_{\mathbf{x}}$ to have explicit expressions of the different phases and of the instantaneous frequency of a n -dimensional AM-FM signals. A simple parametrization uses a generalization to complex spaces of hyperspherical coordinates, generalizing Jones vector to dimensions higher than 2 (see *e.g.* [18]).

In dimension n , vector $\mathbf{p}_{\mathbf{x}}$ has components $p_i, i = 1, \dots, n$ which in an hyperspherical parametrization may be written as

$$\begin{aligned}p_i(t) &= \cos(\alpha_i(t)) e^{j\beta_i(t)} \prod_{k=1}^{i-1} \sin(\alpha_k(t)) \\ \beta_1(t) &= 0 = \alpha_n(t)\end{aligned}$$

where the product term is intended to be equal to 1 for $i = 1$. Since we set $\beta_1(t) = 0 = \alpha_n(t)$ the vector is in dimension $2n - 2$, is of unit norm and hence an element of $\mathbb{CP}^{n-1}$.

For $n = 2$ we recover Jones vector $(\cos(\alpha); \sin(\alpha) \exp(j\beta))$, $\alpha \in [0, \pi], \beta \in [0, 2\pi]$: α is the latitude measured from the North pole, β the longitude.

The berry connection (equivalently the 1-form $A_{\mathbf{x}}$) reads

$$\text{Im}(\mathbf{p}_{\mathbf{x}}^\dagger(t)\dot{\mathbf{p}}_{\mathbf{x}}(t)) = \sum_{i=1}^n \dot{\beta}_i(t) \cos^2(\alpha_i(t)) \prod_{k=1}^{i-1} \sin^2(\alpha_k(t))$$

The second term needed to evaluate the geometrical phase is $\arg(\mathbf{p}_x(t_0)^\dagger \mathbf{p}_x(t))$. A general form for the inner product is

$$\mathbf{p}_x^\dagger(t_0) \mathbf{p}_x(t) = \sum_{i=1}^n \cos(\alpha_i(t_0)) \cos(\alpha_i(t)) e^{j(\beta_i(t) - \beta_i(t_0))} \prod_{k=1}^{i-1} \sin(\alpha_k(t_0)) \sin(\alpha_k(t))$$

from which the argument can be extracted. A general formula for it is awfull, and we will concentrate only on $n = 2$ and 3 since the main applications we focus on in this paper are the analysis of polarization of waves in two or three dimensions. Furthermore, the representation of vector $\mathbf{p}_x$ can be studied in Euclidean geometry by identification of $\mathbb{CP}^1$ as the sphere $\mathcal{S}^2 \in \mathbb{R}^3$. For $n = 2$, the sphere is also called the Poincaré sphere (polarization literature) or the Bloch sphere (quantum mechanics literature). For $n > 2$, as we illustrate for the trivariate case, $\mathcal{S}^2 \in \mathbb{R}^3$ remains of interest, but the evolution of $\mathbf{p}_x$ is represented by the evolution of a constellation of $n - 1$ dots on the sphere, a representation called the Majorana stellar representation. The two following sections detail the bivariate and the trivariate case.

4.4 The bivariate case

Several models for bivariate modulated signals have been proposed, for example in [6], [19], [20]. Many parametrization exists for vector $\mathbf{p}_x$ and as a consequence for φ_x . In essence, the models are equivalent and consist in a set of four parameters.

Euler angle parametrization.

The first parametrization⁴ directly models the ellipse of the polarization vector. The four parameters are denoted as $\{a_x(t), \varphi_x(t), \theta_x(t), \chi_x(t)\}$ such that the complex modulated bivariate signal $\mathbf{x}(t)$, as given in [21], reads:

$$\begin{aligned} \mathbf{x}(t) &= a_x(t) e^{j\varphi_x(t)} \begin{pmatrix} \cos \theta_x(t) \cos \chi_x(t) + j \sin \theta_x(t) \sin \chi_x(t) \\ \sin \theta_x(t) \cos \chi_x(t) - j \cos \theta_x(t) \sin \chi_x(t) \end{pmatrix} \\ &= a_x(t) e^{j\varphi_x(t)} J_3(\theta_x(t)) \begin{pmatrix} \cos \chi_x(t) \\ -j \sin \chi_x(t) \end{pmatrix} \end{aligned} \quad (25)$$

Vector $\mathbf{p}_x(t) \in \mathbb{C}^2$ is completely determined by the two angles $\theta_x(t) \in [-\pi/2; \pi/2]$ and $\chi_x(t) \in [-\pi/4; \pi/4]$, and by $J_3(\theta)$, the rotation matrix of angle θ . Equation (25) is a $\mathbb{C}^2$ version of the quaternion embedding proposed in [20], so that $\theta_x(t)$ and $\chi_x(t)$ are the parameters of the ellipse described by the bivariate signal $\mathbf{x}(t)$ over time (see figure 2 in [20]). $\theta_x(t)$ is the orientation of the ellipse, and $\chi_x(t)$ is the ellipticity of the ellipse. With parametrization (25), one gets⁵:

$$\phi_{d,\mathbf{x},t_0}(t) = \varphi_x(t) - \varphi_x(t_0) + \int_{t_0}^t \dot{\theta}_x(s) \sin 2\chi_x(s) ds \quad (26)$$

⁴Even if Euler angle parametrization is in general relevant of three dimensional problems, we keep this terminology here. The three dimensional case Euler parametrization will be presented as a direct generalization of the two dimensional parametrization.

⁵See A for detailed calculation of this equality.

$$\phi_{g,\mathbf{x},t_0} = \arctan \left(\frac{\sin(\theta_{\mathbf{x}}(t) - \theta_{\mathbf{x}}(t_0)) \sin(\chi_{\mathbf{x}}(t) + \chi_{\mathbf{x}}(t_0))}{\cos(\theta_{\mathbf{x}}(t) - \theta_{\mathbf{x}}(t_0)) \cos(\chi_{\mathbf{x}}(t) - \chi_{\mathbf{x}}(t_0))} \right) - \int_{t_0}^t \dot{\theta}_{\mathbf{x}}(s) \sin 2\chi_{\mathbf{x}}(s) ds$$

for values of the parameters such that the denominator is not zero.

From this result, some comments can be made. If $\theta_{\mathbf{x}}(t)$ remains constant equal to $\theta_{\mathbf{x}}(t_0)$, the geometric phase is constant and equal to zero, or a multiple of π . In this parametrization, $2\theta_{\mathbf{x}}$ represents the longitude on Poincaré sphere, and $\theta_{\mathbf{x}}$ being constant means that vector $\mathbf{p}_{\mathbf{x}}$ follows a vertical great circle on the sphere: The geometric phase is zero along geodesics. However, the geodesic starts at the podal point defined by $\theta_{\mathbf{x}}(t_0), \chi_{\mathbf{x}}(t_0)$, and if at later t it crosses the antipodal point, it will experience a jump of π in the geometric phase. This is explained in the appendix (crossing the antipodal point leads to a change in the branch of an arctan function), but also explained in the preceding section using geometric arguments (when crossing the antipodal point, the shortest geodesic to close the path makes this path the full great circle).

If the elevation (or latitude) $\chi_{\mathbf{x}}(t)$ is constant, then the geometric phase reads

$$\phi_{g,\mathbf{x},t_0}(t) = \arctan(\tan(2\chi_{\mathbf{x}}(t_0)) \tan(\theta_{\mathbf{x}}(t) - \theta_{\mathbf{x}}(t_0)) - \sin(2\chi_{\mathbf{x}}(t_0))(\theta_{\mathbf{x}}(t) - \theta_{\mathbf{x}}(t_0)))$$

It is thus never equal to zero, except except if $\chi(t_0) = 0$, in which case vector $\mathbf{p}_{\mathbf{x}}$ moves along the equator (a great circle) of Poincaré sphere, or if $\chi(t_0) = \pm\pi/4$ meaning that vector $\mathbf{p}_{\mathbf{x}}$ points either the north or the south pole and is static (in this situation last formula is even indefinite). Finally, suppose that $\chi(t) = \chi(t_0) \neq 0$ or $\pm\pi/4$, then for all t_k such that $\theta_{\mathbf{x}}(t_k) = \theta_{\mathbf{x}}(t_0) + k\pi, k \in \mathbb{Z}$, then

$$\phi_{g,\mathbf{x},t_0}(t_k) = k\pi \sin(2\chi_{\mathbf{x}}(t_0))$$

A famous example of a geometric phase, namely the deviation of the direction of oscillation of the Foucault pendulum [10, Chapter 3], has a similar expression: its value accumulated over one day ($t - t_0 = 24$ hours) is $2\pi \sin \ell$, where ℓ is the latitude of the location of the pendulum (2χ in the Poincaré sphere parametrization). In Paris, where $\ell = 48^\circ 52'$, the deviation is thus of 271° after a complete revolution of the Earth, *i.e.* in 24 hours.

Hyperspherical parametrization.

Here, vector $\mathbf{p}_{\mathbf{x}}$ is a Jones vector and the signal reads

$$\mathbf{x}(t) = a_{\mathbf{x}}(t) e^{j\psi_{\mathbf{x}}(t)} \begin{pmatrix} \cos \alpha_1(t) \\ \sin \alpha_1(t) e^{j\beta_2(t)} \end{pmatrix} \quad (28)$$

where $\alpha_1(t) \in [0, \pi], \beta_2(t) \in [0, 2\pi]$ With the Jones vector parametrization, the dynamical and geometrical phases write

$$\phi_{d,\mathbf{x},t_0}(t) = \psi_{\mathbf{x}}(t) - \psi_{\mathbf{x}}(t_0) + \int_{t_0}^t \dot{\beta}_2(s) \sin^2 \alpha_1(s) ds$$

$$\begin{aligned}\phi_{g,\mathbf{x},t_0}(t) &= \frac{\beta_2(t) - \beta_2(t_0)}{2} - \arctan\left(\tan\left(\frac{\beta_2(t) - \beta_2(t_0)}{2}\right) \frac{\cos(\alpha_1(t) + \alpha_1(t_0))}{\cos(\alpha_1(t) - \alpha_1(t_0))}\right) \\ &\quad - \int_{t_0}^t \dot{\beta}_2(s) \sin^2 \theta_1(s) ds\end{aligned}$$

In this parametrization, $2\alpha_1$ is the latitude measured from the North Pole, and β_2 the longitude. We recover that $\phi_g = 0$ if β_2 is constant (vector $\mathbf{p}_{\mathbf{x}}$ follows a geodesic), that $\phi_g = 2\pi \sin^2 \alpha_1(t_0)$ if θ_1 is constant and β_2 does a 2π excursion (vector $\mathbf{p}_{\mathbf{x}}$ traces an isolatitude circle).

4.5 Trivariate case: some results and further interpretations

We begin the exposition by studying the hyperspherical parametrization and the Euler angle parameterizations. This will provide a link with the bivariate case and will allow to interpret the different angles in the parametrization, and opens the way to an intrinsic geometric interpretation. To conclude this section and open some questions, we will introduce the Majorana stellar representation, an elegant way to visualize the evolution of the polarization vector in three dimensions.

Hyperspherical parametrization.

We study three dimensional Jones vector and the associated geometric phase. We thus consider the signal

$$\mathbf{x}(t) = a_{\mathbf{x}}(t) e^{j\varphi_{\mathbf{x}}(t)} \begin{pmatrix} \cos \alpha_1(t) \\ \sin \alpha_1(t) \cos \alpha_2(t) e^{j\beta_2(t)} \\ \sin \alpha_1(t) \sin \alpha_2(t) e^{j\beta_3(t)} \end{pmatrix} \quad (29)$$

We exclude the cases $\alpha_1(t) = \pi/2$ and $\alpha_2(t) = 0$ since they correspond to a two dimensional problem embedded in three dimensions. The angles α_i vary in $[0, \pi]$ and β_i in $[0, 2\pi]$.

The geometric phase can be written as

$$\begin{aligned}\phi_{g,\mathbf{x},t_0}(t) &= \frac{\delta\beta_2(t) + \delta\beta_3(t)}{4} - \arctan\left(\tan\left(\frac{\delta\beta_2(t) + \delta\beta_3(t)}{4}\right) \frac{\cos(\alpha_1(t) + \alpha_1(t_0))}{\cos(\alpha_1(t) - \alpha_1(t_0))}\right) \\ &\quad - \arctan\left(\tan\left(\frac{\delta\beta_3(t) - \delta\beta_2(t)}{4}\right) \frac{\cos(\alpha_2(t) + \alpha_2(t_0))}{\cos(\alpha_2(t) - \alpha_2(t_0))}\right) \\ &\quad - \int_{t_0}^t \dot{\beta}_2(s) \cos^2 \alpha_2(s) \sin^2 \alpha_1(s) ds - \int_{t_0}^t \dot{\beta}_3(s) \sin^2 \alpha_1(s) \sin^2 \alpha_2(s) ds\end{aligned}$$

where $\delta\beta_i(t) = \beta_i(t) - \beta_i(t_0)$. This result is difficult to interpret. We recover however the 2D case if $\alpha_1(t) = \pi/2$ or $\alpha_2(t) = 0$.

Euler angle parametrization.

The parametrization adopted by Lilly in [7] may be preferable to physically understand the different angles. Lilly's model is defined by

$$a_{\mathbf{x}}(t)e^{j\varphi_{\mathbf{x}}(t)}J_3(\gamma_{\mathbf{x}}(t))J_1(\delta_{\mathbf{x}}(t))J_3(\theta_{\mathbf{x}}(t))\begin{pmatrix} \cos \chi_{\mathbf{x}}(t) \\ -j \sin \chi_{\mathbf{x}}(t) \\ 0 \end{pmatrix}$$

$J_1(\theta)$ (resp. J_3) is a rotation of angle θ in the plane (y, z) (resp. the plane (x, y)). Thus the physical interpretation of this model is simple: Define the elliptic motion in the two dimensional plane (x, y) , and then perform two rotations in space to obtain any possible case. We refer the interested readers to [7] for further comments. We make a link with the bivariate case here.

Note that $\mathbf{p}_{\mathbf{x}}(t)$ in Lilly's parametrization can be written as $J_3(\gamma_{\mathbf{x}}(t))J_1(\delta_{\mathbf{x}}(t))(\mathbf{p}_2^\top 0)^\top$, where $\mathbf{p}_2$ is the polarisation vector in the bivariate case, as given in equation (25).

A direct calculation (detailed in the appendix) shows that $\text{Im}(\mathbf{p}_{\mathbf{x}}(t)^\dagger \dot{\mathbf{p}}_{\mathbf{x}}(t)) = \text{Im}(\mathbf{p}_2(t)^\dagger \dot{\mathbf{p}}_2(t)) + 2\dot{\gamma}_{\mathbf{x}}(t) \cos(\delta_{\mathbf{x}}(t)) \text{Im}(p_{21}p_{22}^*)$, $p_{2i}, i = 1, 2$ being the two components of vector $\mathbf{p}_2$. The geometric phase is then found to be

$$\phi_{g3,\mathbf{x},t_0}(t) = \phi_{g2,\mathbf{x},t_0}(t) + \arg\left(\frac{\mathbf{p}_{\mathbf{x}}(t_0)^\dagger \mathbf{p}_{\mathbf{x}}(t)}{\mathbf{p}_2(t_0)^\dagger \mathbf{p}_2(t)}\right) - \int_{t_0}^t \dot{\gamma}_{\mathbf{x}}(s) \cos(\delta_{\mathbf{x}}(s)) \sin(2\chi_{\mathbf{x}}(s)) ds$$

which provides an interesting decomposition of the geometric phase in the trivariate case: It is the sum of the geometric phase due to the two dimensional polarization, of a term measuring the difference in Pancharatnam phases between the 3D and the 2D polarization vectors, and of a term taking into account the zenith and azimuth angles.

If $\gamma_{\mathbf{x}}$ and $\delta_{\mathbf{x}}$ are constant, then $\mathbf{p}_{\mathbf{x}}^\dagger(t_0)\mathbf{p}_{\mathbf{x}} = \mathbf{p}_2^\dagger(t_0)\mathbf{p}_2$. The extra terms vanish and we obtain the fact that $\phi_{g3,\mathbf{x},t_0}(t) = \phi_{g2,\mathbf{x},t_0}(t)$ showing that the problem is indeed two dimensional, and showing that the calculation does not depend on the particular parametrization adopted for $\mathbf{p}_{\mathbf{x}}$. Note however that this does not allow to detect that the problem is indeed two dimensional.

4.5.1 Intrinsic interpretation

A way to couple the two parametrizations is to provide an intrinsic view of the dynamics. Recall that the signal reads,

$$\mathbf{x}(t) = a_{\mathbf{x}}(t)e^{j\varphi_{\mathbf{x}}(t)}\mathbf{p}_{\mathbf{x}}(t) \quad (30)$$

Write $\mathbf{p}_{\mathbf{x}}(t) = \mathbf{p}_{\text{Re}}(t) + j\mathbf{p}_{\text{Im}}(t)$ where $\mathbf{p}_{\text{Re}}$ and $\mathbf{p}_{\text{Im}}$ are real vectors. The real part of $\mathbf{x}(t)$ then describes an ellipse in the plane spanned by $\mathbf{p}_{\text{Re}}$ and $\mathbf{p}_{\text{Im}}$. A normal vector to this plane is $\mathbf{p}_{\text{Re}}(t) \wedge \mathbf{p}_{\text{Im}}(t)$ where $\wedge$ is the wedge product. The unit normal vector

is then defined as

$$\mathbf{n}_x(t) = \frac{\mathbf{p}_{\text{Re}}(t) \wedge \mathbf{p}_{\text{Im}}(t)}{\|\mathbf{p}_{\text{Re}}(t) \wedge \mathbf{p}_{\text{Im}}(t)\|} \quad (31)$$

which can be written in terms of the complex vector $\mathbf{p}_x(t)$ as

$$\mathbf{n}_x(t) = \frac{\text{Im}(\mathbf{p}_x^*(t) \wedge \mathbf{p}_x(t))}{\|\text{Im}(\mathbf{p}_x^*(t) \wedge \mathbf{p}_x(t))\|} \quad (32)$$

Since the plane is constantly rotated, the normal follows the equation

$$\dot{\mathbf{n}}_x(t) = \boldsymbol{\Omega}_x(t) \wedge \mathbf{n}_x(t) \quad (33)$$

where $\boldsymbol{\Omega}_x(t)$ is the instantaneous rotation vector of the plane. We can invert this equation by taking the wedge product with $\mathbf{n}_x(t)$ leading to

$$\mathbf{n}_x(t) \wedge \dot{\mathbf{n}}_x(t) = \mathbf{n}_x(t) \wedge \boldsymbol{\Omega}_x(t) \wedge \mathbf{n}_x(t) \quad (34)$$

$$= \boldsymbol{\Omega}_x(t) - (\boldsymbol{\Omega}_x(t)^\top \mathbf{n}_x(t)) \mathbf{n}_x(t) \quad (35)$$

since $\mathbf{n}_x(t)$ is of unit norm and thanks to the identity $\mathbf{a} \wedge \mathbf{b} \wedge \mathbf{c} = (\mathbf{a}^\top \mathbf{b})\mathbf{c} - (\mathbf{b}^\top \mathbf{c})\mathbf{a}$ for any vectors $\mathbf{a}, \mathbf{b}$ and $\mathbf{c}$. The rotation vector thus satisfies

$$\boldsymbol{\Omega}_x(t) = \mathbf{n}_x(t) \wedge \dot{\mathbf{n}}_x(t) + (\boldsymbol{\Omega}_x(t)^\top \mathbf{n}_x(t)) \mathbf{n}_x(t) \quad (36)$$

and the velocity of $\mathbf{p}_x(t)$ then follows

$$\dot{\mathbf{p}}_x(t) = \boldsymbol{\Omega}_x(t) \wedge \mathbf{p}_x(t) \quad (37)$$

Using the fact that $\mathbf{p}_x(t)$ is orthogonal to $\mathbf{n}_x(t)$ implying $\dot{\mathbf{n}}_x \mathbf{p}_x = -\mathbf{n}_x \dot{\mathbf{p}}_x$, using the above identity, the velocity reduces to

$$\dot{\mathbf{p}}_x(t) = (\mathbf{n}_x^\top \dot{\mathbf{p}}_x(t)) \mathbf{n}_x(t) + \frac{\mathbf{j}(\boldsymbol{\Omega}_x(t)^\top \mathbf{n}_x(t))}{2e} (\mathbf{p}_x^\top \dot{\mathbf{p}}_x(t) \dot{\mathbf{p}}_x(t)^* - \dot{\mathbf{p}}_x(t)) \quad (38)$$

where $e = \|\mathbf{p}_x(t) \wedge \mathbf{p}_x^*(t)\|$ can be called the ellipticity of the ellipse.

Berry connection then reads

$$\text{Im}(\mathbf{p}_x^\dagger(t) \dot{\mathbf{p}}_x(t)) = e \cdot \boldsymbol{\Omega}_x(t)^\top \mathbf{n}_x(t) \quad (39)$$

which provide an complete intrinsic interpretation of the geometric phase: It is the integral over time of the projection of the rotation vector of the plane of the ellipse onto the normal vector to this plane, weighted by the ellipticity.

In the case of Lilly's parametrization, the normal vector is $\mathbf{n}_x(t) = \sin 2\chi_x(t) \cdot (\sin \delta_x(t) \sin \gamma_x(t), -\sin \delta_x(t) \cos \gamma_x(t), \cos \delta_x(t))^\top$. The rotation vector is the sum of the rotation vector associated to the precession and nutation angles $\boldsymbol{\Omega}_{\gamma, \delta}(t) =$

$(\dot{\delta}_{\mathbf{x}}(t) \cos \gamma_{\mathbf{x}}(t), \dot{\delta}_{\mathbf{x}}(t) \sin \gamma_{\mathbf{x}}(t), \dot{\gamma}_{\mathbf{x}}(t))^{\top}$ with the rotation vector $\dot{\theta}(t)\mathbf{n}_{\mathbf{x}}$ of the rotation of the ellipse in its plane. The ellipticity is $e = \sin(2\chi_{\mathbf{x}}(t))$. Finally we obtain in this parametrization

$$\text{Im}(\mathbf{p}_{\mathbf{x}}^{\dagger}(t)\dot{\mathbf{p}}_{\mathbf{x}}(t)) = \sin(2\chi_{\mathbf{x}}(t)) \left(\dot{\theta}_{\mathbf{x}}(t) + \dot{\gamma}_{\mathbf{x}}(t) \cos \delta_{\mathbf{x}}(t) \right) \quad (40)$$

which allows to recover the result of the previous section.

4.5.2 Majorana's stellar representation

To finish and open to future work, we introduce an alternate description developed in 1932 by Ettore Majorana [22] for quantum systems. This representation identifies a J spin as the normalized and symmetrized sum of pairwise tensorial products of $J - 1$ $1/2$ spins. This representation has been introduced for the study of Berry phases by Hannay [23], and further developed by many authors, among whom Kam&Liu [24] has inspired the development here. For generalities and mathematical details, we refer to books of Penrose [25, 26].

We concentrate here on $n = 3$. The idea of Majorana is to use the identity between $\mathbb{C}^3$ and the symmetrized tensorial product $\text{Sym}^2(\mathbb{C}^2)$. Let $\mathbf{e}_1, \mathbf{e}_2$ be the canonical basis of $\mathbb{C}^2$. We construct a basis of $\text{Sym}^2(\mathbb{C}^2)$ by setting $\mathbf{S} = (\mathbf{e}_1 \otimes \mathbf{e}_1, (\mathbf{e}_1 \otimes \mathbf{e}_2 + \mathbf{e}_2 \otimes \mathbf{e}_1)/\sqrt{2}, \mathbf{e}_2 \otimes \mathbf{e}_2)$ which spans $\text{Sym}^2(\mathbb{C}^2)$, $\otimes$ standing for the tensorial product.

Let $(\mathbf{f}_1, \mathbf{f}_2, \mathbf{f}_3)$ be the canonical basis of $\mathbb{C}^3$. The so-called spin basis or spherical basis in $\mathbb{C}^3$ is given by $\mathbf{h}_0 = -(\mathbf{f}_1 - \mathbf{j}\mathbf{f}_2)/\sqrt{2}, \mathbf{h}_1 = \mathbf{f}_3, \mathbf{h}_2(\mathbf{f}_1 + \mathbf{j}\mathbf{f}_2)/\sqrt{2}$. The isomorphism between $\mathbb{C}^3$ and $\text{Sym}^2(\mathbb{C}^2)$ can be defined by identifying this last basis with the basis $\mathbf{S}$.

Any vector $\mathbf{y}$ of $\mathbb{C}^3$ can then be written as $\mathbf{x}_1 \otimes \mathbf{x}_2 + \mathbf{x}_2 \otimes \mathbf{x}_1$, where $\mathbf{x}_1$ and $\mathbf{x}_2$ are elements of $\mathbb{C}^2$:

Obviously $\mathbf{x}_1 \otimes \mathbf{x}_2 + \mathbf{x}_2 \otimes \mathbf{x}_1$ is in $\mathbb{C}^3$. If $\mathbf{x}_i = \alpha_i \mathbf{e}_1 + \beta_i \mathbf{e}_2$, then the coordinates of $\mathbf{y}$ in basis $\mathbf{S}$ are $(y_0, y_1, y_2) = (2\alpha_1\alpha_2, \sqrt{2}(\alpha_1\beta_2 + \alpha_2\beta_1), 2\beta_1\beta_2)$, which can be expressed as $\sqrt{2}(\beta_1\beta_2 - \alpha_1\alpha_2, \mathbf{j}(\alpha_1\alpha_2 + \beta_1\beta_2), \alpha_1\beta_2 + \alpha_2\beta_1)$ in the canonical basis of $\mathbb{C}^3$.

The converse is more tedious to show, but the fundamental reason is the fact that $\mathbb{C}$ is algebraically closed. Indeed, associate to $\mathbf{y}$ the second order polynomial $P(z) = y_0 z^2 - \sqrt{2}y_1 z + y_2$ where (y_0, y_1, y_2) are the coordinates of $\mathbf{y}$ in the spin basis or equivalently in basis $\mathbf{S}$. This polynomial can be factorized⁶ as $P(z) = 2(\alpha_1 z - \beta_1)(\alpha_2 z - \beta_2)$. The roots $z_i = \beta_i/\alpha_i$ can be projected back on $\mathcal{S}^2$ by the inverse stereographic projection. The two vectors pointing on the sphere can then be lifted to two vectors of $\mathbb{C}^2$! Polynomial $P(z)$ is called Majorana's polynomial.

If we use the stereographic projection with respect to the North pole down to the equatorial plane, a Jones vector $(\cos(\theta), \sin(\theta)e^{j\chi})$ is projected to the complex number $\tan(\theta)e^{j\chi}$, and conversely, any complex number z is projected back to the Jones vector $(\cos(\arctan|z|), \sin(\arctan|z|)e^{j\arg(z)})$. Thus complex numbers such that $|z| < 1$ are projected onto the south hemisphere, whereas complex numbers with $|z| > 1$ go back to the northern hemisphere. As usual in stereographic projection the North pole is projected to infinity. The vector can also be written $(1 + |z|^2)^{-1/2}(1, -z)$. Majorana

⁶The cases $\alpha_1 = 0$ and/or $\alpha_2 = 0$ are excluded from the discussion, but really mean that the problem is not three dimensional but rather two or one.

constellation is covariant to a rotation of the three dimensional space, and thus reflects the internal parameters of the polarization.

4.5.3 The geometry of Majorana's stellar representation

Let $\mathbf{p} \in \mathbb{C}^3$, $\mathbf{p}_1$ and $\mathbf{p}_2$ in $\mathbb{C}^2$, all with unit norm. Note that $\mathbf{p}_i$ projects by π on the two dimensional sphere on a point or *star* which can be represented by a euclidean unit vector $\mathbf{u}_i$ which satisfies $\pi(\mathbf{p}_i) = \mathbf{p}_i \mathbf{p}_i^\dagger = (\mathbf{I} + \mathbf{u}_i \cdot \boldsymbol{\sigma})/2$, where $\boldsymbol{\sigma}$ is the basis of Pauli matrices ⁷.

$$\mathbf{p} = \frac{1}{\sqrt{2A_2}} (\mathbf{p}_1 \otimes \mathbf{p}_2 + \mathbf{p}_2 \otimes \mathbf{p}_1) = \frac{1}{\sqrt{2A_2}} \tilde{\mathbf{p}} \quad (41)$$

and the normalizing constant reads

$$2A_2 = \tilde{\mathbf{p}}^\dagger \tilde{\mathbf{p}} = 2 \left(1 + \frac{1 + \mathbf{u}_1 \cdot \mathbf{u}_2}{2} \right) = 3 + \mathbf{u}_1 \cdot \mathbf{u}_2 \quad (42)$$

This is shown using Pauli representation that proves $|\mathbf{p}_1^\dagger \mathbf{p}_2|^2 = (1 + \mathbf{u}_1 \cdot \mathbf{u}_2)/2$. Now as shown in Appendix E, it is possible to relate Berry connection (or equivalently the linear form $A_{\mathbf{p}}(\dot{\mathbf{p}})$) in $\mathbb{C}^3$ to Berry connections $A_{\mathbf{p}_i}(\dot{\mathbf{p}}_i)$ in $\mathbb{C}^2$ and the vector $\mathbf{u}_i$ pointing on the Poincaré sphere. This link is provided by

$$\text{Im}(\mathbf{p}^\dagger \dot{\mathbf{p}}) = \text{Im}(\mathbf{p}_1^\dagger \dot{\mathbf{p}}_1) + \text{Im}(\mathbf{p}_2^\dagger \dot{\mathbf{p}}_2) + \frac{1}{2} \frac{(\dot{\mathbf{u}}_1 - \dot{\mathbf{u}}_2) \cdot (\mathbf{u}_1 \wedge \mathbf{u}_2)}{3 + \mathbf{u}_1 \cdot \mathbf{u}_2}$$

If $\mathbf{p}_i$ is parametrized as a Jones vector as $(\cos(\alpha_i), \sin(\alpha_i)e^{j\gamma_i})$, the Berry connection of each vectors writes $\text{Im}(\mathbf{p}_i^\dagger \dot{\mathbf{p}}_i) = \dot{\gamma}_i \sin^2(\alpha_i)$, and vectors $\mathbf{u}_i$ have spherical coordinates $(\sin(2\alpha_i) \cos(\gamma_i), \sin(2\alpha_i) \sin(\gamma_i), \cos(2\alpha_i))$. The previous equation along with the calculation of $\arg(\mathbf{p}^\dagger(t_0)\mathbf{p}(t))$ allows to evaluate the geometric phase.

A beautiful result appears if $\mathbf{p}(t)$ is T -periodic. In this case, the roots of Majorana's polynomial are T -periodic and the vectors $\mathbf{p}_i$ as well. Thus after an interval on length T , the different vectors are in phase with the origins, meaning $\arg(\mathbf{p}^\dagger(t_0)\mathbf{p}(t_0 + T)) = 0 = \arg(\mathbf{p}_i^\dagger(t_0)\mathbf{p}_i(t_0 + T))$, $i = 1, 2$. Therefore for a closed loop, we obtain

$$\varphi_{g,\mathbf{x},t_0} = \varphi_{g,\mathbf{x}_1,t_0} + \varphi_{g,\mathbf{x}_2,t_0} - \frac{1}{2} \int_{t_0}^{t_0+T} \frac{(\dot{\mathbf{u}}_1 - \dot{\mathbf{u}}_2) \cdot (\mathbf{u}_1 \wedge \mathbf{u}_2)}{3 + \mathbf{u}_1 \cdot \mathbf{u}_2} dt$$

where we see the contribution from each star is completed by a contribution linked to the interaction between these two stars.

We can study the conditions under which the interaction contribution is zero:

- if $\mathbf{u}_1 = \mathbf{u}_2$: We end up with a two dimensional vector $\mathbf{p}$, and we are back to the bivariate case. This occurs if $\theta_1(t) = \pi/2$ or $\theta_2(t) = 0$. In this case, Majorana's polynomial is of degree 1, and there is only one star;
- if $\dot{\mathbf{u}}_1 = \dot{\mathbf{u}}_2$: This situation occurs if the great circle connecting the two stars experiences a rotation and the stars do not move along this circle;

⁷Pauli matrices are described in appendix E

- if $\dot{\mathbf{u}}_1 - \dot{\mathbf{u}}_2 \perp \mathbf{u}_1 \wedge \mathbf{u}_2$: The two stars slide along the great circle connecting them.

In all others cases, the interaction term is present. Kam&Liu in [24] explicit the following form shown in the appendix as well. Let $\mathbf{R} = (\mathbf{u}_1 + \mathbf{u}_2)/2$ and $\mathbf{r} = \mathbf{u}_1 - \mathbf{u}_2$, and $\tilde{\mathbf{R}}, \tilde{\mathbf{r}}$ their normalized counterparts. Then $\mathbf{u}_1 = \mathbf{R} + \mathbf{r}/2$ and $\mathbf{u}_2 = \mathbf{R} - \mathbf{r}/2$. Then Berry connection writes

$$\text{Im}(\dot{\mathbf{p}}^\dagger \dot{\mathbf{p}}) = \text{Im}(\dot{\mathbf{p}}_1^\dagger \dot{\mathbf{p}}_1) + \text{Im}(\dot{\mathbf{p}}_2^\dagger \dot{\mathbf{p}}_2) - \frac{1}{2} f(\psi) \cos(\psi) (\tilde{\mathbf{r}} \wedge \dot{\tilde{\mathbf{r}}}) \cdot \tilde{\mathbf{R}}$$

where $2\psi = \arccos(\mathbf{u}_1 \cdot \mathbf{u}_2)$ and $f(\psi) = \sin^2(\psi)/(1 + \cos^2(\psi))$. Thus ψ is the angle between the $\mathbf{u}_i$'s and $\mathbf{R}$, or equivalently half the angle between the vectors pointing to the two stars, and $(\tilde{\mathbf{r}} \wedge \dot{\tilde{\mathbf{r}}}) \cdot \tilde{\mathbf{R}}$ is the infinitesimal angle between $\mathbf{r}$ and $\dot{\mathbf{r}}$.

This concludes the theoretical developments we wanted to present. We now turn to more pragmatic aspects: estimation issues in the next section, more interpretations on synthetic cases and potential applications in signal processing in section 6.

5 Estimation of the geometric phase

In this section, we consider the problem of estimating the geometric phase of a discretized multivariate signal $\{\mathbf{x}(t_k)\}_{k=0}^K$ for a set of time instants $\{t_k\}_{k=0}^K$, with $\mathbf{x}(t_k) \in \mathbb{C}^n$. The goal is to estimate the geometric phase accumulated by this signal when reaching the time t_k . Obviously, the measurement is always dependent of the initial instant of measure, and we always consider that it is t_0 in the sequel. For this purpose, we introduce the Bargmann invariant which is well known in quantum mechanics [14] and propose a recursive estimator inspired by the work in [27]. **We also describe the influence of the sampling frequency rate on the geometric phase estimation accuracy.** Recall that we consider the case of normalized signals, *i.e.* $\mathbf{x}^\dagger(t_k)\mathbf{x}(t_k) = 1 \quad \forall t_k$.

Note that in what follows, we consider signals sampled uniformly. However, the theory presented works with non-uniform sampling as well.

5.1 Bargmann invariant

A way to estimate the geometric phase is to use its connexion to the so-called *Bargmann invariant*, which was initially introduced in quantum mechanics, in the study of invariants for the Wigner theorem [28]. The use of Bargmann invariant to compute the geometric phase was detailed in [14]. In the context of signal processing it reads as follows: the geometric phase accumulated at time instant t_k since time t_0 by the multivariate signal $\{\mathbf{x}(t_k)\}_{k=1}^K$ is given as:

$$\phi_{g,\mathbf{x},t_0}(t_k) = -\arg \mathbf{B}_{\mathbf{x}}^{0:k} \quad (43)$$

where the Bargmann invariant $\mathbf{B}_{\mathbf{x}}^{i:j}$, with $j > i$, reads:

$$\mathbf{B}_{\mathbf{x}}^{i:j} = \mathbf{x}^\dagger(t_i) \mathbf{x}(t_{i+1}) \left[\prod_{l=i+1}^{j-1} \mathbf{x}^\dagger(t_l) \mathbf{x}(t_{l+1}) \right] \mathbf{x}^\dagger(t_j) \mathbf{x}(t_i) \quad (44)$$

A demonstration (directly inspired from [14]) of the relation given in equation (43) can be found in Appendix C.

The Bargmann invariant is a $U(1)^{\otimes K}$ -invariant: Given a smooth function $\alpha(t)$, the discrete multivariate time series $\tilde{\mathbf{x}}$ given as:

$$\tilde{\mathbf{x}}(t_k) = e^{j\alpha(t_k)} \mathbf{x}(t_k) \quad \forall t_k \quad (45)$$

has the same Bargmann invariant as $\mathbf{x}$. This reads for any pair (i, j) with $j > i$ as:

$$\mathbf{B}_{\mathbf{x}}^{i:j} = \mathbf{B}_{\tilde{\mathbf{x}}}^{i:j} \quad (46)$$

Hence, two multivariate signals being two sections of the fiber bundle with the same projection in the projective space have the same Bargmann invariant. This is a gauge invariant property that is shared by the geometric phase. This makes the relation between the Bargmann invariant and the geometric phase a rather geometrically intuitive one.

When considering unit norm multivariate signals (*i.e.* signals with $\|\mathbf{x}(t_k)\|^2 = 1, \forall t_k$), Bargmann invariant takes the form:

$$\mathbf{B}_{\mathbf{x}}^{i:j} = \text{Tr}(\rho_{\mathbf{x}}(t_i) \rho_{\mathbf{x}}(t_{i+1}) \dots \rho_{\mathbf{x}}(t_j)) \quad (47)$$

with $\rho_{\mathbf{x}}(t_k) = \mathbf{x}(t_k) \mathbf{x}^\dagger(t_k)$. This expression of the Bargmann invariant shows that the geometric phase accumulated along a path is the argument of the trace of a product of hermitian matrices. The hermiticity of the product is not guaranteed by multiplication of the projectors $\rho_{\mathbf{x}}(t_k)$, and the eigenvalues may be complex valued: This results in a non-null argument for the trace, *i.e.* appearance of a geometric phase. Taking $j \log \mathbf{B}_{\mathbf{x}}^{0:k}$, and using the expression on the right hand side in equation (44), we get a direct expression of the geometric phase as the sum of pairwise angles between (normalized) signal values, *i.e.*:

$$\phi_{g, \mathbf{x}, t_0}(t_k) = -\arg \mathbf{B}_{\mathbf{x}}^{0:k} = -\sum_{l=0}^{k-1} \arg \mathbf{x}(t_l)^\dagger \mathbf{x}(t_{l+1}) - \arg \mathbf{x}(t_k)^\dagger \mathbf{x}(t_0) \quad (48)$$

which is the expression used in optics [27] for the computation of the geometric phase. Thus, expression (44) and (48) provide effective, and equivalent, ways to compute the geometric phase of a discretized multivariate signal.

It is also possible to define a recursive estimator for the geometric phase, which allows to obtain its estimation at time instant t_{k+1} given the one at t_k . The expression

for this estimator reads:

$$\phi_{g,\mathbf{x},t_0}(t_{k+1}) = \phi_{g,\mathbf{x},t_0}(t_k) + \arg \left(\frac{\mathbf{x}^\dagger(t_{k+1})\mathbf{x}(t_k)\mathbf{x}^\dagger(t_k)\mathbf{x}(t_0)}{\mathbf{x}^\dagger(t_{k+1})\mathbf{x}(t_0)} \right) \quad (49)$$

where a natural choice for the initialization is to set $\phi_{g,\mathbf{x},t_0}[t_0] = 0$. The derivation to obtain the expression (49) is simple and provided in Appendix C. In the sequel, we will make use of both estimators indistinctively in the illustrations section and we consider the geometric phase concept in noise-free scenarios. More detailed study of geometric phase estimators is left for future works.

5.2 Estimation of the geometric phase: influence of sampling frequency

Consider a continuous-time multivariate signal $\mathbf{x}(t)$ and its sampled version $\{\mathbf{x}(t_k)\}_{k \in \mathbb{Z}}$ with $t_k = kT_e$, where $1/T_e = F_e$ is the sampling frequency. Let us fix a finite time horizon, say $\Delta = [0, 1]$, and assume that the *continuous* time signal $\mathbf{x}(t)$ describes a continuous path $\mathcal{C}$ in projective space $\mathbb{CP}^{n-1}$. A natural question is whether the geometric phase acquired by $\mathbf{x}(t)$ during the horizon Δ and associated to the path $\mathcal{C}$ can be related to the geometric phase of the *discrete* signal $\{\mathbf{x}(t_k)\}$ over the same horizon Δ .

As recalled in Appendix C and in [14, 29], the expression of the Bargmann invariant associated to the discrete signal $\{\mathbf{x}(t_k)\}$ defines a geometric phase corresponding to a polygon in the projective space $\mathbb{CP}^{n-1}$. More precisely, recalling that $\rho_{\mathbf{x}}(t_k) = \mathbf{x}(t_k)\mathbf{x}^\dagger(t_k)$, then the geometric phase associated to $\rho_{\mathbf{x}}(t_1), \dots, \rho_{\mathbf{x}}(t_m)$ corresponds to that of a m -vertex polygon defining a path $\mathcal{C}_{T_e}$ in $\mathbb{CP}^{n-1}$, such that successive $\rho_{\mathbf{x}}(t_k)$ (modulo m) are connected by (free) geodesics [29, Eq. (2.9)]. As the sampling period T_e decreases (*i.e.*, sampling frequency increases), the polygon vertices become denser, and the discrete path approaches the continuous curve $\mathcal{C}$.

Fig. 4 illustrates this phenomenon for a bivariate signal $\mathbf{x}(t)$ with $\rho_{\mathbf{x}}(0) = \rho_{\mathbf{x}}(1)$ (path $\mathcal{C}$ is a loop) and $\rho_{\mathbf{x}}(t)$ describes an iso-latitude trajectory in $\mathbb{CP}^1 \equiv \mathcal{S}^2$. In this case, the geometric phase of the continuous signal is available in closed form, *i.e.*, $\phi_{\mathbf{x},0}(1) = \pi(1 - \cos \alpha)$ where α is the co-latitude. We evaluate for increasing values of sampling frequencies the geometric phase of the discrete signal using (43); we observe rapid convergence of the discrete geometric phase towards its continuous part, which falls below 10^{-3} relative error at $F_e \simeq 100$ Hz and below 10^{-4} relative error at $F_e \simeq 10^3$ Hz – convergence is quadratic, as shown in Appendix C.

6 Illustrations

This section presents several examples of the appearance of a geometric phase for multivariate signals. First, a case highlighting a few properties of the geometric phase is presented, in the bivariate case, for a specific polarization dynamics (*i.e.* a specific path in the projective space) of the signal. Then, a second example demonstrates how the geometric phase affects the instantaneous frequency interpretation/concept. This phenomenon is also illustrated in the case of a bivariate signal. The third example

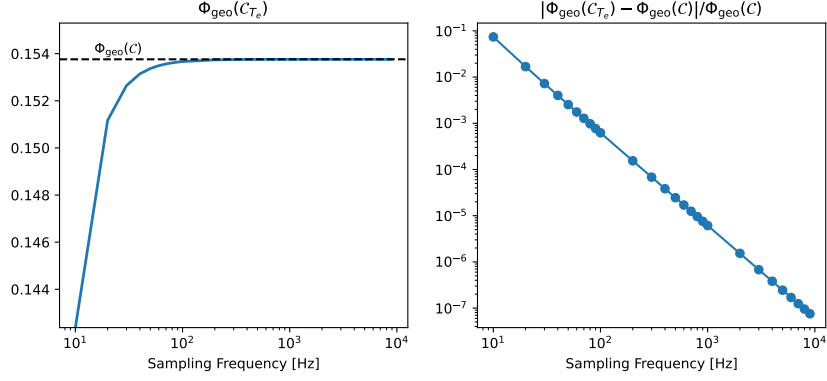


Fig. 4 Convergence of the geometric phase computed from discrete samples its continuous counterpart as a function of the sampling frequency F_e .

concerns multivariate signals of higher dimension and relates to an array processing scenario: we introduce a case where the geometric phase reveals imperfections affecting of a linear array of sensors.

6.1 Geodesic triangle and spherical cap on the Poincaré sphere

Central to the values and the dynamics of geometric phase is the geometry of the path in the projective space. Sections 6.1.1 and 6.1.2 illustrate some of the properties of the geometric phase driven by this path.

6.1.1 Geodesic triangle path

As a first example, we consider the case of a monochromatic bivariate signal $\mathbf{x}(t)$ with constant amplitude and which coherence matrix $\rho_{\mathbf{x}}(t)$ follows a path made of two geodesic triangles. Figure 5 shows the trajectory followed by $\rho_{\mathbf{x}}(t)$ in the projective space (which is $\mathcal{S}^2$ in the bivariate case). The initial point (at time $t = 0s$) is 'A' and arrows show the order with which the different points/locations are visited: 'A' $\rightarrow$ 'B' $\rightarrow$ 'C' $\rightarrow$ 'A' $\rightarrow$ 'D' $\rightarrow$ 'C' $\rightarrow$ 'A'. The blue trajectory ('A' $\rightarrow$ 'B' $\rightarrow$ 'C' $\rightarrow$ 'A') is followed firstly, from time $t = 0s$ to $t = 3s$, while the yellow trajectory ('A' $\rightarrow$ 'D' $\rightarrow$ 'C' $\rightarrow$ 'A') is followed from $t = 3s$ to $t = 6s$. The blue path is traversed counterclockwise while the yellow one is traversed clockwise. The full trajectory thus consists in two consecutive geodesic triangle paths traveled in opposite directions.

This trajectory followed in the projective space induces a geometric phase for the bivariate signal $\mathbf{x}(t)$. In Figure 6, the two components $x_1(t)$ and $x_2(t)$ of the bivariate signal $\mathbf{x}(t)$ are displayed in the two upper rows, while the geometric phase accumulated along the path is displayed in the lower row. Along the time axis, the various points traversed along the path are given. Remembering that the geometric phase is computed with respect to the initial point 'A' at time $t = 0s$, its variations along time can be understood as follows:

- From 'A' to 'B': the path is a geodesic (a great circle) from point 'A' on the equator to point 'B' also on the equator. As a consequence, the geometric phase accumulated is zero from $t = 0s$ to $t = 1s$.
- From 'B' to 'C': from position 'B', the complete path (starting from 'A' the initial point) is no longer a geodesic as there is a right angle turn. As a consequence, the geometric phase accumulated from $t = 0s$ to any point between 'B' and 'C' (*i.e.* for $1s < t \leq 2s$) is no longer null. As the area surrounded by the path is not closed, the computation of the geometric phase consists in closing it with a geodesic. The area is thus a geodesic triangle with non-equal edges which values is given by the Girard theorem **Est-ce qu'on fait une annexe pour justifier le comportement en artan ?** which fits the geometric phase curve in Figure 6 for $1s < t \leq 2s$. When reaching position 'C', the geodesic triangle covers a surface of a $1/16$ of the sphere surface, *i.e.* $4\pi/16$). This corresponds to a geometric phase of half this area, which is $\pi/8 \simeq -0.39$ as can be seen on the lower row of Figure 6.
- From 'C' to 'A': as the edge between 'C' and 'A' is a geodesic, the geometric phase does not accumulate along this portion of the path, so that it remains identical to what it reaches at position 'C', *i.e.* $\pi/8$. This causes the plateau in lower row of Figure 6 between $t = 2s$ and $t = 3s$. At this point, half of the path has been traversed.
- From 'A' to 'D', then to 'C' and back to 'A': from $t = 3s$, the first geodesic triangle has been traversed and the second one is traversed during the last $3s$. This second part of the path consists in the same shape, but traveled in opposite direction. The consequence is that the variations of the geometric phase in the interval starts with a plateau, then an increase with the same Girard theorem behaviour and finally a plateau during the last second, with a geometric phase having reached the zero value.

The behaviour of the geometric phase on this path highlights several of its properties listed in Section 3.8. For example, the fact that following a geodesic in the projective space does accumulate zero geometric phase can be seen on several parts of this. One needs to understand that when saying that, it is always relative to the original/reference point, so that a path made of several geodesic arcs (just like here as can be seen in Figure 5) may not have a null geometric phase. It is the total path, relative to the reference point that should be a geodesic, for the geometric phase to stay null. Then, the direction on the path is important as it controls the sign of the geometric phase accumulated, allowing potentially (as shown in this example) the geometric phase to vanish at some point after having reached high values. This property is important as it means that the interpretation of the geometric phase should be made on a time interval (and initial point) and not for its value at a given time instant.

NLB: Changer la notation dans les figures pour les phases.

This first example gives some indications on the cases where a geometric phase can be accumulated over time for a bivariate signal. The computation of this phase can be made using an estimator as given in 5. The effect of the phase on the signal may not be obvious at first sight, for example on $x_1(t)$ and $x_2(t)$ as displayed in the first two rows of Figure 6. The next section illustrates how some important observables/parameters of a bivariate modulated signal can be affected by the existence of a geometric phase.

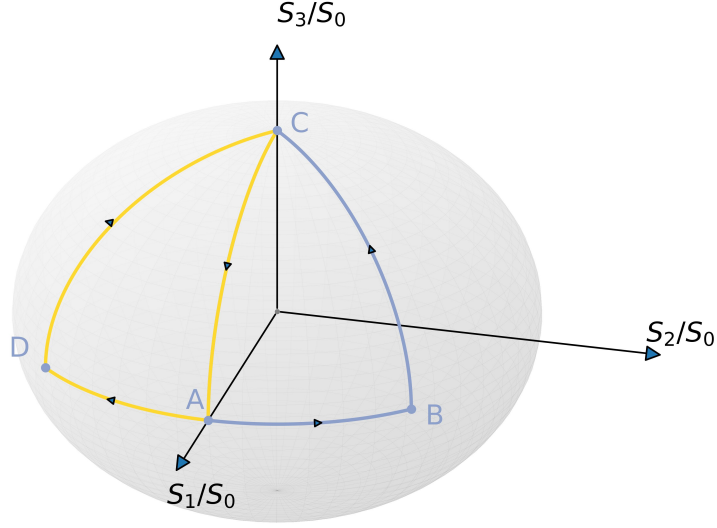


Fig. 5 Path followed by $\rho_{\mathbf{x}}(t)$ on the Poincaré Sphere (*i.e.* S^2) between time $t = 0s$ at initial location 'A' and time $t = 6s$ at point 'A'. The blue path 'ABCA' is traversed counterclockwise while the yellow path 'ADCA' is traversed clockwise.

6.1.2 Spherical cap path

Still in the bivariate case, we now consider a trajectory for the coherence matrix which is an iso-latitude closed loop on the Poincaré sphere, as depicted on the right-hand side of Figure 7. The starting and ending points of the trajectory are identical and indicated by a light blue dot. During the time interval $[0, 1]$, the bivariate signal has thus its coherence matrix $\rho_{\mathbf{x}}(t)$ that follows a path with constant co-elevation parameter $\chi_{\mathbf{x}}(t) = \pi/6$, and attitude parameter $\theta_{\mathbf{x}}(t)$ that takes various expressions, namely πt , $\pi(4(t - \frac{1}{2})^3 + \frac{1}{2})$, $\pi((\frac{t}{4} - \frac{1}{8})^{1/3} + \frac{1}{2})$, πt^3 and $\pi t^{1/3}$. For each of one of these evolutions, the geometric phase is computed and displayed on the left-hand side of Figure 7. In all cases, the initial value of the geometric phase is 0, and its final value,

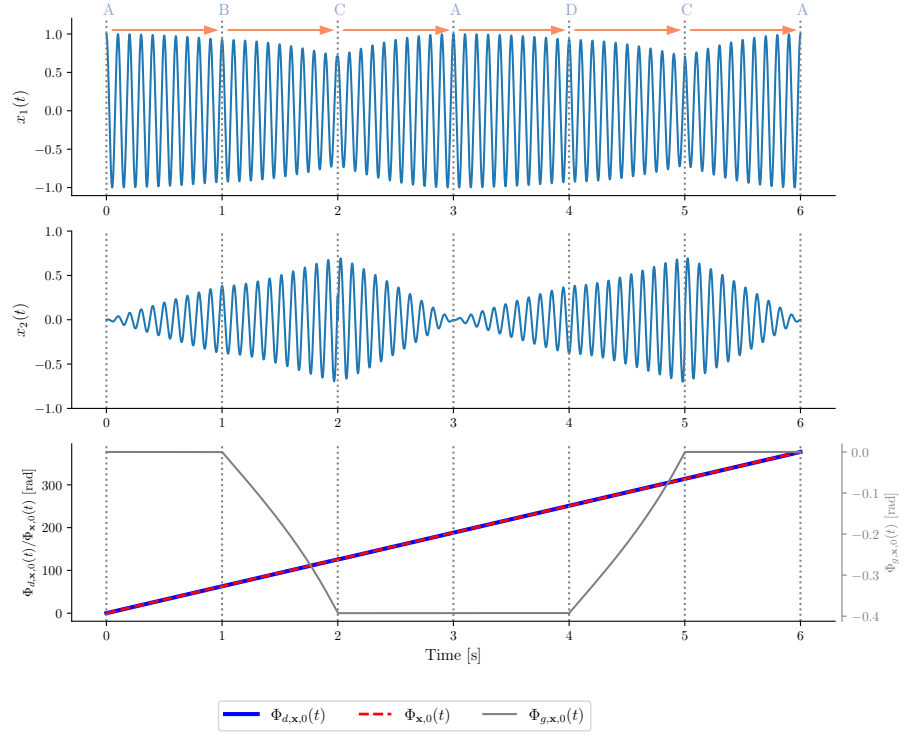


Fig. 6 Components $x_1(t)$ (top) and $x_2(t)$ (middle) of the bivariate signal and the accumulated geometric phase (bottom) during the time interval taken to traverse the path in the projective space depicted in Figure 5. Positions on the projective space are label at time they are reached.

accumulated along the paths during the time interval $[0, 1]$ is $\pi/2$. As explained in 4.4, this value of the geometric phase corresponds to half the area of the cap delimited by the path. Here, the cap has a surface of π so that $\phi_{g,x,0}(1) = \pi/2$. The cap with such an area (of π) corresponds to a co-elevation parameter $\chi_x(t) = \pi/4$ with the classical parametrization of the Poincaré sphere [20].

The various behaviour of the geometric phase accumulated during the time interval are displayed in Figure 7. The way $\theta_x(t)$ varies with time dictates the geometric phase accumulation along time. As the same loop (starting and arrival point) is followed for each case, the total accumulated phase is the same once the path is entirely parcouré. This illustrates two properties of the geometric phase: first it is relative to the initial point, and second the velocity at which the path is followed does not influence the result, as long as the final point is the same. The differences in velocities displayed for $\theta_x(t)$ show that for example at time $0.5s$, three curves have a value of $\pi/4$, so that they are at the location on the path on the sphere. The behaviour are different thanks to the dynamics of $\theta_x(t)$. One can thus say that the value of the geometric phase is really dependent on the path length in the projective space and does not depend on the velocity used to go from the initial point to the final point.

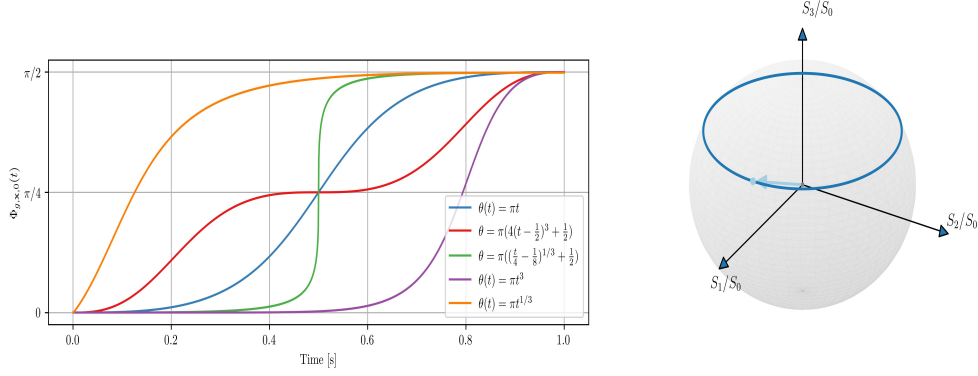


Fig. 7 Geometric phase (left hand-side) for an iso-latitude loop path on the Poincaré sphere (right hand side) followed at different velocities. The closed loop on the Poincaré sphere is at constant co-elevation $\chi_{\mathbf{x}} = \pi/4$ and with various $\theta_{\mathbf{x}}(t)$. The initial and final values of the geometric phase are identical for all $\theta_{\mathbf{x}}(t)$, but their evolution depend on the dynamics of the azimuth angle $\theta_{\mathbf{x}}(t)$.

6.2 Geometric phase effect on instantaneous frequency

We consider a type of bivariate signal $\mathbf{x}(t)$ that is narrow-banded and with polarization parameters $\theta_{\mathbf{x}}(t)$ and $\chi_{\mathbf{x}}(t)$ fluctuating (with predefined law) over time. This is a modulated bivariate signal. As was detailed previously, in the bivariate case, the projective space $\mathbb{CP}^{n-1}$ is the 2-sphere $\mathcal{S}^2$ in $\mathbb{R}^3$ which can be parametrized with $\theta_{\mathbf{x}}(t)$ and $\chi_{\mathbf{x}}(t)$. These parameters are the one used in equation (25) for a modulated bivariate signal. In this example, we set the following time dependency for the parameters of $\mathbf{x}(t)$

$$\begin{cases} a_{\mathbf{x}}(t) = 1 \\ \phi_{\mathbf{x}}(t) = 2\pi \frac{(\nu_1 - \nu_0)}{2} t^2 + 2\pi \nu_0 t \\ \theta_{\mathbf{x}}(t) = 2\pi \nu_{\theta} t^{\alpha} - \frac{\pi}{4} \\ \chi_{\mathbf{x}}(t) = 2\pi \nu_{\chi} t^{\beta} + \frac{\pi}{16} \end{cases} \quad (50)$$

From the expression of $\phi_{\mathbf{x}}(t)$ in (50) one can see that $\mathbf{x}(t)$ is a linear chirp in frequency, while its parameters $\theta_{\mathbf{x}}(t)$ and $\chi_{\mathbf{x}}(t)$ have power-law variations with respect to time. In the simulations presented below, the frequencies have been set to $\nu_0 = 100$, $\nu_1 = 120$, $\nu_{\theta} = 2$ and $\nu_{\chi} = 1$.

In Figures 8, 9, and 10 the time frequency representation of $\mathbf{x}(t)$ is given for three sets of values, respectively $\alpha = 0$ and $\beta = 0$ for Figure 8, $\alpha = 1$ and $\beta = 0.5$ for Figure 9, and $\alpha = 2$ and $\beta = 1.5$ for Figure 10. The time-frequency representations are computed using the BiSPy toolbox [30] and are composed of four images corresponding to the Stokes time-frequency representation introduced in [20] to analyze non-stationary bivariate signals. The S_0 component encodes the energy (just as standard time-frequency representation), while S_1 , S_2 and S_3 solely encode the polarization information related to the parameters $\theta_{\mathbf{x}}(t)$ and $\chi_{\mathbf{x}}(t)$. Details and exact expressions can be found in [20]. In Figure 8, the polarization information is constant in the bandwidth of the signal and during the whole time interval. As $\theta_{\mathbf{x}}(t)$ and $\chi_{\mathbf{x}}(t)$ are both constants, there is no geometric phase acquired over time for this case and therefore

the linear frequency is unaltered (*i.e.* it stays linear on the whole time interval, see S_0 in the upper right plot of Figure 8).

On the opposite, in Figures 9 and 10, parameters $\theta_{\mathbf{x}}(t)$ and $\chi_{\mathbf{x}}(t)$ are fluctuating with time, according to (50), with two different sets of values (α, β) . The fluctuations of these parameters can be seen on S_1 , S_2 and S_3 as they are functions of them (see [20] for detailed expressions). In addition, an extra phenomenon happens in the S_0 component in Figures 9 and 10 (upper left corners): the instantaneous frequency, which should be vary linearly with time as in the model (50), exhibits fluctuations around the value it has when $\theta_{\mathbf{x}}(t)$ and $\chi_{\mathbf{x}}(t)$ are null/constant. This effect is due to the existence of a geometric phase, and this effect is amplified as time grows. More precisely, the deviations from the straight slope (from 100 Hz to 250 Hz in the time-frequency plane) that can be seen in Figures 9 and 10 is due to the *geometric frequency* term as described in section 2.4. As a consequence, denoting $f_{\mathbf{x}(\alpha, \beta)}(t)$ the instantaneous frequency of the signal from the model (50) one get the following expression:

$$f_{\mathbf{x}}(t) = f_{\mathbf{x}(0,0)}(t) + \nu_{\theta} \alpha t^{\alpha-1} \sin \left(4\pi \nu_{\chi} t^{\beta} + \frac{\pi}{8} \right) \quad (51)$$

where $f_{\mathbf{x}(0,0)}(t)$ is the instantaneous frequency of the signal with constant $\theta_{\mathbf{x}}(t)$ and $\chi_{\mathbf{x}}(t)$ parameters (the type of signal displayed in Figure 8). In the simple case of a linear chirp, we have that $f_{\mathbf{x}(0,0)}(t) = (\nu_1 - \nu_0)t + \nu_0$. The perturbation due to the geometric phase (*i.e.* the geometric frequency) is not dependent of the actual instantaneous frequency of the signal $\nu_{\mathbf{x}(0,0)}(t)$, but only depend on trajectory in the projective space $\mathbb{CP}^{n-1}$.

The results displayed in this section illustrate the way a geometric phase can affect a central parameter/observable of modulated bivariate (and more generally multivariate) signals: its instantaneous frequency. Such a phenomenon was not foreseen by authors in [4, 6] as they only considered constant values $\mathbf{p}(t) = \mathbf{p}_0$ in the model depicted in (11). However, as recalled earlier, Lilly in [7] described the effect of possible fluctuating angles, but did not identify it with a geometric phase. Letting this parameter change with time allows to model a much larger class of multivariate signals, but the time dependencies of $\mathbf{p}(t)$ comes with the possible appearance of a geometric phase that can perturbate the instantaneous frequency evaluation (in a scenario where one wants to estimate this frequency from an observed signal). Monitoring (through its evaluation/estimation) the geometric phase allows potentially evaluate its contribution to the instantaneous frequency estimation. The geometric phase effect should thus be taken into account systematically to avoid any mismatch with the "true" instantaneous frequency. Note also that the fluctuations of the instantaneous frequency due to the geometric phase carry some informations on the $\mathbf{p}(t)$ vector. In the bivariate case, this can be understood as polarization fluctuations and could be of interest to estimate in applications where polarization fluctuations carry information on the source that generates the bivariate signal (see [21] for an example on seismology data).

This example shows that the geometric phase has potential influence on some central quantities of multivariate modulated signals, and thus should be evaluated in the process of analysing time-varying properties of non stationnary signals.

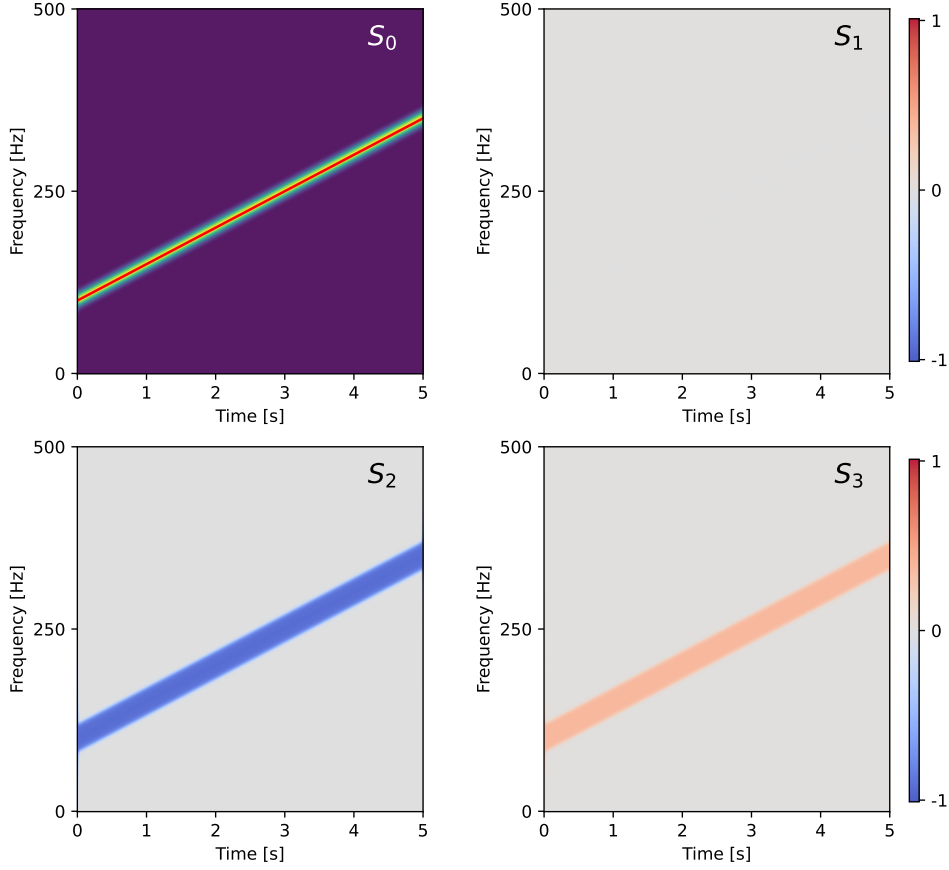


Fig. 8 Time frequency representation of $\mathbf{x}(t)$ as in (50) for $\alpha = 0$ and $\beta = 0$.

6.3 Geometric phase effects in array processing

In this section, we consider the case where the multivariate nature of the signal comes from the multiple measurements/signals of an array of sensors. The multivariate signal $\mathbf{x}(t) \in \mathbb{C}^{N_s}$ is made of the recordings of the N_s sensors of the array. In the sequel, we only detail cases including linear arrays, but a generalization to arbitrarily shaped arrays could be made. In the examples below, it is shown how geometric phases can appear when Uniform Linear Arrays (ULA) of sensors suffer from imperfections or are subject to perturbations. ULAs are used on a daily basis in a large range applications such as seismology, communications, acoustics (from medical ultrasound imaging to oceanography) or astrophysics to name just a few.

Consider a set of N_s sensors dispatched along a line and equally spaced with an inter-sensor distance d . The multivariate signal received on the antenna is $\mathbf{x}(t) =$

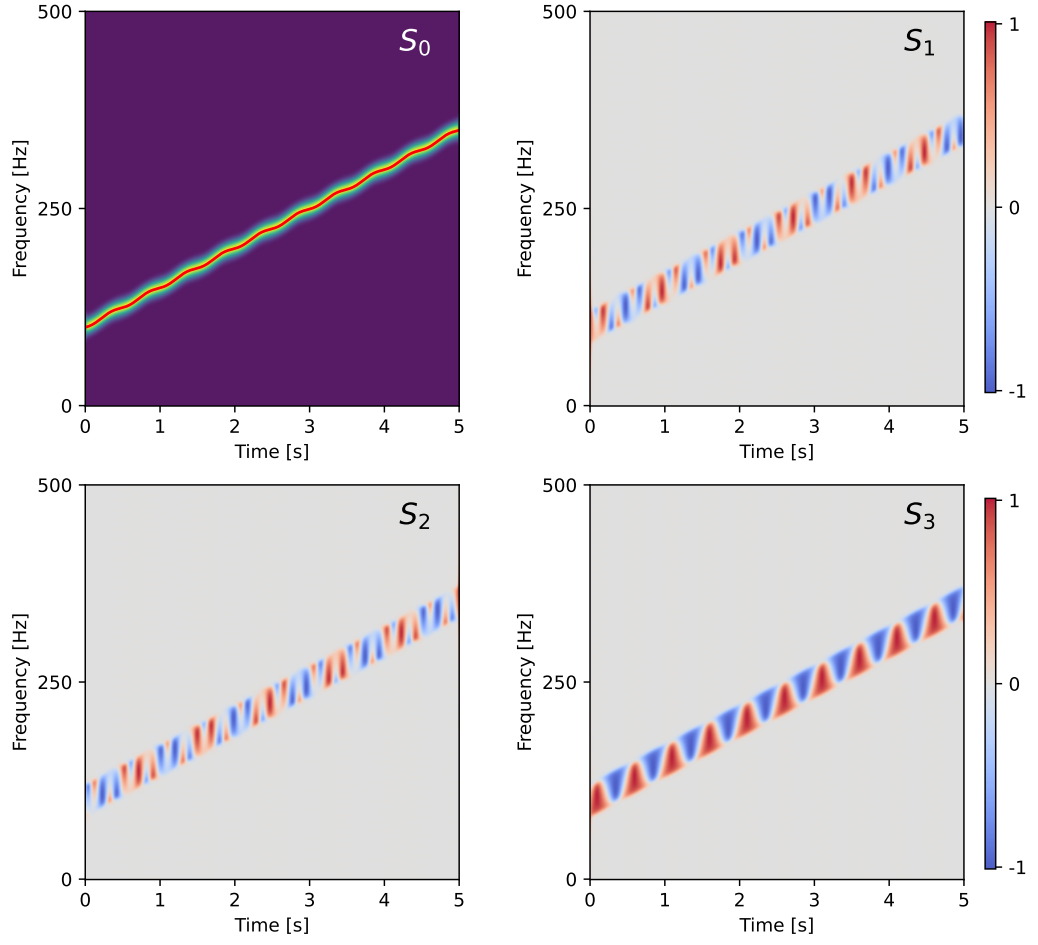


Fig. 9 Time frequency representation of $\mathbf{x}(t)$ as in (50) for $\alpha = 1$ and $\beta = 1.5$

$[x_1(t), x_2(t), \dots, x_{N_s}(t)]^T \in \mathbb{C}$, where $x_k(t)$ is the signal recorded by the k^{th} sensor of the array. The signal is supposed assumed narrow-banded (a classical assumption in array processing) with central frequency ν_0 , so that it can be written like:

$$\mathbf{x}(t) = a_{\mathbf{x}}(t)e^{-j2\pi\nu_0 t}\mathbf{s}(\tau) \quad (52)$$

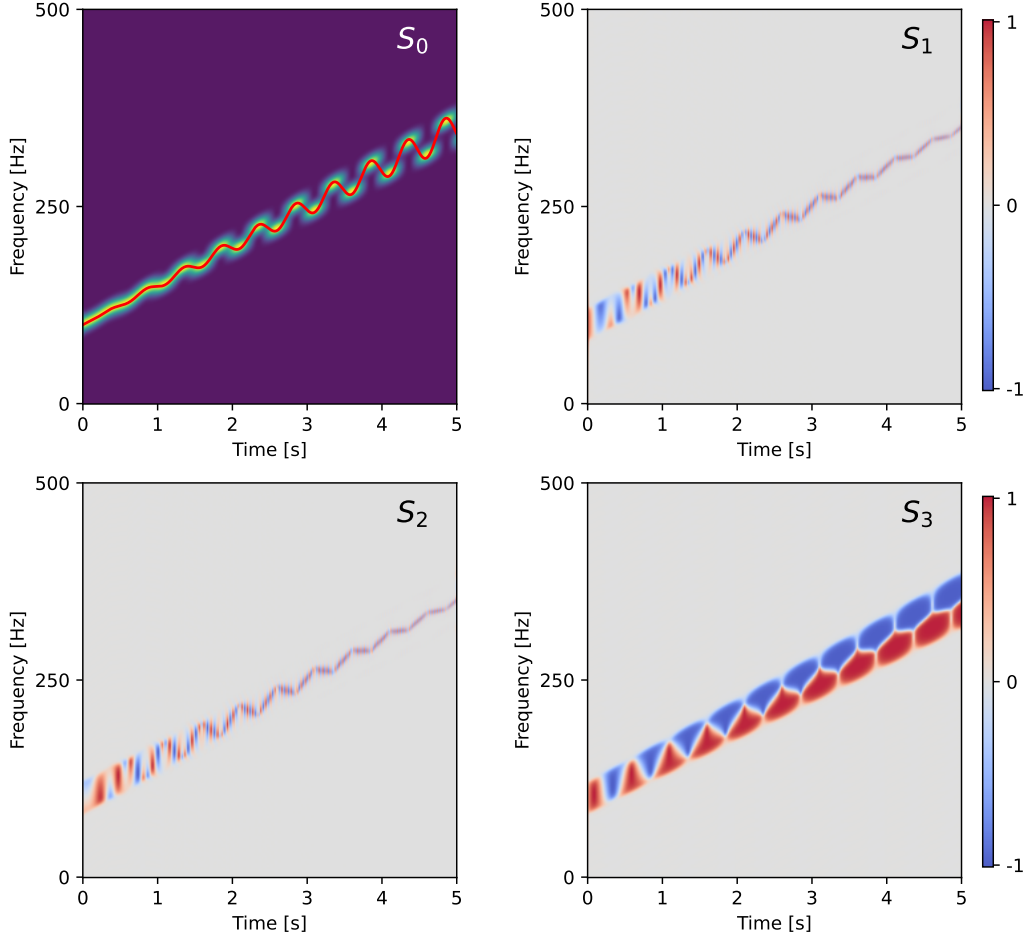


Fig. 10 Time frequency representation of $\mathbf{x}(t)$ as in (50) for $\alpha = 2$ and $\beta = 1.5$.

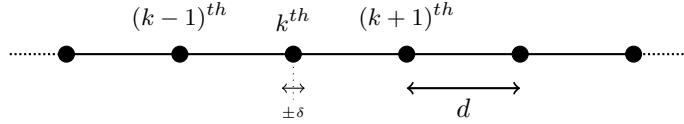
where the normalized *steering vector* $\mathbf{s}(\tau) \in \mathbb{C}^{N_s}$ associated to a source located in the far-field of the array and impinging on it with an angle θ is given by:

$$\mathbf{s}(\tau) = \frac{1}{\sqrt{N_s}} \begin{bmatrix} 1 \\ e^{j2\pi\nu_0\tau} \\ \vdots \\ e^{j2\pi\nu_0(N_s-1)\tau} \end{bmatrix} \quad (53)$$

where $\tau = (d/c) \cos \theta$ is the time delay between two adjacent sensors for a source located at an "angle of arrival" θ and where c is the wave velocity in the propagation medium. The steering vector is a central quantity in array processing as it encodes the spatial signature of the source on the array. It is used in many applications such as beamforming, direction of arrival estimation, source separation, etc. Here again, we will consider cases with no amplitude variations, *i.e.* $a_{\mathbf{x}}(t) = 1 \forall t$. The steering vector expression in (53) is for the ULA case where the inter-sensor distance d is constant across the array.

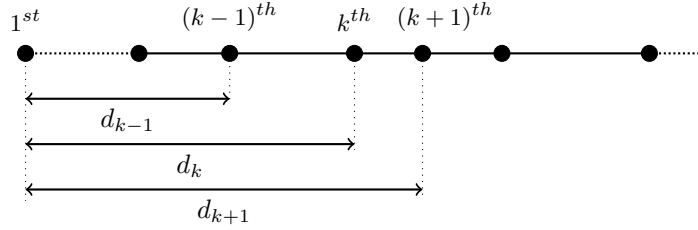
In the sequel, the following two scenarios are detailed:

- *i)* one sensor is perturbed during the measurement, so that the inter-sensor distance d is fluctuating for the k^{th} sensor with regards to its neighbors ($(k-1)^{th}$ and $(k+1)^{th}$ sensors) as depicted below:



where δ corresponds to the fluctuations of the k^{th} sensor position.

- *ii)* the angle of arrival $\theta(t)$ is varying with time, while the inter-sensor distance is constant during the measurement, but this inter-sensor distance may not be equal to d (a case of mis-placement of the antenna sensor leading to a non-uniform linear array, *i.e.* NULA, case). The distance between the first sensor of the array and the $(k+1)^{th}$ sensor is denoted d_k as illustrated below⁸:



Both scenarios are relevant in practice. For example, the first one can be encountered in seismology when the sensors are deployed on the ground and thus are subject to potential local ground motion, while the second one can be encountered in communications or tracking when the source is moving with respect to the array while imperfections affect the array shape such that it is no longer an ULA. In both scenarios, the steering vector can fluctuate with time, meaning that the time delay is a function of time, *i.e.* $\tau(t)$. As a consequence, the antenna signal $\mathbf{x}(t)$ now takes the form:

$$\mathbf{x}(t) = e^{-j2\pi\nu_0 t} \mathbf{s}(\tau(t)) \quad (54)$$

⁸We make use of this notation for the NULA as it allows to see the ULA as a special case. As d_k is the distance between the first and k^{th} , in the ULA case we have that $d_k = kd$. This notation is useful for comparison of geometric phases expressions in ULA and NULA cases.

where as said previously, $a(t)$ is set to 1 $\forall t$. The multivariate complex-valued signal $\mathbf{x}(t)$ of dimension N_s has a projection in $\mathbb{CP}^{N_s-1}$ given as

$$\rho_{\mathbf{x}}(t) = \frac{\mathbf{s}(\tau(t))\mathbf{s}^\dagger(\tau(t))}{\|\mathbf{s}(\tau(t))\|} \quad (55)$$

As $\tau(t)$ is fluctuating with time, $\rho_{\mathbf{x}}(t)$ follows a path in $\mathbb{CP}^{N_s-1}$ which induces a geometric phase for the steering vector. We now consider the two different scenarios and show how geometric phases can appear in these scenarios.

Scenario i)

Here, the inter-sensor distance d is fluctuating for the k^{th} sensor with regards to his neighbors. This can be modeled by letting $\tau(t)$ fluctuate with time as $\tau(t) = \tau_0 + \delta\tau(t)$ where $\tau_0 = (d/c)\cos\theta$ is the nominal time delay between two adjacent sensors and $\delta\tau(t)$ is a time-varying perturbation. The geometric phase accumulated during the measurement (time interval $[t_0, t]$) takes the following expression:

$$\begin{aligned} \phi_{g,\mathbf{x},t_0}(t) = \arg \left(\frac{\sin \left(2\pi\nu_0(k-1)\frac{\cos\theta}{c}(\delta\tau(t) - \delta\tau(t_0)) \right)}{N_s - 1 + \cos \left(2\pi\nu_0(k-1)\frac{\cos\theta}{c}(\delta\tau(t) - \delta\tau(t_0)) \right)} \right) \\ - \frac{2\pi\nu_0}{N_s}(k-1)\frac{\cos\theta}{c}(\delta\tau(t) - \delta\tau(t_0)) \end{aligned} \quad (56)$$

where one can see the dependency on the position of the perturbed sensor. As an illustration, choosing a perturbation of type $\delta\tau(t) = \alpha d \cos(2\pi\nu_1 t)$, with $|\alpha| \ll 1$ mimics a resonance phenomenon of the sensor due to some possible malfunction or external parasitic harmonic source. In Figure 11, the geometric phase is displayed (together with total and dynamical phase) for a ULA of $N_s = 15$ sensors, a carrying frequency $\nu_0 = 40$ Hz, an inter-sensor distance $d = 1$ m, a wave velocity $c = 1$ m.s⁻¹, an angle of arrival $\theta = 85.5^\circ$ and a perturbation with $\alpha = 0.1$ and $\nu_1 = 10$ Hz.

The geometric phase exhibits fluctuations with time due to the perturbation of the inter-sensor distance for the 5^{th} sensor. The amplitude of these fluctuations depends on the position $k = 5$ of the perturbed sensor, as can be seen in the expression of $\phi_{g,\mathbf{x},t_0}(t)$ (56). In this example, we set $k = 5$, which corresponds to the middle sensor of the array. The value of the geometric phase thus reveals the presence of a perturbation on the array, its location and frequency, and could be used to monitor/compensate this perturbation during the measurement. Interestingly, expression (56) generalizes naturally to the case of multiple sensors malfunctioning. The resulting geometric phase is simply the linear combination of the geometric phases for different values of k , with possible perturbation frequencies ν_k . While in the case of a single perturbation, the perturbation is monochromatic (as displayed in Figure 11), the geometric phase would be polychromatic in the case of multiple sensors affected by different perturbations.

Scenario ii)

In this second scenario, we study the effect of fluctuations over time of the direction of arrival $\theta(t)$. We provide a general expression (the NULA) and then shows the ULA

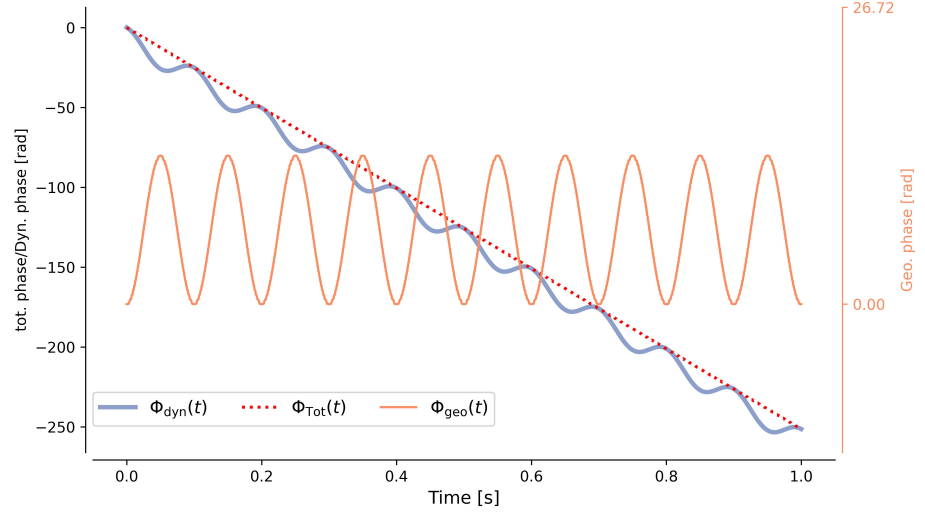


Fig. 11 Geometric phase (orange) accumulated along time for the case where the intersensor distance is fluctuating for the k^{th} sensor of the array (Scenario i) **NLB:Refaire la figure (label des axes et notation phases)**

version. Consider, as described earlier, a linear array made of N_s sensors, with intersensor distance between the first sensor and the k^{th} denoted by d_k . In such case, the geometric phase accumulated during the measurement (time interval $[t_0, t]$) takes the following expression:

$$\phi_{g,x,t_0}(t) = \arg \left(\sum_{k=0}^{N_s-1} \left(e^{2\pi j \nu_0 \frac{\cos \theta(t) - \cos \theta(t_0)}{c}} \right)^{d_k} \right) - 2\pi \frac{\nu_0 L}{N_s c} (\cos \theta(t) - \cos \theta(t_0)) \quad (57)$$

where we set $d_0 = 0$ and $L = \sum_{k=1}^{N_s-1} d_k$ is the sum of the distances from the first sensor to each sensor of the array. This expression is for a general linear array. Interestingly, for the special case of a ULA, where $d_k = k.d$, the geometric phase given in (57) is exactly equal to zero. This comes from the fact that when $d_k = k.d$, then $L = dN_s(N_s - 1)/2$ and the first term on the right hand side of (57) is equal to $\arg \left(\frac{1}{N_s} \frac{1-q^{N_s}}{1-q} \right)$, where $q = e^{j2\pi \nu_0 \frac{d}{c} (\cos \theta(t) - \cos \theta(t_0))}$. The fact that for a real number η $\arg \left(\frac{1-e^{j\eta n}}{1-e^{j\eta}} \right) = \eta(n-1)/2$, shows that the geometric case vanishes for the ULA case.

From this scenario, one can conclude that for a linear array, if the angle of arrival is evolving with time⁹, then the geometric phase is potentially non-null. Remarkably, the case where the geometric phase is exactly equal to zero is when the distance between each sensor is d . However, expression (57) is also null when θ is constant ($\theta(t_0) = \theta(t)$), so that the geometric phase needs some fluctuations of the incoming wave direction to

⁹This can even be generalized to a time fluctuation of τ in the model as stated in (54), as having θ varying with time is one way to make τ vary with time. c or d could also be varying with time (this last case was considered in a specific way in scenario i).

be non-zero. Potential application of this interesting property of the geometric phase of an array processing could be some type of calibration (test of the good positioning of the sensors) of an array for example.

NLB:Rajouter une remarque sur l'ordre de grandeur des variations de la phase géométrique en fonction de la longueur d'onde

The two scenarios presented here are for linear arrays. As stated previously, the case of non-linear arrays could be considered as well as it is widely used in applications.

7 Conclusion and future work

Appendix A Expression of the geometric phase for a bivariate signal

The geometric phase acquired during the interval $[t, t_0]$ is

$$\phi_{g,\mathbf{x},\mathbf{t}_0} = \arg \mathbf{p}_{\mathbf{x}}^\dagger(t_0)\mathbf{p}_{\mathbf{x}}(t) - \int_{t_0}^t \text{Im}(\mathbf{p}_{\mathbf{x}}^\dagger(s)\dot{\mathbf{p}}_{\mathbf{x}}(s)) ds$$

Omitting t and $\mathbf{x}$ for the sake of clarity, and recalling that the expression of $\mathbf{p}(\mathbf{t})$ is given in (25), we write

$$\begin{aligned} \mathbf{p}^\dagger(t)\dot{\mathbf{p}}(t) &= [\cos\theta(t)\cos\chi(t) - \mathbf{j}\sin\theta(t)\sin\chi(t), \sin\theta(t)\cos\chi(t) + \mathbf{j}\cos\theta(t)\sin\chi(t)] \\ &\cdot \begin{bmatrix} -\dot{\theta}(t)\sin\theta(t)\cos\chi(t) - \dot{\chi}(t)\cos\theta(t)\sin\chi(t) + \mathbf{j}\dot{\theta}(t)\sin\chi(t) + \mathbf{j}\dot{\chi}(t)\sin\theta(t)\cos\chi(t) \\ \dot{\theta}(t)\cos\theta(t)\cos\chi(t) - \dot{\chi}(t)\sin\theta(t)\sin\chi(t) + \mathbf{j}\dot{\theta}(t)\sin\theta(t)\sin\chi(t) - \mathbf{j}\dot{\chi}(t)\cos\theta(t)\cos\chi(t) \end{bmatrix} \end{aligned} \quad (\text{A1})$$

so that the imaginary part takes the form:

$$\begin{aligned} \text{Im}(\mathbf{p}^\dagger(t)\dot{\mathbf{p}}(t)) &= \dot{\theta}(t)\cos^2\theta(t)\cos\chi(t)\sin\chi(t) + \dot{\chi}(t)\cos\theta(t)\sin\theta(t)\cos^2\chi(t) \\ &\quad + \dot{\theta}(t)\sin^2\theta(t)\cos\chi(t)\sin\chi(t) + \dot{\chi}(t)\cos\theta(t)\sin\theta(t)\sin^2\chi(t) \\ &\quad + \dot{\theta}(t)\sin^2\theta(t)\cos\chi(t)\sin\chi(t) - \dot{\chi}(t)\sin\theta(t)\cos\theta(t)\cos^2\chi(t) \\ &\quad + \dot{\theta}(t)\cos^2\theta(t)\cos\chi(t)\sin\chi(t) - \dot{\chi}(t)\cos\theta(t)\sin\theta(t)\sin^2\chi(t) \end{aligned} \quad (\text{A2})$$

which simplifies into:

$$\text{Im}(\mathbf{p}^\dagger(t)\dot{\mathbf{p}}(t)) = \dot{\theta}(t)\sin 2\chi(t) \quad (\text{A3})$$

We now evaluate $\mathbf{p}_{\mathbf{x}}^\dagger(t_0)\mathbf{p}_{\mathbf{x}}(t)$ to find¹⁰

$$\begin{aligned} \mathbf{p}_{\mathbf{x}}^\dagger(t_0)\mathbf{p}_{\mathbf{x}}(t) &= \cos\theta_0\cos\chi_0\cos\theta\cos\chi + \sin\theta_0\sin\chi_0\sin\theta\sin\chi \\ &\quad + \mathbf{j}(\cos\theta_0\cos\chi_0\sin\theta\sin\chi - \sin\theta_0\sin\chi_0\cos\theta\cos\chi) \\ &\quad + \sin\theta_0\cos\chi_0\sin\theta\cos\chi + \cos\theta_0\sin\chi_0\cos\theta\sin\chi \\ &\quad + \mathbf{j}(-\sin\theta_0\cos\chi_0\cos\theta\sin\chi + \cos\theta_0\sin\chi_0\sin\theta\cos\chi) \\ &= \cos\theta_0\cos\theta\cos(\chi_0 - \chi) + \sin\theta_0\sin\theta\cos(\chi_0 - \chi) \\ &\quad + \mathbf{j}(-\sin\theta_0\cos\theta\sin(\chi + \chi_0) + \cos\theta_0\sin\theta\sin(\chi + \chi_0)) \\ &= \cos(\theta_0 - \theta)\cos(\chi_0 - \chi) + \mathbf{j}\sin(\theta - \theta_0)\sin(\chi + \chi_0) \end{aligned}$$

If $\theta = \theta_0$ at all time, the inner product is real, and the argument of $\mathbf{p}_{\mathbf{x}}^\dagger(t_0)\mathbf{p}_{\mathbf{x}}(t)$ is either 0 or π , depending on the sign of $\cos(\chi_0 - \chi)$. Recall that in this parametrization χ_0 varies in $[-\pi/4; \pi/4]$, but $\chi_{\mathbf{x}}(t)$ can take any value in $\mathbb{R}$. so that the argument of $\mathbf{p}_{\mathbf{x}}^\dagger(t_0)\mathbf{p}_{\mathbf{x}}(t)$ is a multiple of π . This explains the π jump experienced by the geometric phase when the state of the signal follows a geodesic and passes the antipodal point with respect to the initial state.

¹⁰We again omit t and $\mathbf{x}$, and (t_0) is replaced by $_0$ for the sake of clarity.

If $\theta \neq \theta_0$, the geometric phase is given by

$$\phi_{g,\mathbf{x},\mathbf{t}_0}(t) = \arctan\left(\frac{\sin(\theta - \theta_0)\sin(\chi + \chi_0)}{\cos(\theta - \theta_0)\cos(\chi - \chi_0)}\right) - \int_{t_0}^t \dot{\theta}_{\mathbf{x}}(s) \sin 2\chi_{\mathbf{x}}(s) ds$$

plus or minus a multiple of π depending on the branch of the arctangent.

Appendix B Instantaneous frequency as the first instantaneous moment of its energy density

Similarly to the univariate case, it is possible to show that the instantaneous frequency of a bivariate signal is the first moment of the energy density. This property is well known in time-frequency analysis [1, 2]. Here, we derive the calculation to show that the first moment of the energy density of a bivariate signal exhibits an extra term (in addition to the instantaneous phase) which is precisely the geometric phase. The derivation we use is based on complex vector valued signal. In [17, Section 1.4 & Appendix], similar results were presented using the quaternion formalism.

Consider an n -dimensional complex vector valued signal $\mathbf{x}(t) \in \mathbb{C}^n$, and its Fourier transform $\mathbf{X}(\nu) \in \mathbb{C}^n$. The first instantaneous moment $\nu_{\mathbf{x}}(t)$ is defined implicitly by

$$\int \nu \|\mathbf{X}(\nu)\|^2 d\nu = \int \nu_{\mathbf{x}}(t) \|\mathbf{x}(t)\|^2 dt \quad (\text{B4})$$

The left-hand side of this equation can be re-expressed as:

$$\int \nu \|\mathbf{X}(\nu)\|^2 d\nu = -\frac{j}{2\pi} \int \mathbf{x}^\dagger(\nu) \dot{\mathbf{x}}(\nu) d\nu = -\frac{j}{2\pi} \int \frac{\mathbf{x}^\dagger(\nu) \dot{\mathbf{x}}(\nu)}{|\mathbf{x}(t)|^2} \|\mathbf{x}(t)\|^2 dt \quad (\text{B5})$$

where we used the basic property Fourier transform for derivatives. Direct identification shows that the first instantaneous moment is given by

$$\nu_{\mathbf{x}}(t) = \frac{1}{2\pi} \text{Im} \frac{\mathbf{x}^\dagger(\nu) \dot{\mathbf{x}}(\nu)}{\|\mathbf{x}(t)\|^2} \quad (\text{B6})$$

In the bivariate case, this reduces to

$$\nu_{\mathbf{x}}(t) = -j \frac{x_1^*(t) \dot{x}_1(t) + x_2^*(t) \dot{x}_2(t)}{|x_1(t)|^2 + |x_2(t)|^2} \quad (\text{B7})$$

For unit norm signals, as shown above for parametrization given in (25), this reduces to

$$2\pi\nu_{\mathbf{x}}(t) = \dot{\phi}(t) + \dot{\theta}(t) \sin 2\chi(t) \quad (\text{B8})$$

This expression shows that for the bivariate case, the first instantaneous moment contains the frequency $\dot{\phi}(t)$ as well as the geometric term $\dot{\theta}(t) \sin 2\chi(t)$. Consequences of this extra geometric term are illustrated in Section 6.

Appendix C Bargmann invariant and the estimation of the geometric phase

Estimation and Bargmann invariant. Given n samples $\mathbf{x}(t_i)$ of the signal $x(t)$ on interval $[t_1, t_n]$ we develop here an estimation of its geometrical phase on this interval. We assume here that $\mathbf{x}(t)$ is of unit norm. If not, the proof below adapts easily by renormalizing. Consider the approximation of $\mathbf{x}(t)$ by $\tilde{\mathbf{x}}(t)$ on $[t_1, t_2]$ which interpolates the samples $\mathbf{x}(t_i)$, $\forall i = 1, \dots, n$ by geodesic arcs.

The geometric phase associated to $\tilde{\mathbf{x}}(t)$ is given as:

$$\begin{aligned} \phi_{g, \tilde{\mathbf{x}}, t_1}(t_n) &= \phi_{\tilde{\mathbf{x}}, t_1}(t_n) - \phi_{d, \tilde{\mathbf{x}}, t_1}(t_n) \\ &= \arg(\tilde{\mathbf{x}}^\dagger(t_1)\tilde{\mathbf{x}}(t_n)) - \sum_{r=1}^{n-1} \phi_{d, \tilde{\mathbf{x}}, t_r}(t_{r+1}) \\ &= \arg(\tilde{\mathbf{x}}^\dagger(t_1)\tilde{\mathbf{x}}(t_n)) - \sum_{r=1}^{n-1} \arg(\tilde{\mathbf{x}}^\dagger(t_r)\tilde{\mathbf{x}}(t_{r+1})) - \underbrace{\phi_{g, \tilde{\mathbf{x}}, t_r}(t_{r+1})}_{=0 \text{ on a geodesic}} \\ &= -\arg(\tilde{\mathbf{x}}^\dagger(t_1)\tilde{\mathbf{x}}(t_2)\tilde{\mathbf{x}}^\dagger(t_2))\tilde{\mathbf{x}}(t_3) \cdots \tilde{\mathbf{x}}^\dagger(t_{n-1})\tilde{\mathbf{x}}(t_n)\tilde{\mathbf{x}}^\dagger(t_n)\tilde{\mathbf{x}}(t_1)) \\ &= -\arg \text{Tr}(\tilde{\mathbf{x}}(t_1)\tilde{\mathbf{x}}^\dagger(t_1)\tilde{\mathbf{x}}(t_2)\tilde{\mathbf{x}}^\dagger(t_2))\tilde{\mathbf{x}}(t_3) \cdots \tilde{\mathbf{x}}^\dagger(t_{n-1})\tilde{\mathbf{x}}(t_n)\tilde{\mathbf{x}}^\dagger(t_n)) \end{aligned}$$

But $\tilde{\mathbf{x}}(t_i) = \mathbf{x}(t_i)$ because $\tilde{\mathbf{x}}(t)$ on $[t_1, t_2]$ interpolates the samples $\mathbf{x}(t_i)$. Let Bargman invariant $\mathbf{B}_x^{1:n}$ be defined as

$$\mathbf{B}_x^{1:n} = \text{Tr}(\rho_x(t_1)\rho_x(t_2) \cdots \rho_x(t_{n-1})\rho_x(t_n))$$

We have shown

$$\phi_{g, \tilde{\mathbf{x}}, t_1}(t_n) = -\arg \mathbf{B}_x^{1:n} \approx \phi_{g, \mathbf{x}, t_1}(t_n)$$

Recursive estimator. From the calculations above, we can obtain a recursive formulation of the estimator of the geometric phase. We saw that

$$\phi_{g, \tilde{\mathbf{x}}, t_1}(t_n) = -\sum_{r=1}^{n-1} \arg(\mathbf{x}^\dagger(t_r)\mathbf{x}(t_{r+1})) - \arg(\mathbf{x}^\dagger(t_n)\mathbf{x}(t_1))$$

This can be expanded as

$$\phi_{g, \tilde{\mathbf{x}}, t_1}(t_n) = -\arg(\mathbf{x}^\dagger(t_{n-1})\mathbf{x}(t_1)) - \sum_{r=1}^{n-2} \arg(\mathbf{x}^\dagger(t_r)\mathbf{x}(t_{r+1})) \quad (\text{C9})$$

$$+ \arg(\mathbf{x}^\dagger(t_{n-1})\mathbf{x}(t_n)) - \arg(\mathbf{x}^\dagger(t_n)\mathbf{x}(t_1)) \quad (\text{C10})$$

$$= \phi_{g, \hat{x}, t_1}(t_{n-1}) - \arg \frac{\mathbf{x}^\dagger(t_{n-1})\mathbf{x}(t_n)\mathbf{x}^\dagger(t_n)\mathbf{x}(t_1)}{\mathbf{x}^\dagger(t_{n-1})\mathbf{x}(t_1)} \quad (\text{C11})$$

$$(\text{C12})$$

where we use the additivity of arguments and a simple rearrangement of the terms to obtain the last line expression.

Error of estimation: A simple example.

Error of estimation: A simple proof.

We give here a simplified proof in support of the $1/n^2$ behavior of the estimator. The geometric phase corresponds to the circulation of a 1-form along a curve. We consider this more general problem. Let A_x be a 1-form on $\mathcal{M}$, *i.e.* $\forall x \in \mathcal{M}$, $A_x : T_x \mathcal{M} \rightarrow \mathbb{R}, v \mapsto A_x(v)$. We suppose for the sake of simplicity that $\mathcal{M}$ is an N -dimensional Riemannian manifold embedded in a Euclidean space.

Let $\mathcal{C}$ be defined by $\gamma : [0, T] \in \mathbb{R} \rightarrow \mathcal{M}$. The estimation of the geometric phase based on Bargmann invariant amounts to approximate $\mathcal{C}$ sampled at t_k by an interpolating curve $\mathcal{C}_n$ where each segment $\mathcal{C}_k^{k+1}$ is a geodesic arc joining the points $\gamma(t_k)$ and $\gamma(t_{k+1})$.

If the curve is not closed, Bargman invariant implicitly closes it by the geodesic linking the extremities. We thus study the approximation of the geometric phase calculated on $\mathcal{C} \cup \mathcal{G}$ by the geometric phase calculated on $\mathcal{C}_n \cup \mathcal{G}$.

We want to evaluate the error

$$\begin{aligned} \Delta_n &= \oint_{\mathcal{C} \cup \mathcal{G}} A_x - \oint_{\mathcal{C}_n \cup \mathcal{G}} A_x = \int_{\mathcal{C}} A_x - \int_{\mathcal{G}} A_x - \int_{\mathcal{C}_n} A_x + \int_{\mathcal{G}} A_x = \oint_{\mathcal{C} \cup \mathcal{C}_n} A_x \\ &= \sum_{k=0}^{n-1} \oint_{\mathcal{C}|_{[t_k, t_{k+1}]} \cup \mathcal{C}_k^{k+1}} A_x = \sum_{k=0}^{n-1} \Delta_k \end{aligned}$$

We thus need to evaluate the circulation of A_x along the edges of the small sets $\mathcal{C}|_{[t_k, t_{k+1}]} \cup \mathcal{C}_k^{k+1}$. Let a surface $\mathcal{S}_k$ such that $\partial \mathcal{S}_k = \mathcal{C}|_{[t_k, t_{k+1}]} \cup \mathcal{C}_k^{k+1}$. Thanks to Stokes theorem, we have

$$\Delta_k = \iint_{\mathcal{S}_k} dA_x$$

where dA_x is the exterior derivatives of A_x . If n is sufficiently large¹¹ the domain $\mathcal{S}_k$ is very small: it can be approximated by $\tilde{\mathcal{S}}_k = \tilde{\mathcal{C}}|_{[t_k, t_{k+1}]} \cup \tilde{\mathcal{C}}_k^{k+1}$ where $\tilde{\mathcal{C}}|_{[t_k, t_{k+1}]}$ is a cubic approximation $\mathcal{C}|_{[t_k, t_{k+1}]}$, and $\tilde{\mathcal{C}}_k^{k+1}$ the geodesic that closes it. Then we have the approximation

$$\Delta_k = \iint_{\mathcal{S}_k} dA_x \approx \iint_{\tilde{\mathcal{S}}_k} dA_x$$

¹¹This point should be developed beyond this heuristics. A rigorous derivation should take into account the injection radius of the manifold, which gives the size of the domain in which the exponential map is a diffeomorphism. We let this for future work.

We reparametrize $\gamma(s)$ such that $\gamma(0) = 0$ and $\|\dot{\gamma}(s)\| = 1$ for $s \in [0, \delta_k]$ where $\delta_k = (t_{k+1} - t_k)$. We make use of the local Frenet frame $e_j(s), j = 1, \dots, N$ at $\gamma(0)$. The cubic approximation of $\tilde{\mathcal{C}}|_{[t_k, t_{k+1}]}$ can be parametrized by $\tilde{\gamma}(s) = (s - s^3/6\kappa_g^2)e_1(s) + (\kappa_g s^2/2 + \dot{\kappa}_g/6s^3)e_2(s) + \alpha s^3 e_3(s) + O(s^4)$, α being a constant we do not need, and where $\kappa_g = \|\nabla \dot{\gamma}|_{s=0}\|$ the geodesic curvature of the curve at t_k . We denote by $x_i(s)$ the coordinate of the curve along $e_i(s)$. Thus $x_1(s) = (s - s^3/6\kappa_g^2)$, $x_2(s) = (\kappa_g s^2/2 + \dot{\kappa}_g/6s^3)$ and $x_3(s) = \alpha s^3$. Hence $dx_1(s) = (1 - s^2/2\kappa_g^2)ds$, $dx_2(s) = (\kappa_g s + \dot{\kappa}_g/2s^2)ds$ and $dx_3(s) = 3\alpha s^2$.

Thus, limited to the space generated by (e_1, e_2, e_3) , we have

$$\Delta_k \approx \iint_{\tilde{\mathcal{S}}_k} dA_x(e_1, e_2) dx_1 \wedge dx_2 + \iint_{\tilde{\mathcal{S}}_k} dA_x(e_2, e_3) dx_2 \wedge dx_3 + \iint_{\tilde{\mathcal{S}}_k} dA_x(e_1, e_3) dx_1 \wedge dx_3$$

Assuming $dA_x(e_i, e_j)$ constant on the infinitesimal $\tilde{\mathcal{S}}_k$, each individual integral above reduces to

$$\begin{aligned} \iint_{\tilde{\mathcal{S}}_k} dA_x(e_i, e_j) dx_i \wedge dx_j &= dA_x(e_i(t_k), e_j(t_k)) \iint_{\tilde{\mathcal{S}}_k} dx_i \wedge dx_j \\ &= \frac{dA_x(e_i(t_k), e_j(t_k))}{2} \oint_{\partial \tilde{\mathcal{S}}_k} (x_i dx_j - x_j dx_i) \end{aligned}$$

Analyzing the dominant terms in s in each of $(x_i dx_j - x_j dx_i)$, it appears that the dominant term in the previous calculation is provided by the (1,2) term. Since along the geodesic in $\mathbb{R}^2$, x_1 is proportional to x_2 , the contribution to the circulation comes only from the circulation along $\tilde{\mathcal{C}}|_{[t_k, t_{k+1}]}$ and thus

$$\begin{aligned} \Delta_k &\approx dA_x(e_i(t_k), e_j(t_k)) \int_0^{\delta_k} \left(x_1(s) \frac{dx_2(s)}{ds} - x_2 \frac{dx_1(s)}{ds} \right) ds \\ &= dA_x(e_i(t_k), e_j(t_k)) \kappa_g(t_k) \frac{\delta_k^3}{12} + O(\delta_k^4) \end{aligned}$$

To simplify, suppose that $\delta_k = T/n$, then we finish with

$$\begin{aligned} \Delta_n &= \sum_{k=0}^{n-1} \Delta_k = \frac{T^3}{12n^3} \sum_{k=0}^{n-1} \left(dA_x(e_i(t_k), e_j(t_k)) \kappa_g(t_k) + O\left(\frac{1}{n^4}\right) \right) \\ &\approx \frac{T^2}{12n^2} \int_0^T dA_x(e_i(t), e_j(t)) \kappa_g(t) dt + O\left(\frac{1}{n^3}\right) \end{aligned}$$

where we have approximated the sum by the Riemann integral. This confirms the $1/n^2$ behavior of the estimate of the geometric phase. A more general proof is the object of future works, along with a robustness analysis against the possible presence of observation noise.

Appendix D Decomposition of the 3D geometric phase into the 2D corrected by fluctuations of the elliptic plane

In the sequel we omit all unnecessary indexes variables such $\mathbf{x}$ or (t) , and t_0 will be represented by a 0. We have $\mathbf{p} = J_3(\gamma)J_1(\delta)(\mathbf{p}_2^\top 0)^\top$ so that its derivative writes

$$\dot{\mathbf{p}} = \dot{J}_3(\gamma)J_1(\delta) \begin{pmatrix} \mathbf{p}_2 \\ 0 \end{pmatrix} + J_3(\gamma)\dot{J}_1(\delta) \begin{pmatrix} \mathbf{p}_2 \\ 0 \end{pmatrix} + J_3(\gamma)J_1(\delta) \begin{pmatrix} \dot{\mathbf{p}}_2 \\ 0 \end{pmatrix}$$

The derivative of the matrices are given by

$$\dot{J}_3(\gamma) = \dot{\gamma} \begin{pmatrix} -\sin(\gamma) & -\cos(\gamma) & 0 \\ \cos(\gamma) & -\sin(\gamma) & 0 \\ 0 & 0 & 0 \end{pmatrix} = -\dot{\gamma} \begin{pmatrix} \cos(\gamma - \frac{\pi}{2}) & -\sin(\gamma - \frac{\pi}{2}) & 0 \\ \sin(\gamma - \frac{\pi}{2}) & \cos(\gamma - \frac{\pi}{2}) & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

(and likewise for J_1). Furthermore, $J_3^\dagger(\gamma) = J_3(-\gamma)$ so that $J_3(-\gamma)J_3(\gamma) = \mathbf{I}$ and

$$J_3^\dagger(\gamma)\dot{J}_3(\gamma) = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

(and likewise for J_1). Hence we have

$$\begin{aligned} \mathbf{p}^\dagger \dot{\mathbf{p}} &= (\mathbf{p}_2^\dagger 0) \left(-\dot{\gamma}J_1(-\delta) \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} J_1(\delta) \begin{pmatrix} \mathbf{p}_2 \\ 0 \end{pmatrix} - \dot{\delta} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix} \begin{pmatrix} \mathbf{p}_2 \\ 0 \end{pmatrix} + \begin{pmatrix} \dot{\mathbf{p}}_2 \\ 0 \end{pmatrix} \right) \\ &= -\dot{\gamma}(\mathbf{p}_2^\dagger 0) \begin{pmatrix} 0 & \cos(\delta) & -\sin(\delta) \\ -\cos(\delta) & 0 & 0 \\ \sin(\delta) & 0 & 0 \end{pmatrix} \begin{pmatrix} \mathbf{p}_2 \\ 0 \end{pmatrix} + \mathbf{p}_2^\dagger \dot{\mathbf{p}}_2 \\ &= -\dot{\gamma} \cos(\delta)(p_{21}^* p_{22} - p_{21} p_{22}^*) + \mathbf{p}_2^\dagger \dot{\mathbf{p}}_2 \end{aligned}$$

The term $-(p_{21}^* p_{22} - p_{21} p_{22}^*)$ is shown to be equal to $j \sin(2\chi)$ where χ is the excentricity of the ellipse described by $\mathbf{p}_2$.

The same kind of calculations allows to obtained

$$\begin{aligned} \mathbf{p}_0^\dagger \mathbf{p} &= \cos(\gamma - \gamma_0)p_{21}^*(0)p_{21} - \sin(\gamma - \gamma_0)(\cos(\delta_0)p_{22}^*(0)p_{21} - \cos(\delta)p_{21}^*(0)p_{22}) \\ &\quad + (\cos(\delta_0)\cos(\delta)\cos(\gamma - \gamma_0) + \sin(\delta_0)\sin(\delta))p_{22}^*(0)p_{22} \end{aligned}$$

Thus if γ and δ are constant, we verify that $\mathbf{p}(t_0)^\dagger \mathbf{p}(t) = \mathbf{p}_2(t_0)^\dagger \mathbf{p}_2(t)$.

Appendix E Majorana stellar representation

Pauli matrices. At this point and to prepare the following, it is interesting to link $\mathbb{CP}^1$ to $\mathcal{S}^2$ explicitly with the help of Pauli matrices. Let $\mathbf{p} \in \mathbb{C}^2$ with unit norm. Then

$\pi(\mathbf{p}) = \mathbf{p}\mathbf{p}^\dagger$ lies in $\mathbb{CP}^1$ and can be parametrized with Pauli matrices: There exists a vector $\mathbf{u} \in \mathbb{R}^3$ with unit norm such that $\pi(\mathbf{p}) = (\mathbf{I} + \mathbf{u} \cdot \boldsymbol{\sigma})/2$ where $\mathbf{u} \cdot \boldsymbol{\sigma} = \sum_i u_i \sigma_i$, and where the Pauli matrices are defined by

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -j \\ j & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

These matrices are traceless, orthogonal with respect to $\text{Tr}(\sigma_i \sigma_j)$ and satisfy $\sigma_1 \sigma_2 = j \sigma_3, \sigma_1 \sigma_3 = -j \sigma_2, \sigma_2 \sigma_3 = j \sigma_1$. Vector $\mathbf{u}$ defines a dot on the Poincaré sphere. For the Jones vector above, $\mathbf{u}$ is written in spherical coordinate as $(\sin 2\theta \cos \chi, \sin(2\theta) \sin \chi, \cos(2\theta))$. Therefore when the two dimensional complex vector $\mathbf{p}(t)$ traces a curve in $\mathbb{C}^2$, its projection by π traces a curve in $\mathbb{CP}^1$ which can be identified with the curve traced on the sphere by the associated vector $\mathbf{u}(t)$.

The geometry of Majorana's constellation for $n = 3$. Let $\mathbf{p} \in \mathbb{C}^3$ and $\mathbf{p}_1$ and $\mathbf{p}_2$ in $\mathbb{C}^2$. $\mathbf{p}_i$ projects by π on the 2 dimensional sphere by a dot (or star) represented by euclidean vector $\mathbf{u}_i$ which satisfies $\pi(\mathbf{p}_i) = \mathbf{p}_i \mathbf{p}_i^\dagger = (\mathbf{I} + \mathbf{u}_i \cdot \boldsymbol{\sigma})/2$, where $\boldsymbol{\sigma}$ is the basis of Pauli matrices. Then

$$\mathbf{p} = \frac{1}{\sqrt{2A_2}} \tilde{\mathbf{p}} \quad \text{where} \quad \tilde{\mathbf{p}} = \mathbf{p}_1 \otimes \mathbf{p}_2 + \mathbf{p}_2 \otimes \mathbf{p}_1 \quad (\text{E13})$$

and

$$2A_2 = \tilde{\mathbf{p}}^\dagger \tilde{\mathbf{p}} = 1 + \frac{1 + \mathbf{u}_1 \cdot \mathbf{u}_2}{2} = \frac{3 + \mathbf{u}_1 \cdot \mathbf{u}_2}{2} \quad (\text{E14})$$

This can be shown using Pauli representation and $|\mathbf{p}_1^\dagger \mathbf{p}_2|^2 = (1 + \mathbf{u}_1 \cdot \mathbf{u}_2)/2$. Berry's connection is given by $jA_{\mathbf{p}}(\dot{\mathbf{p}}) = j\mathbf{p}^\dagger \dot{\mathbf{p}}$. We have

$$\begin{aligned} \dot{\mathbf{p}} &= -\frac{\dot{A}_2}{2\sqrt{2}A_2^{3/2}} \tilde{\mathbf{p}} + \frac{1}{\sqrt{2A_2}} \dot{\tilde{\mathbf{p}}} \\ &= -\frac{\dot{A}_2}{2\sqrt{2}A_2^{3/2}} \tilde{\mathbf{p}} + \frac{1}{\sqrt{2A_2}} (\dot{\mathbf{p}}_1 \otimes \mathbf{p}_2 + \dot{\mathbf{p}}_2 \otimes \mathbf{p}_1 + \mathbf{p}_1 \otimes \dot{\mathbf{p}}_2 + \mathbf{p}_2 \otimes \dot{\mathbf{p}}_1) \end{aligned}$$

leading to (recall $(\mathbf{p}_1 \otimes \mathbf{p}_2)^\dagger (\mathbf{y}_1 \otimes \mathbf{y}_2) = (\mathbf{p}_1^\dagger \mathbf{y}_1)(\mathbf{p}_2^\dagger \mathbf{y}_2)$)

$$\mathbf{p}^\dagger \dot{\mathbf{p}} = -\frac{\dot{A}_2}{4A_2} + \frac{1}{2A_2} \left(2\mathbf{p}_1^\dagger \dot{\mathbf{p}}_1 + 2\mathbf{p}_2^\dagger \dot{\mathbf{p}}_2 + 2(\mathbf{p}_1^\dagger \mathbf{p}_2)(\mathbf{p}_2^\dagger \dot{\mathbf{p}}_1) + 2(\mathbf{p}_2^\dagger \mathbf{p}_1)(\mathbf{p}_1^\dagger \dot{\mathbf{p}}_2) \right)$$

Now, since $2A_2 = (\mathbf{p}_1 \otimes \mathbf{p}_2 + \mathbf{p}_2 \otimes \mathbf{p}_1)^\dagger (\mathbf{p}_1 \otimes \mathbf{p}_2 + \mathbf{p}_2 \otimes \mathbf{p}_1) = 2 + 2|\mathbf{p}_2^\dagger \mathbf{p}_1|^2$, we have

$$\dot{A}_2 = \left(\mathbf{p}_2^\dagger \mathbf{p}_1 (\dot{\mathbf{p}}_1^\dagger \mathbf{p}_2 + \mathbf{p}_1^\dagger \dot{\mathbf{p}}_2) + \mathbf{p}_1^\dagger \mathbf{p}_2 (\dot{\mathbf{p}}_2^\dagger \mathbf{p}_1 + \mathbf{p}_2^\dagger \dot{\mathbf{p}}_1) \right)$$

and

$$\mathbf{p}^\dagger \dot{\mathbf{p}} = \frac{1}{A_2} \left(\mathbf{p}_1^\dagger \dot{\mathbf{p}}_1 + \mathbf{p}_2^\dagger \dot{\mathbf{p}}_2 + \frac{1}{2}(\mathbf{p}_1^\dagger \mathbf{p}_2)(\mathbf{p}_2^\dagger \dot{\mathbf{p}}_1 - \dot{\mathbf{p}}_2^\dagger \mathbf{p}_1) + \frac{1}{2}(\mathbf{p}_2^\dagger \mathbf{p}_1)(\mathbf{p}_1^\dagger \dot{\mathbf{p}}_2 - \dot{\mathbf{p}}_1^\dagger \mathbf{p}_2) \right)$$

Let us explicitly evaluate $(\mathbf{p}_1^\dagger \mathbf{p}_2)(\mathbf{p}_2^\dagger \dot{\mathbf{p}}_1 - \dot{\mathbf{p}}_2^\dagger \mathbf{p}_1)$ using Pauli matrices and real vectors $\mathbf{u}_i$.

We have $\mathbf{p}_1^\dagger \mathbf{p}_2 \mathbf{p}_2^\dagger \dot{\mathbf{p}}_1 = \text{Tr} \left(\mathbf{p}_2 \mathbf{p}_2^\dagger \dot{\mathbf{p}}_1 \mathbf{p}_1^\dagger \right)$. But $\dot{\mathbf{p}}_1 \mathbf{p}_1^\dagger = \overbrace{\mathbf{p}_1 \mathbf{p}_1^\dagger}^{\dot{\mathbf{p}}_1 \mathbf{p}_1^\dagger} - \mathbf{p}_1 \dot{\mathbf{p}}_1^\dagger$ so that $\dot{\mathbf{p}}_1 \mathbf{p}_1^\dagger (\mathbf{p}_1 \mathbf{p}_1^\dagger) = \overbrace{\mathbf{p}_1 \mathbf{p}_1^\dagger (\mathbf{p}_1 \mathbf{p}_1^\dagger)}^{\dot{\mathbf{p}}_1 \mathbf{p}_1^\dagger} - \mathbf{p}_1 \dot{\mathbf{p}}_1^\dagger (\mathbf{p}_1 \mathbf{p}_1^\dagger)$ and this leads to $\dot{\mathbf{p}}_1 \mathbf{p}_1^\dagger = \overbrace{\mathbf{p}_1 \mathbf{p}_1^\dagger (\mathbf{p}_1 \mathbf{p}_1^\dagger)}^{\dot{\mathbf{p}}_1 \mathbf{p}_1^\dagger} + \mathbf{p}_1^\dagger \dot{\mathbf{p}}_1 (\mathbf{p}_1 \mathbf{p}_1^\dagger)$ because $\mathbf{p}_1$ is of unit norm. Thus

$$\begin{aligned} \mathbf{p}_1^\dagger \mathbf{p}_2 \mathbf{p}_2^\dagger \dot{\mathbf{p}}_1 &= \text{Tr} \left(\mathbf{p}_2 \mathbf{p}_2^\dagger \dot{\mathbf{p}}_1 \mathbf{p}_1^\dagger \right) \\ &= \text{Tr} \left(\mathbf{p}_2 \mathbf{p}_2^\dagger \overbrace{\mathbf{p}_1 \mathbf{p}_1^\dagger}^{\dot{\mathbf{p}}_1 \mathbf{p}_1^\dagger} (\mathbf{p}_1 \mathbf{p}_1^\dagger) \right) + \mathbf{p}_1^\dagger \dot{\mathbf{p}}_1 \text{Tr} \left(\mathbf{p}_2 \mathbf{p}_2^\dagger \mathbf{p}_1 \mathbf{p}_1^\dagger \right) \\ &= \frac{1}{8} \text{Tr} \left((\mathbf{I} + \mathbf{u}_2 \cdot \sigma)(\dot{\mathbf{u}}_1 \cdot \sigma)(\mathbf{I} + \mathbf{u}_1 \cdot \sigma) \right) + \mathbf{p}_1^\dagger \dot{\mathbf{p}}_1 |\mathbf{p}_2^\dagger \mathbf{p}_1|^2 \end{aligned}$$

Recall that $(\mathbf{u}_1 \cdot \sigma)(\mathbf{u}_2 \cdot \sigma) = \mathbf{u}_1 \cdot \mathbf{u}_2 + j(\mathbf{u}_1 \wedge \mathbf{u}_2) \cdot \sigma$ and that $\text{Tr} \sigma_i = 0$. Then we have

$$\begin{aligned} \mathbf{p}_1^\dagger \mathbf{p}_2 \mathbf{p}_2^\dagger \dot{\mathbf{p}}_1 &= \frac{1}{8} \text{Tr} \left((\mathbf{I} + \mathbf{u}_2 \cdot \sigma)(\dot{\mathbf{u}}_1 \cdot \sigma + \dot{\mathbf{u}}_1 \cdot \mathbf{u}_1 \mathbf{I} + j(\dot{\mathbf{u}}_1 \wedge \mathbf{u}_1) \cdot \sigma) \right) + \mathbf{p}_1^\dagger \dot{\mathbf{p}}_1 |\mathbf{p}_2^\dagger \mathbf{p}_1|^2 \\ &= \frac{1}{4} (\dot{\mathbf{u}}_1 \cdot \mathbf{u}_1 + \mathbf{u}_2 \cdot \dot{\mathbf{u}}_1 + j \mathbf{u}_2 \cdot (\dot{\mathbf{u}}_1 \wedge \mathbf{u}_1)) + \mathbf{p}_1^\dagger \dot{\mathbf{p}}_1 |\mathbf{p}_2^\dagger \mathbf{p}_1|^2 \end{aligned}$$

Likewise we get for the other terms

$$\begin{aligned} \mathbf{p}_1^\dagger \mathbf{p}_2 \dot{\mathbf{p}}_2^\dagger \mathbf{p}_1 &= \overline{\mathbf{p}_2^\dagger \mathbf{p}_1 \dot{\mathbf{p}}_2^\dagger} \\ &= \frac{1}{4} (\dot{\mathbf{u}}_2 \cdot \mathbf{u}_2 + \mathbf{u}_1 \cdot \dot{\mathbf{u}}_2 - j \mathbf{u}_1 \cdot (\dot{\mathbf{u}}_2 \wedge \mathbf{u}_2)) + \overline{\mathbf{p}_2^\dagger \dot{\mathbf{p}}_2} |\mathbf{p}_2^\dagger \mathbf{p}_1|^2 \\ \mathbf{p}_2^\dagger \mathbf{p}_1 \mathbf{p}_1^\dagger \dot{\mathbf{p}}_2 &= \frac{1}{4} (\dot{\mathbf{u}}_2 \cdot \mathbf{u}_2 + \mathbf{u}_1 \cdot \dot{\mathbf{u}}_2 + j \mathbf{u}_1 \cdot (\dot{\mathbf{u}}_2 \wedge \mathbf{u}_2)) + \mathbf{p}_2^\dagger \dot{\mathbf{p}}_2 |\mathbf{p}_2^\dagger \mathbf{p}_1|^2 \\ \mathbf{p}_2^\dagger \mathbf{p}_1 \dot{\mathbf{p}}_1^\dagger \mathbf{p}_2 &= \overline{\mathbf{p}_1^\dagger \mathbf{p}_2 \mathbf{p}_2^\dagger \dot{\mathbf{p}}_1} \\ &= \frac{1}{4} (\dot{\mathbf{u}}_1 \cdot \mathbf{u}_1 + \mathbf{u}_2 \cdot \dot{\mathbf{u}}_1 - j \mathbf{u}_2 \cdot (\dot{\mathbf{u}}_1 \wedge \mathbf{u}_1)) + \overline{\mathbf{p}_1^\dagger \dot{\mathbf{p}}_1} |\mathbf{p}_2^\dagger \mathbf{p}_1|^2 \end{aligned}$$

Then $\mathbf{p}^\dagger \dot{\mathbf{p}}$ can be shown to read

$$\mathbf{p}^\dagger \dot{\mathbf{p}} = \frac{1}{A_2} \left(\mathbf{p}_1^\dagger \dot{\mathbf{p}}_1 + \mathbf{p}_2^\dagger \dot{\mathbf{p}}_2 + \frac{1}{4} j(\dot{\mathbf{u}}_1 - \dot{\mathbf{u}}_2) \cdot (\mathbf{u}_1 \wedge \mathbf{u}_2) + |\mathbf{p}_2^\dagger \mathbf{p}_1|^2 j \text{Im} \left(\mathbf{p}_1^\dagger \dot{\mathbf{p}}_1 + \mathbf{p}_2^\dagger \dot{\mathbf{p}}_2 \right) \right)$$

Writing $1/A_2 = 1 - (1 - A_2)/A_2$ and noting that $A_2 = |\mathbf{p}_2^\dagger \mathbf{p}_1|^2$ allows to write

$$\mathbf{p}^\dagger \dot{\mathbf{p}} = \mathbf{p}_1^\dagger \dot{\mathbf{p}}_1 + \mathbf{p}_2^\dagger \dot{\mathbf{p}}_2 + \frac{1}{A_2} \left(\frac{1}{4} \mathbf{j}(\dot{\mathbf{u}}_1 - \dot{\mathbf{u}}_2) \cdot (\mathbf{u}_1 \wedge \mathbf{u}_2) + |\mathbf{p}_2^\dagger \mathbf{p}_1|^2 \left(\mathbf{j} \text{Im} \left(\mathbf{p}_1^\dagger \dot{\mathbf{p}}_1 + \mathbf{p}_2^\dagger \dot{\mathbf{p}}_2 \right) - \mathbf{p}_1^\dagger \dot{\mathbf{p}}_1 + \mathbf{p}_2^\dagger \dot{\mathbf{p}}_2 \right) \right)$$

and finally

$$\text{Im}(\mathbf{p}^\dagger \dot{\mathbf{p}}) = \text{Im}(\mathbf{p}_1^\dagger \dot{\mathbf{p}}_1) + \text{Im}(\mathbf{p}_2^\dagger \dot{\mathbf{p}}_2) + \frac{1}{2} \frac{(\dot{\mathbf{u}}_1 - \dot{\mathbf{u}}_2) \cdot (\mathbf{u}_1 \wedge \mathbf{u}_2)}{3 + \mathbf{u}_1 \cdot \mathbf{u}_2}$$

Let $\mathbf{R} = (\mathbf{u}_1 + \mathbf{u}_2)/2$ and $\mathbf{r} = \mathbf{u}_1 - \mathbf{u}_2$, and $\tilde{\mathbf{R}}, \tilde{\mathbf{r}}$ their normalized counterparts. Then $\mathbf{u}_1 = \mathbf{R} + \mathbf{r}/2$ and $\mathbf{u}_2 = \mathbf{R} - \mathbf{r}/2$. Then the time derivative of $\tilde{\mathbf{r}}$ can be written as

$$\dot{\tilde{\mathbf{r}}} = \frac{\dot{\mathbf{r}}}{\|\mathbf{r}\|} - \tilde{\mathbf{r}} \frac{\dot{\mathbf{r}}^\top \mathbf{r}}{\mathbf{r}^\top \mathbf{r}}$$

and thus

$$\begin{aligned} (\dot{\mathbf{u}}_1 - \dot{\mathbf{u}}_2) \cdot (\mathbf{u}_1 \wedge \mathbf{u}_2) &= (\mathbf{r} \wedge \mathbf{R}) \cdot \dot{\mathbf{r}} \\ &= \|\mathbf{r}\| \|\mathbf{R}\| (\tilde{\mathbf{r}} \wedge \tilde{\mathbf{R}}) \cdot (\|\mathbf{r}\| \dot{\tilde{\mathbf{r}}} + \gamma \tilde{\mathbf{r}}) \\ &= \|\mathbf{r}\|^2 \|\mathbf{R}\| (\tilde{\mathbf{r}} \wedge \tilde{\mathbf{R}}) \cdot \dot{\tilde{\mathbf{r}}} \end{aligned}$$

Then, expliciting the different norms leads to

$$\frac{1}{2} \frac{(\dot{\mathbf{u}}_1 - \dot{\mathbf{u}}_2) \cdot (\mathbf{u}_1 \wedge \mathbf{u}_2)}{3 + \mathbf{u}_1 \cdot \mathbf{u}_2} = f(\psi) \cos(\psi) (\tilde{\mathbf{r}} \wedge \tilde{\mathbf{R}}) \cdot \dot{\tilde{\mathbf{r}}}$$

where $2\psi = \arccos(\mathbf{u}_1 \cdot \mathbf{u}_2)$ and $f(\psi) = \sin^2(\psi)/(1 + \cos^2(\psi))$.

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